
Software Requirements Specification

for

Study Group Management System

Version 1.0 approved

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March 15, 2026

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1. Introduction

1.1 Purpose

The purpose of this SRS is to document the complete functional and non-functional requirements for the **Study Group Management System**. This document is intended to be used by the development team as a blueprint for system design, implementation, and testing, and by stakeholders to understand the scope and expected functionality of the final product. This SRS clearly defines what the **Study Group Management System** platform is expected to do, covering all user interactions from secure user authentication and profile management to the creation and organization of study groups, the sharing of educational resources, and real-time collaboration through integrated text and video chat features.

1.2 Document Conventions

The following conventions are used throughout this document:

Convention	Meaning	Example
SGMS	Study Group Management System	SGMS shall allow users to select seats.

[BR-XXX]	Business Rule	The system shall enforce [BR-01]: Tickets cannot be refunded after the show starts.
[REQ-XXX]	Functional Requirement	[REQ-1.1] Allow customer login.
[NFR-XXX]	Non-Functional Requirement	[NFR-3.1] The system shall respond within 3 seconds.
<i>Italics</i>	Important terms or defined concepts	The <i>Patron</i> is the primary end-user.

1.3 Project Vision and Product Scope

1.3.1 Vision Statement

For students and lifelong learners who are seeking a more interactive and collaborative way to study, the Study Group Management System is a secure web application that allows users to create study groups, share resources, and participate in real-time discussions. Unlike traditional isolated study methods, our product provides a synchronous learning environment with integrated text and video chat, ensuring that members can collaborate efficiently and effectively from anywhere.

1.3.2 Product Scope

The Study Group Management System (SGMS) will cover the entire process of organizing study groups and facilitating academic collaboration.

The system scope includes:

- **Authentication & User Management Module:** Manages user identity: registration, login/logout, JWT token lifecycle, account status control, and profile management.
- **Group Management Module:** Handles study group CRUD, member join-request workflow, and admin-level group status control.
- **Session & Attendance Management Module:** Manages study session scheduling, live video rooms (LiveKit), and member attendance tracking.
- **Resource Module:** Provides a per-group file library organised into folders, with star ratings and Cloudinary-backed uploads.
- **Community & Engagement Module:** Enables group discussions (posts, threaded comments, voting), in-app notifications, and content reporting
- **Analytics & Administration Module:** Delivers role-adaptive statistics and charts for members, leaders, and admins.

The system scope exclude:

- **LMS Integration:** Direct synchronization or integration with external Learning Management Systems (e.g., Canvas, Blackboard).
- **Native Mobile Support:** Development of native iOS or Android applications (the focus remains on a responsive web-based experience).
- **Payment Processing:** Any monetization or payment gateway integration for group memberships or shared resources.

1.4 References

The following documents were used as reference materials for this SRS:

Project Charter for Study Group Management System 1.0

COS SRS.docx (Template for SRS Structure)1.0

Stakeholder Interviews Summary

2. Overall Description

2.1 Product Perspective

The Study Group Management System (SGMS) is a standalone web application designed to streamline collaborative learning by providing a centralized platform for creating study groups, managing educational resources, and multi-channel communication. It is intended to consolidate scattered learning materials and discussion threads into a single, organized environment. The system is independent of existing enterprise learning management systems (ELMS) and focuses on user-driven group dynamics.

2.2 Product Functions (Features)

The SGMS provides two main user interfaces and a set of core services:

Function Category	ID	Description (What the system does)

Authentication & Account	F1.0	Account Access: The system allows visitors to create a new account, sign in with their credentials, stay signed in across browser sessions, and sign out securely.
	F1.1	Profile Management: The system allows logged-in users to view their personal profile and update their display name, photo, and bio at any time.
User Management (Admin)	F2.0	User Directory: The system allows logged-in users to view any member's public profile, and allows administrators to browse and search the full user list with filtering by account status.
	F2.1	User Account Administration: The system allows administrators to activate, suspend, deactivate, or permanently remove any user account on the platform.
Study Groups	F3.0	Browse & Discover Groups: The system allows any visitor to browse all available study groups and view a group's full profile, including its members, sessions, resources, and discussions. Logged-in users

		also see their membership status and can access a personal list of all groups they belong to.
	F3.1	Group Setup & Management: The system allows logged-in users to create a new study group and automatically become its leader. Leaders can update the group's information or permanently close the group. Administrators can archive or restore any group.
	F3.2	Group Membership: The system allows users to request to join a group, withdraw a pending request, and allows group leaders to approve or decline membership requests.
Study Sessions	F4.0	Browse Sessions: The system allows users to browse study sessions with filtering by group, status, or upcoming dates, and view the full details of any session including its schedule, description, and attendee list.
	F4.1	Session Management: The system allows group members to create and schedule new sessions or start one immediately. Leaders and organisers can edit

		session details, conclude or cancel ongoing sessions, and permanently remove sessions.
	F4.2	Live Participation & Notification: The system allows group members to join an ongoing session's live video call directly from the platform, and allows organisers to send in-app notifications to all group members about a session.
Attendance	F5.0	Attendance Tracking: The system automatically records a member as present when they join a live session. Leaders can also manually record each member's attendance status (Present, Absent, Late, or Excused) and view the full attendance roster for any session.
Resources & Folders	F6.0	Resource Library: The system allows members to browse all study materials shared within their groups, view full resource details, and filter materials by folder. Groups can organise resources into named folders that members can create and rename.
	F6.1	Resource Actions: The system allows members to upload and share files with their group, remove

		materials they uploaded, and bookmark useful resources with a star for quick access.
Discussions (Posts & Comments)	F7.0	Discussion Board: The system allows group members to browse, create, edit, and delete discussion posts (with optional file attachments) within a group. Members can upvote or downvote posts to surface the most relevant content.
	F7.1	Comments & Replies: The system allows members to post threaded comments and replies on any discussion, edit or delete their own comments, mention other members with @, and upvote or downvote comments.
Notifications	F8.0	Notification Centre: The system delivers in-app notifications to users for relevant platform activity. Users can view a full notification feed, see a badge count of unread items, and mark individual or all notifications as read.
Admin Interface	F9.0	Platform Overview & Analytics: The system provides administrators with a dashboard of key platform metrics and charts, and allows drill-down

		into per-group statistics including session activity and member growth.
	F9.1	Platform Content Oversight: The system provides administrators with full visibility of all users, groups, sessions, and resources across the platform, with the ability to manage statuses and take moderation actions.
	F9.2	Content Reporting & Moderation: The system allows any logged-in user to flag inappropriate content for review, and provides administrators with a dedicated queue to review and resolve submitted reports.
	F9.3	Content Reporting & Moderation: The system allows any logged-in user to flag inappropriate content for review, and provides administrators with a dedicated queue to review and resolve submitted reports.

2.3 User Classes and Their Use Cases

The SGMS identifies three primary user classes:

User Class	Description	Key Characteristics & Permissions
Guest (Unauthenticated)	Any visitor who has not logged in. Can browse public content but cannot interact.	Goals: Discover the platform, browse available study groups and sessions, decide whether to register. Skills: Basic web browsing; no technical skills required. Permissions: Permissions: F1.1 (register & login only), F3.1 (browse only), F4.1 (browse only)
Member (Student)	A registered user who has joined one or more study groups with the MEMBER role. The default role after registration.	Goals: Join groups relevant to their studies, attend scheduled sessions, access and upload shared learning resources, participate in group discussions, and track personal attendance. Skills: Basic computer literacy; comfortable with file uploads and chat interfaces. Permissions: F1.1, F1.2, F3.1, F3.2 (create only), F3.3 (request & cancel

		only), F4.1, F4.3 (join meeting only), F5.1 (self check-in only), F6.1, F6.2 (upload & star only), F7.1, F7.2, F8.1, F9.3 (submit report only)
Leader (Group Leader)	A MEMBER who has the LEADER role within one or more groups (granted at group creation or by promotion). Has elevated management authority within their groups.	Goals: Organize and run study groups effectively; schedule and conduct sessions; manage membership; curate resources; moderate discussions; review attendance data. Skills: Organisational skills; comfortable managing schedules and member lists; basic understanding of content moderation. Permissions: All Member permissions, with expanded access on - F3.2 (edit & delete own group), F3.3 (approve & reject requests), F4.2, F4.3, F5.1, F6.2
Admin (Platform Administrator)	A super-user with full platform authority. Typically a system operator	Goals: Maintain platform health and safety; manage user accounts; monitor groups and sessions globally; resolve content reports;

	or staff member. Accesses a dedicated admin portal entirely separate from the student/leader UI.	access platform-wide analytics. Skills: Platform administration; data interpretation; moderate technical proficiency. Permissions: All Leader permissions, plus — F2.1 (full), F2.2, F3.2, F9.1, F9.2, F9.3
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2.4 Operating Environment

This section describes the environment in which the software will operate.

- Client Side:** React 19 (SPA) + Vite 6; React Router v7; Zustand v5 + TanStack Query v5 (state management); React Hook Form + Zod (forms); TailwindCSS v4; Recharts v2 (charts); LiveKit components (real-time video); Axios v1 (HTTP); served via Nginx on port 80.
- Server Side:** Node.js 20+ (ESM); Express 4; RESTful JSON API at /api; JWT auth (access 15 min / refresh 7 days, hashed in DB); role-based middleware; express-validator; Helmet + CORS + bcrypt security; Morgan logging; Prisma Client v6 ORM; served on port 5000.
- Database:** PostgreSQL 16; Prisma 6 (schema-first migrations, 20 models); soft-delete via deletedAt pattern; hosted on Neon (cloud) or Docker (self-hosted).
- Third-Party Services:** Cloudinary (file storage, signed direct-upload); LiveKit (real-time video rooms, server-side token generation).

- **Runtime Environment:** Node.js v20+ / npm v10+; Docker + Docker Compose v3.8 (optional full-stack or DB-only mode); config via .env files.

2.5 Design and Implementation Constraints

The project must adhere to the following constraints:

- **Technology Stack:** Must use **Node.js, ReactJS, and Monolithic Architecture** as defined in the project scope.
- **Security:** All financial transactions and user data must be secured using appropriate encryption and protocols (SSL/TLS).
- **Performance:** The system must be able to handle peak traffic during popular movie releases without significant degradation in response time (detailed in Section 6.0 Quality Attributes).

2.6 Assumptions and Dependencies

2.6.1 Assumptions

- Users have access to hardware (webcam/microphone) for video-based study sessions.
- Database availability: PostgreSQL (Neon or Docker) is continuously available with < 100 ms latency to the backend server.
- The internet connection between the student and the server is stable during real-time communication.

2.6.2 Dependencies

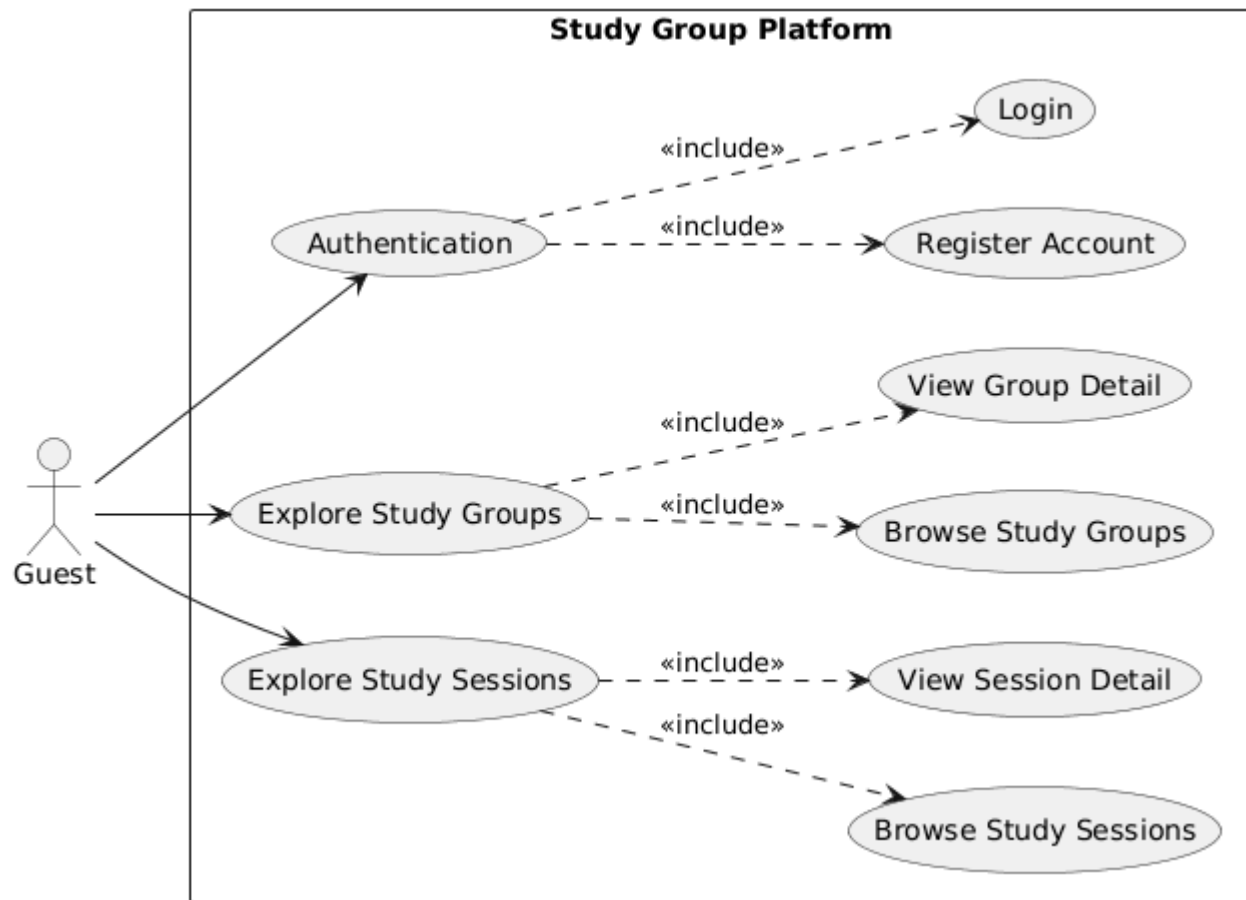
- **External Services:** PostgreSQL 16 (critical — all data); Cloudinary v2 (high — file uploads); LiveKit v2 (medium — live video only)

- **Node.js Ecosystem:** Dependent on specific npm packages (Sequelize, Express, Handlebars, Bcrypt).
- **Database Engine:** Dependent on a PostgreSQL-compatible database for data persistence.

3. Functional Requirements

3.1 Use Cases for Guest

Guests are unauthenticated users who can browse study groups and sessions, view public information, and register or log in to access additional features.

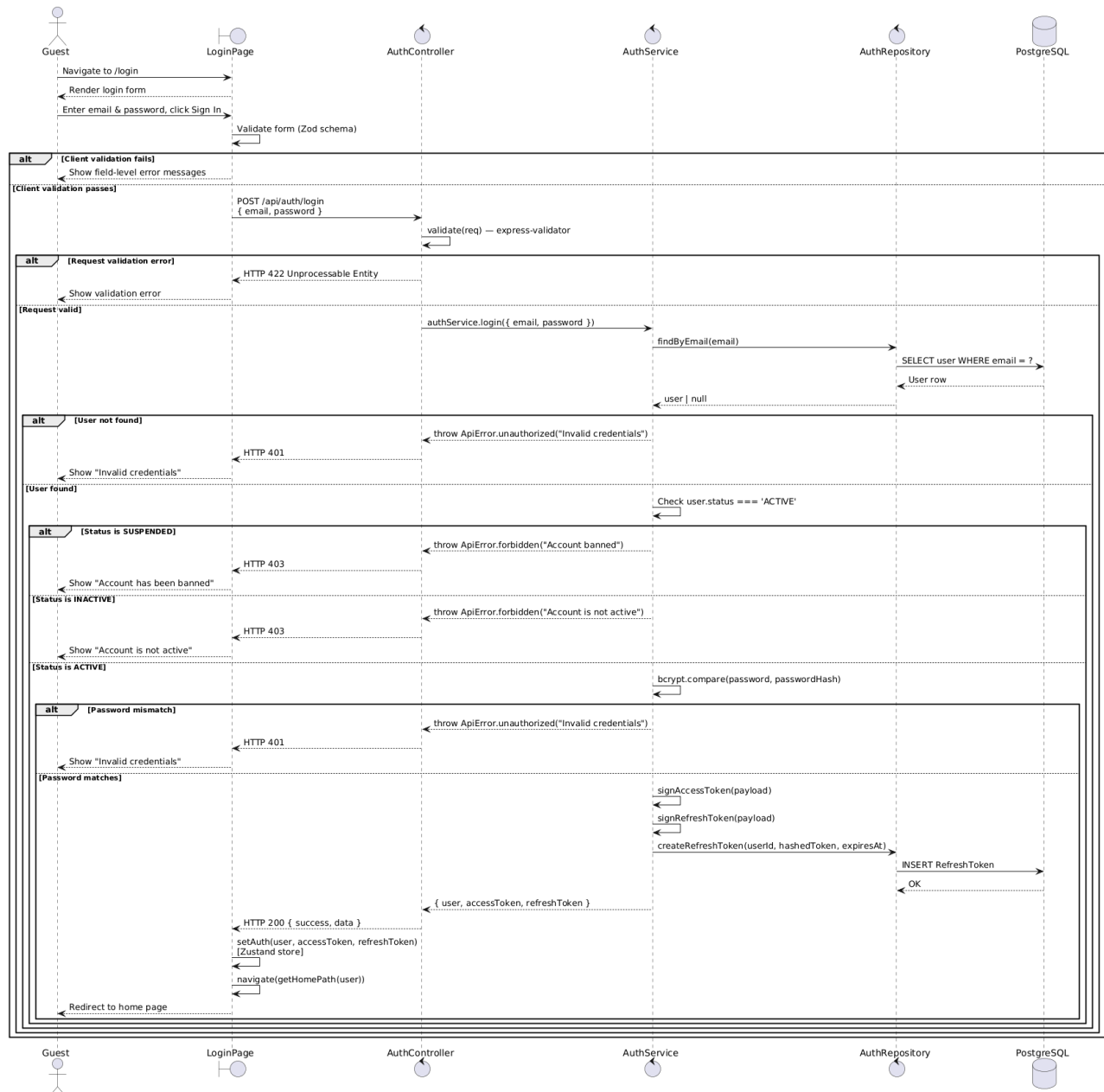


UC-G-01: Login

Item	Description
Use Case ID	UC-G-01

Use Case Name	Login
Description	Allows a registered user to sign in to the platform using their email address and password and be directed to their personalised home page.
Actor(s)	Guest (New User)
Priority	High
Trigger	Guest clicks the "Sign in" button or link on any public page.
Pre-Conditions	1. The Guest is not currently signed in. 2. The Guest has a registered account on the platform.
Post-Conditions	1. The Guest is successfully signed in and their session is maintained. 2. The system redirects the user to their role-appropriate home page.
Basic Flow	1. Guest navigates to the Login page. 2. System displays the login form with email address and password fields. 3. Guest enters their registered email address and password. 4. Guest clicks Sign in. 5. System validates that the email is in a correct format and the password field is not empty. 6. System verifies the email and password against the registered account.

	7. System confirms the account is active. 8. System signs the user in and preserves the session. 9. System redirects the user to their role-appropriate home page.
Alternative Flow	Navigate to Register: 1. Guest clicks the "Register" link on the Login page. 2. System navigates to the Register page.
Exception Flow	<i>Invalid Input:</i> 1. Guest submits the form with an invalid email format or an empty password. 2. System displays validation error messages next to the affected fields. 3. The sign-in request is not submitted. <i>Wrong Credentials:</i> 1. Guest enters an email or password that does not match any account. 2. System displays "Invalid credentials" below the form. 3. The user remains on the Login page. <i>Account Inactive:</i> 1. Guest attempts to sign in with a deactivated account. 2. System displays a message indicating the account is not active. 3. Access is denied.

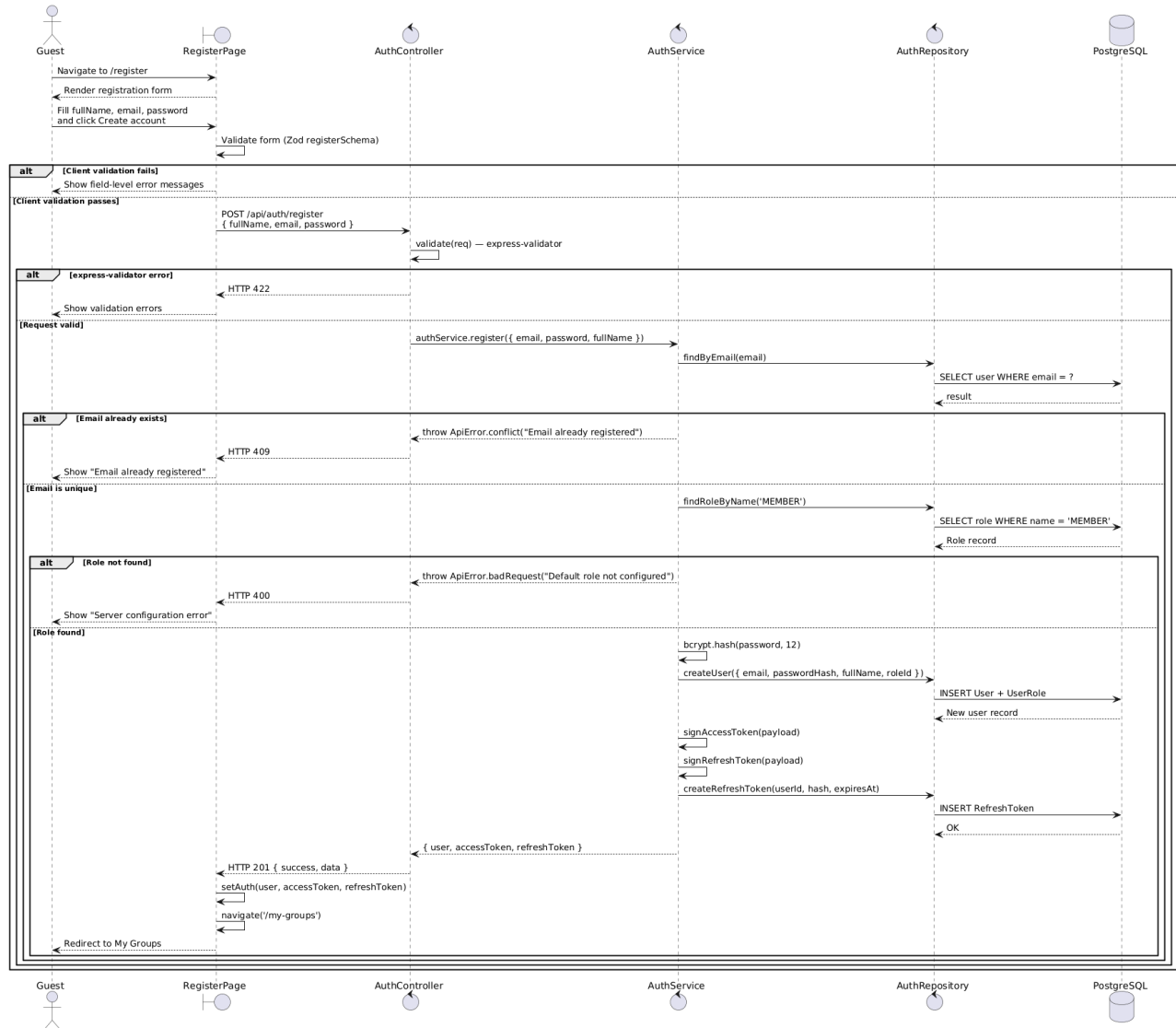


UC-G-02: Register Account

Item	Description
Use Case ID	UC-G-02
Use Case Name	Register Account

Description	Allows a new visitor to create a StudyHub account by providing their full name, email address, and password. Upon successful registration, the user is automatically signed in with the default Member role.
Actor(s)	Guest
Priority	High
Trigger	Guest clicks the "Register" link or button on any public page.
Pre-Conditions	1. The Guest is not currently signed in. 2. The Guest does not already have an account registered with the same email address.
Post-Conditions	1. A new account is created and assigned the Member role. 2. The user is automatically signed in. 3. The system redirects the user to the My Groups page.
Basic Flow	1. Guest navigates to the Register page. 2. System displays the registration form with fields for full name, email address, and password. 3. Guest fills in all required fields. 4. Guest clicks Create account. 5. System validates the entered information: full name must be 2–100 characters; email must be in a valid format; password must be at least 8 characters and contain at least one uppercase letter and one number. 6. System checks that the email address is not already in use.

	7. System creates the new account and assigns the Member role. 8. System automatically signs the user in. 9. System redirects the user to the My Groups page.
Alternative Flow	<i>Navigate to Login:</i> 1. Guest clicks the "Sign in" link. 2. System navigates to the Login page.
Exception Flow	<i>Validation Failure:</i> 1. Guest submits the form with missing or invalid information. 2. System displays field-level error messages (e.g., "Password must be at least 8 characters with an uppercase letter and a number"). 3. The account is not created. <i>Email Already Registered:</i> 1. Guest submits a form using an email address already associated with an existing account. 2. System displays "Email already registered". 3. The account is not created.

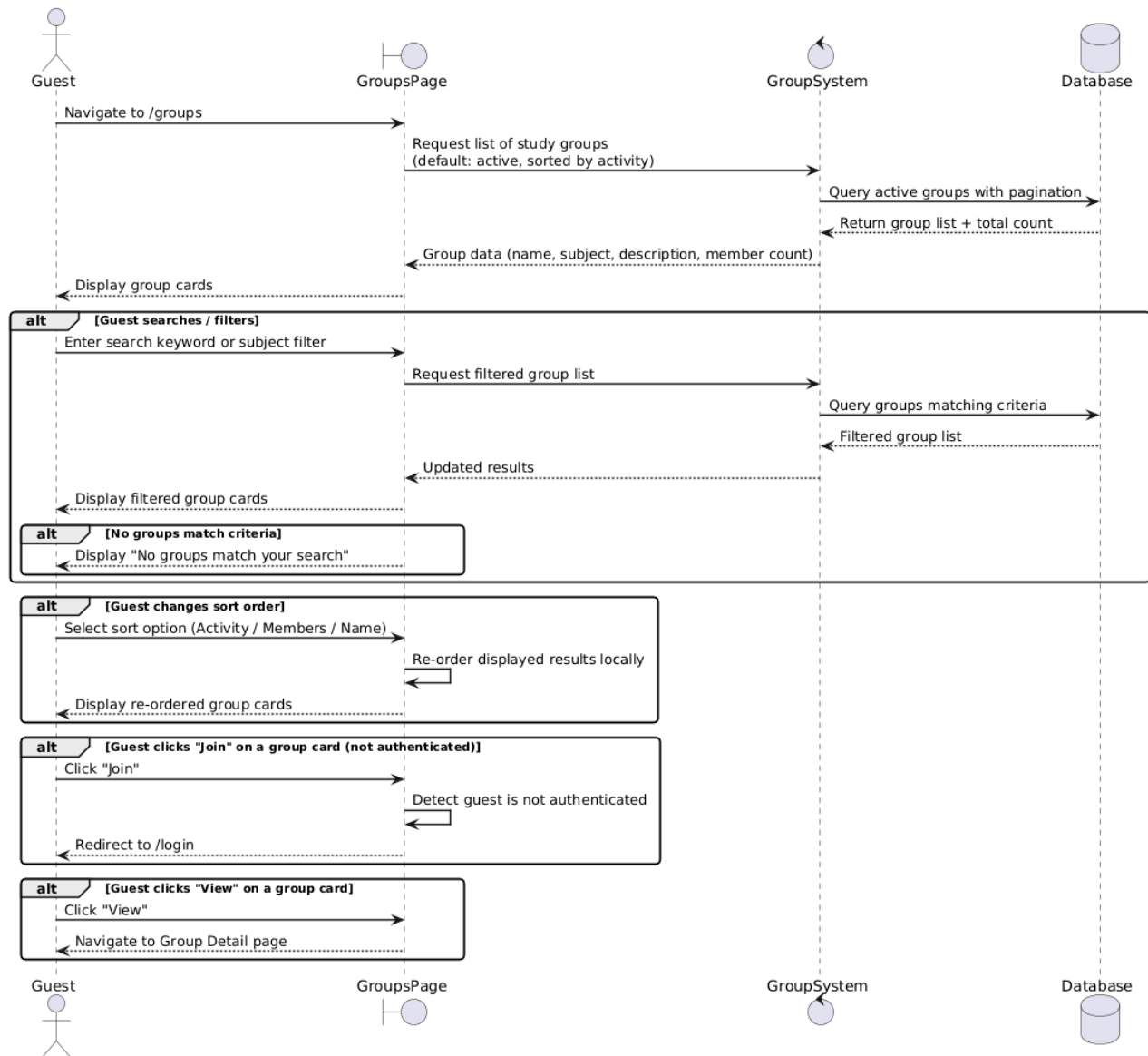


UC-G-03: Browse Study Groups

Item	Description
Use Case ID	UC-G-03
Use Case Name	Browse Study Groups

Description	Allows a guest to view a paginated list of all active study groups on the platform. The guest may search by name or subject, filter by subject area, and sort the results.
Actor(s)	Guest
Priority	High
Trigger	Guest navigates to the Study Groups page.
Pre-Conditions	1. The platform has at least one active study group.
Post-Conditions	1. A list of active study groups matching the guest's search or filter criteria is displayed.
Basic Flow	1. Guest navigates to the Study Groups page. 2. System retrieves and displays a list of active study groups, showing each group's name, subject, description, and member count. 3. Guest may enter a keyword in the search field to filter groups by name or subject. 4. System updates the displayed list to show only matching groups. 5. Guest may enter text in the subject filter field to further narrow results.

	6. Guest may change the sort order (by activity, by member count, or by name). 7. System re-orders the displayed results accordingly. 8. Guest selects a group card to view its details.
Alternative Flow	<i>No Results:</i> 1. Guest enters search terms that match no groups. 2. System displays a "No groups match your search" message. <i>Attempt to Join Without Authentication:</i> 1. Guest clicks the "Join" button on a group card. 2. System detects the guest is not logged in. 3. System redirects the guest to the Login page.
Exception Flow	<i>System Unavailable:</i> 1. System fails to retrieve the list of groups. 2. System displays an error or loading state. 3. Guest may refresh the page and try again.



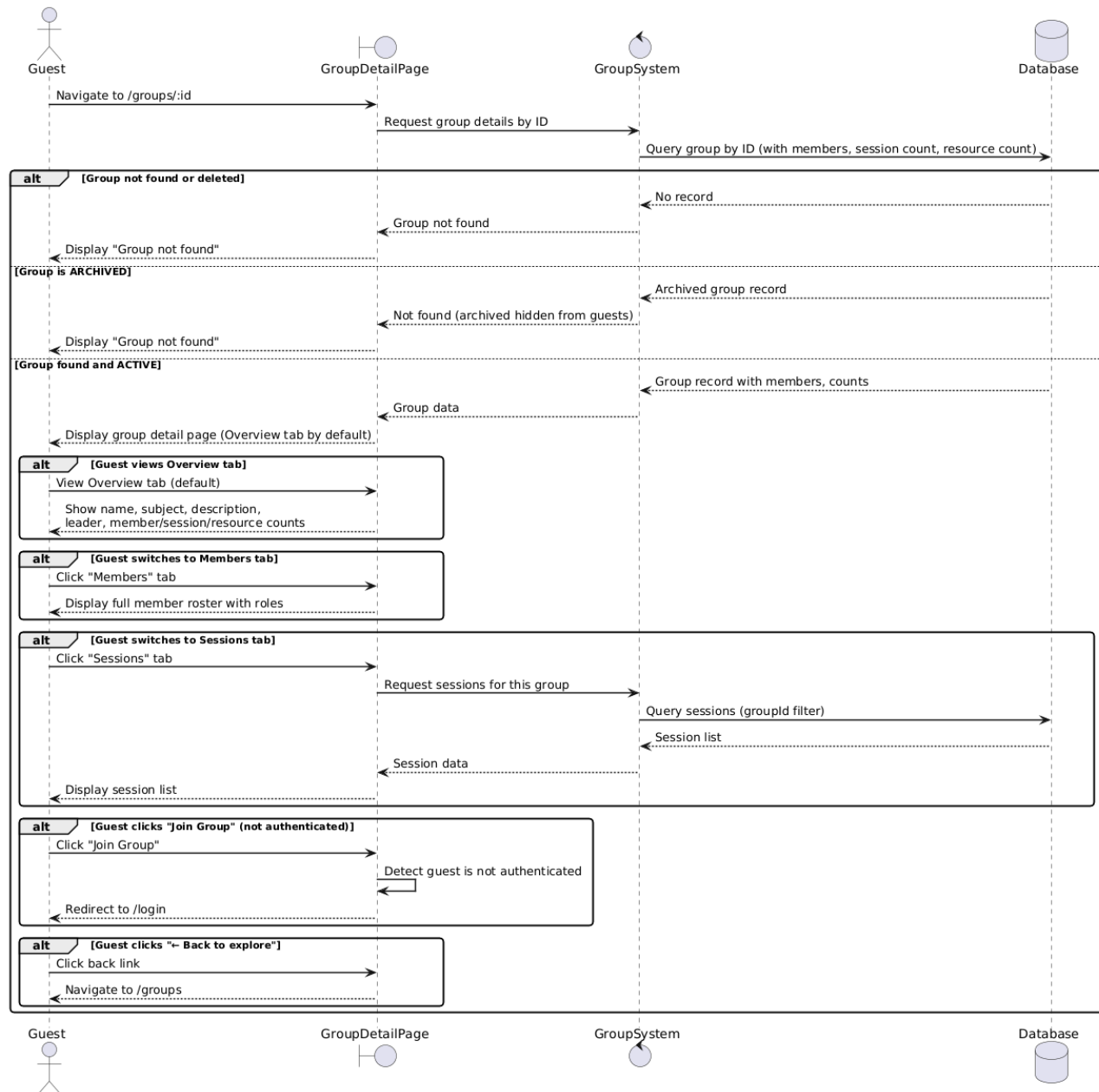
UC-G-04: View Group Detail

Item	Description
Use Case ID	UC-G-04
Use Case Name	View Group Detail

Description	Allows a guest to view detailed information about a specific study group, including its description, member roster, scheduled sessions, shared resources, and discussion board in a read-only capacity.
Actor(s)	Guest
Priority	High
Trigger	Guest selects a specific group from the Study Groups list or navigates directly to the group's detail URL.
Pre-Conditions	1. The selected study group exists and has an Active status.
Post-Conditions	1. The group's full detail page is displayed with Overview, Members, Sessions, Resources, and Discussions tabs. 2. No data is modified.
Basic Flow	1. Guest selects a group from the Study Groups list. 2. System retrieves and displays the group's detail page. 3. System shows the group name, subject, description, leader name, member count, session count, and resource count in the Overview tab. 4. Guest may switch to the Members tab to view the full member roster with roles. 5. Guest may switch to the Sessions tab to view scheduled and past sessions. 6. Guest may switch to the Resources tab to browse shared study materials.

	7. Guest may switch to the Discussions tab to read existing discussion posts and comments. 8. Guest may click "Join Group" to request membership (requires authentication).
Alternative Flow	<i>Navigate Back:</i> 1. Guest clicks "← Back to explore". 2. System returns the guest to the Study Groups list. <i>Attempt to Join Without Authentication:</i> 1. Guest clicks "Join Group". 2. System detects the guest is not logged in. 3. System redirects the guest to the Login page.
Exception Flow	<i>Group Not Found:</i> 1. Guest navigates to a group URL that does not exist or has been deleted. 2. System displays a "Group not found" message. <i>Group is Archived:</i> 1. Guest attempts to access an archived group. 2. System treats the group as not found and displays a "Group not found" message. <i>System Unavailable:</i> 1. System fails to retrieve group data.

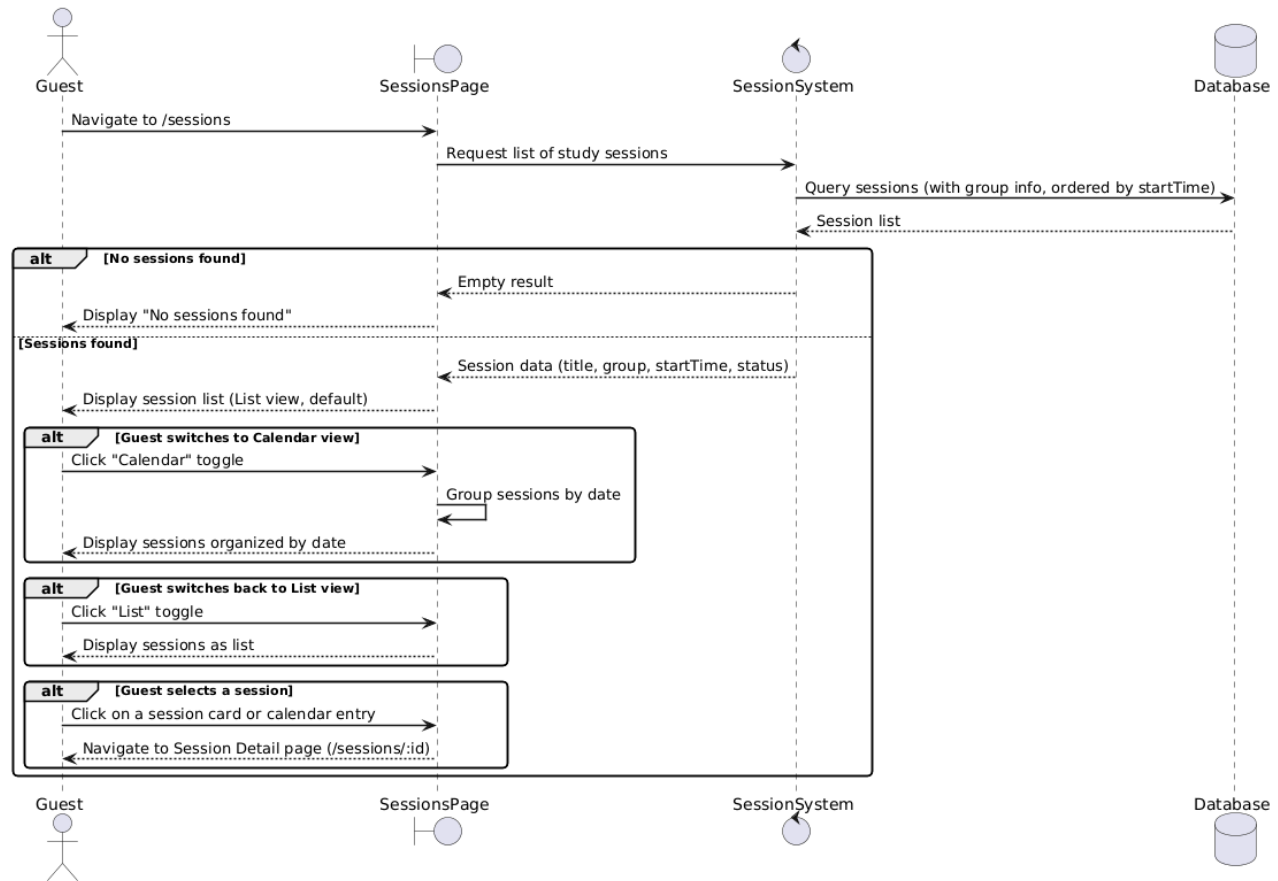
2. System displays an error state.
3. Guest may refresh the page.



UC-G-05: Browse Study Sessions

Item	Description
Use Case ID	UC-G-05
Use Case Name	Browse Study Sessions
Description	Allows a guest to view a list of publicly available study sessions across all groups. Sessions can be browsed in list or calendar view, and may be filtered by group or status.
Actor(s)	Guest
Priority	Medium
Trigger	Guest navigates to the Study Sessions page.
Pre-Conditions	1. At least one study session exists in the system..
Post-Conditions	1. A list of study sessions is displayed with their title, group name, start time, and status. 2. No data is modified.
Basic Flow	1. Guest navigates to the Study Sessions page. 2. System retrieves and displays a list of study sessions, showing each session's title, associated group name, scheduled start time, and current status (Scheduled, In Progress, Completed, or Cancelled). 3. Guest may toggle between a list view and a calendar view

	(grouped by date). 4. Guest may select a specific session to view its full details.
Alternative Flow	<i>Calendar View:</i> 1. Guest clicks the "Calendar" toggle button. 2. System groups and displays sessions organized by their scheduled date. 3. Guest selects a session entry to view its detail page. <i>No Sessions Available:</i> 1. System finds no sessions matching the current query. 2. System displays a "No sessions found" message.
Exception Flow	<i>System Unavailable:</i> 1. System fails to retrieve the session list. 2. System displays an error or loading state. 3. Guest may refresh the page and try again.

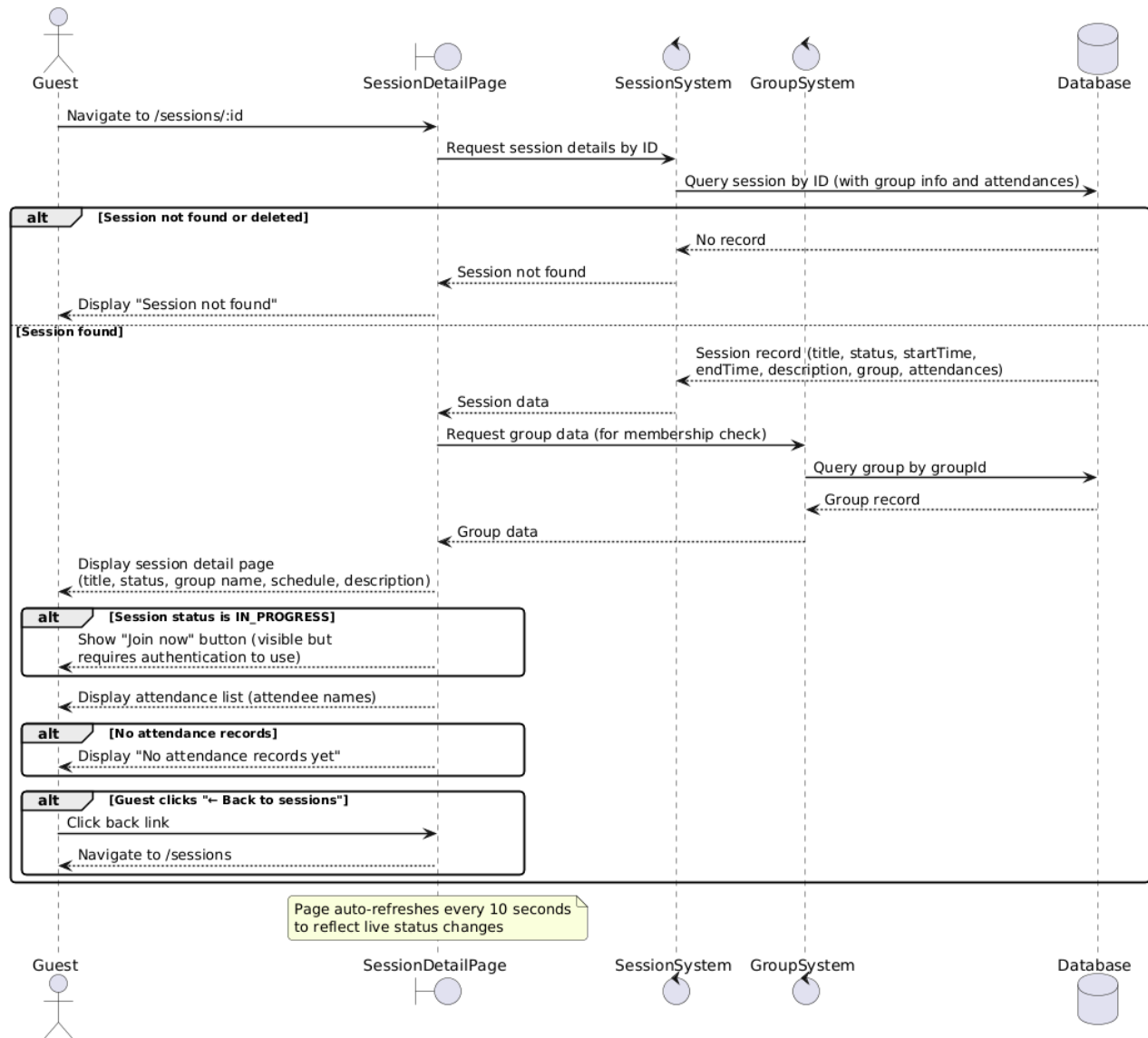


UC-G-06: View Session Detail

Item	Description
Use Case ID	UC-G-06
Use Case Name	View Session Detail
Description	Allows a guest to view the complete details of a specific study session, including its title, description, associated group, schedule (start and end time), current status, and the list of attendees.

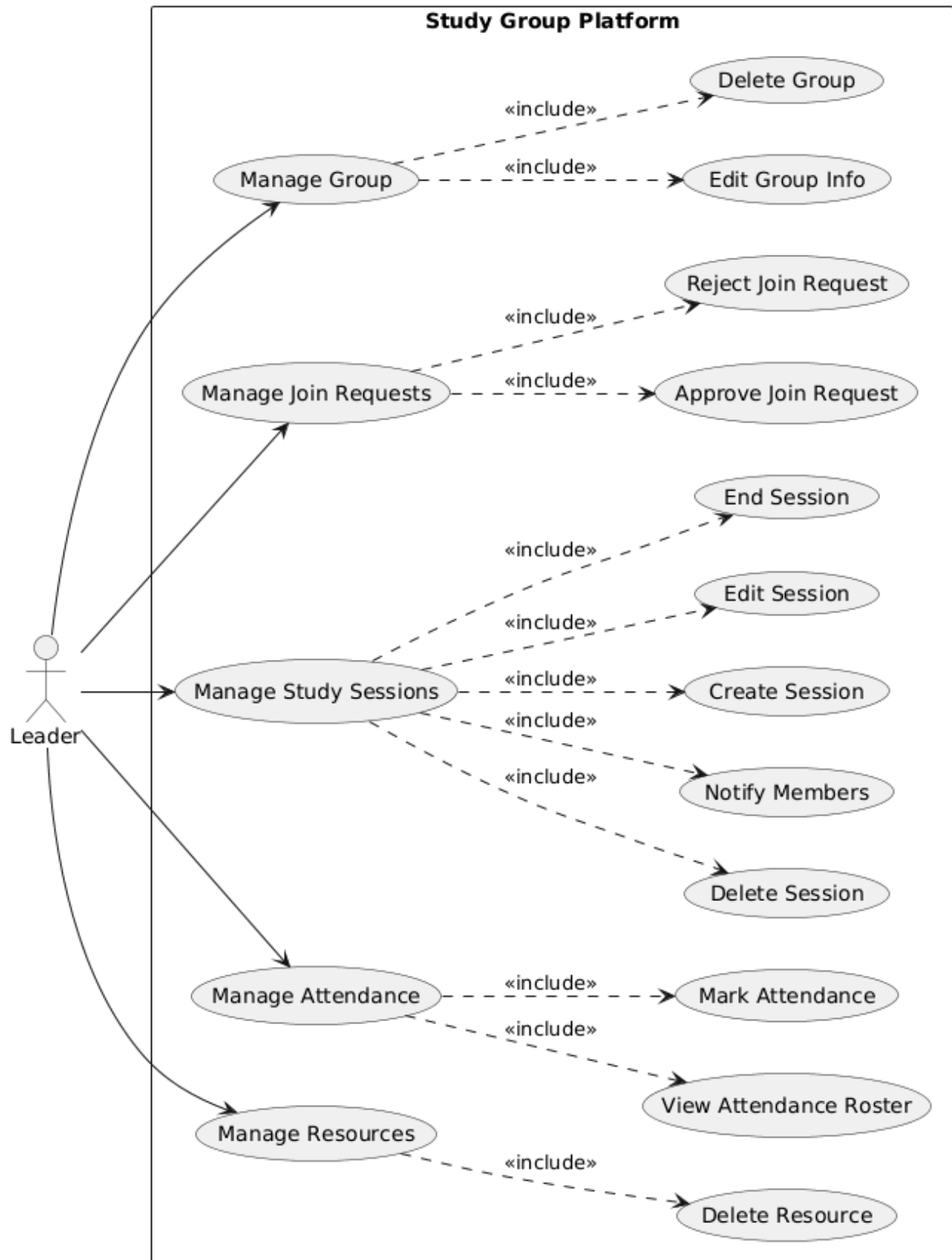
Actor(s)	Guest
Priority	Medium
Trigger	Guest selects a specific session from the Study Sessions list or navigates directly to the session's detail URL.
Pre-Conditions	1. The selected study session exists and has not been deleted.
Post-Conditions	1. The session's full details are displayed. 2. No data is modified.
Basic Flow	1. Guest selects a session from the Study Sessions list. 2. System retrieves and displays the session detail page. 3. System shows the session title, current status badge, associated group name, and description. 4. System displays the session schedule: start time and end time (or "Ongoing" if no end time is set). 5. System displays the attendance list showing the names of users who have attended. 6. Guest may click "← Back to sessions" to return to the session list.
Alternative Flow	<i>Navigate Back:</i> 1. Guest clicks "← Back to sessions". 2. System returns the guest to the Study Sessions list. <i>In-Progress Session (Join Button Visible):</i>

	1. Session status is "In Progress". 2. System displays a "Join now" button (accessible only to authenticated group members). 3. Guest who is not authenticated sees the button but would be redirected to login upon clicking.
Exception Flow	<i>Session Not Found:</i> 1. Guest navigates to a session URL that does not exist or has been deleted. 2. System displays a "Session not found" message. <i>System Unavailable:</i> 1. System fails to retrieve session data. 2. System displays an error state. 3. Guest may refresh the page.



3.2 Use Cases for Leaders

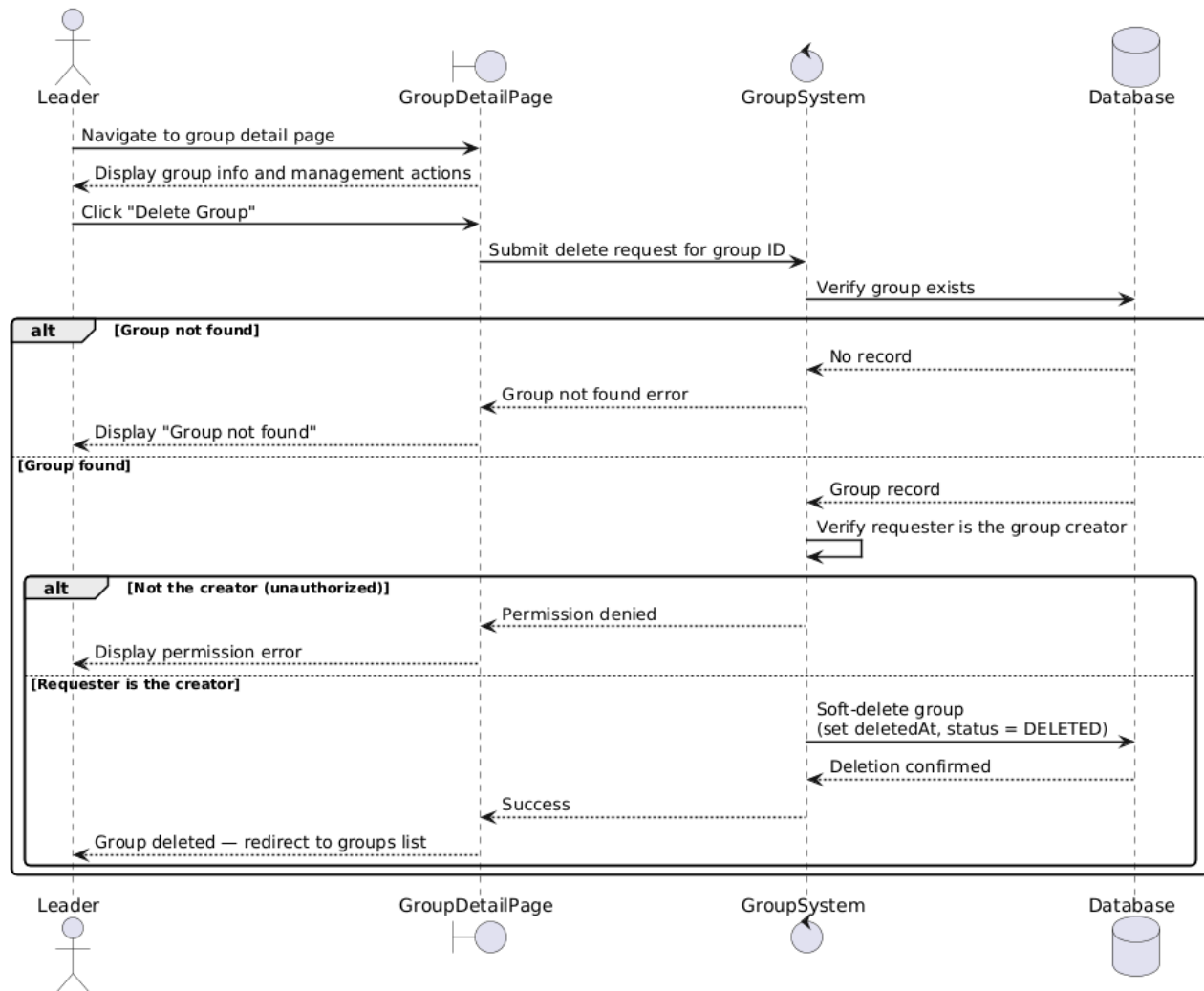
Leaders are members who manage study groups, handle membership requests, organize study sessions, track attendance, and oversee group resources.



UC-L-01: Delete Group

Item	Description
Use Case ID	UC-L-01
Use Case Name	Delete Group
Description	Allows a Leader to permanently remove (soft-delete) a study group they created, making it inaccessible to all members and the public.
Actor(s)	Leader
Priority	High
Trigger	Leader chooses to delete a group from the group management interface.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader is the creator of the group to be deleted. 3. The target group exists and has not already been deleted.
Post-Conditions	1. The group is marked as deleted and removed from all public listings. 2. The group is no longer accessible to members or guests.
Basic Flow	1. Leader navigates to the group management page. 2. System displays group information and available management actions. 3. Leader selects the option to delete the group.

	4. System verifies that the Leader is the group creator. 5. System marks the group as deleted. 6. System confirms the deletion and removes the group from all listings.
Alternative Flow	<i>Cancel Deletion:</i> 1. Leader decides not to proceed with deletion. 2. Leader dismisses the action. 3. System takes no action and the group remains unchanged.
Exception Flow	<i>Unauthorized Attempt:</i> 1. A user who is not the group creator attempts to delete the group. 2. System denies the action and displays a permission error. 3. Group remains unchanged. <i>Group Not Found:</i> 1. Leader attempts to delete a group that no longer exists. 2. System displays a "Group not found" error. <i>System Error:</i> 1. System encounters an unexpected failure during deletion. 2. System displays an error message. 3. Leader may retry the action.

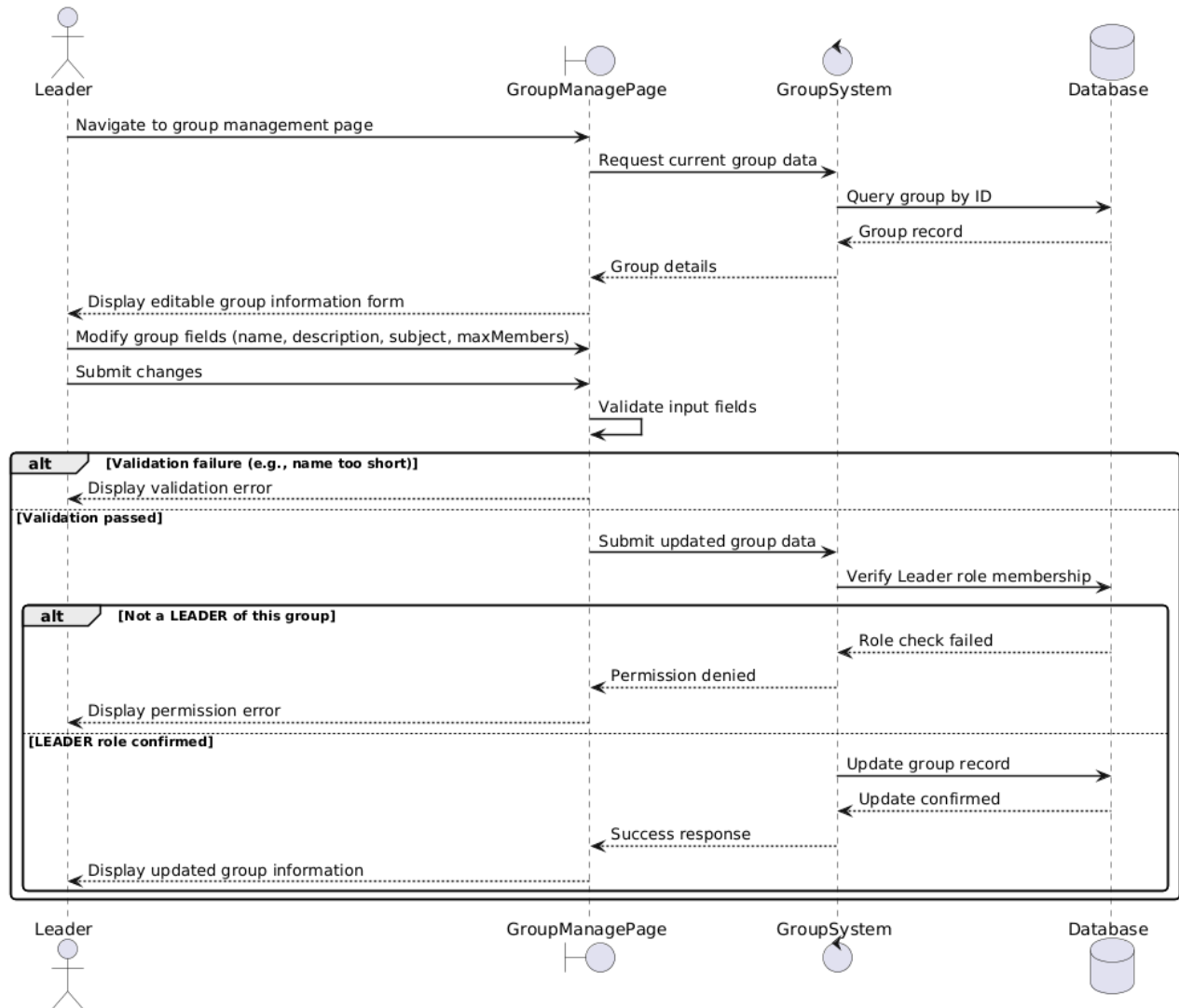


UC-L-02: Edit Group Info

Item	Description
Use Case ID	UC-L-02
Use Case Name	Edit Group Info
Description	Allows a Leader to update the details of a study group they lead, including the group name, description, subject, and maximum member capacity.

Actor(s)	Leader
Priority	High
Trigger	Leader submits updated group information through the group management interface.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader holds the LEADER role within the target group. 3. The target group exists and is active.
Post-Conditions	1. The group's information is updated in the system. 2. The updated details are immediately reflected in the group listing and detail page.
Basic Flow	1. Leader navigates to the group management page. 2. System displays the current group information in an editable form. 3. Leader modifies one or more fields (name, description, subject, or maximum members). 4. Leader submits the changes. 5. System validates the entered data. 6. System verifies the Leader's role within the group. 7. System saves the updated group information. 8. System confirms the update and displays the refreshed group details.
Alternative Flow	<i>Cancel Edit:</i> 1. Leader decides not to save changes.

	2. Leader dismisses the form. 3. System discards the unsaved changes and the group remains unchanged.
Exception Flow	<i>Unauthorized Attempt:</i> 1. A user who is not a LEADER of the group submits the update. 2. System denies the action and displays a permission error. 3. Group information remains unchanged. <i>Validation Failure:</i> 1. Leader submits a group name shorter than 3 characters or longer than 100 characters. 2. System displays a validation error. 3. Changes are not saved. <i>System Error:</i> 1. System encounters an unexpected failure during the save operation. 2. System displays an error message. 3. Leader may retry the action.



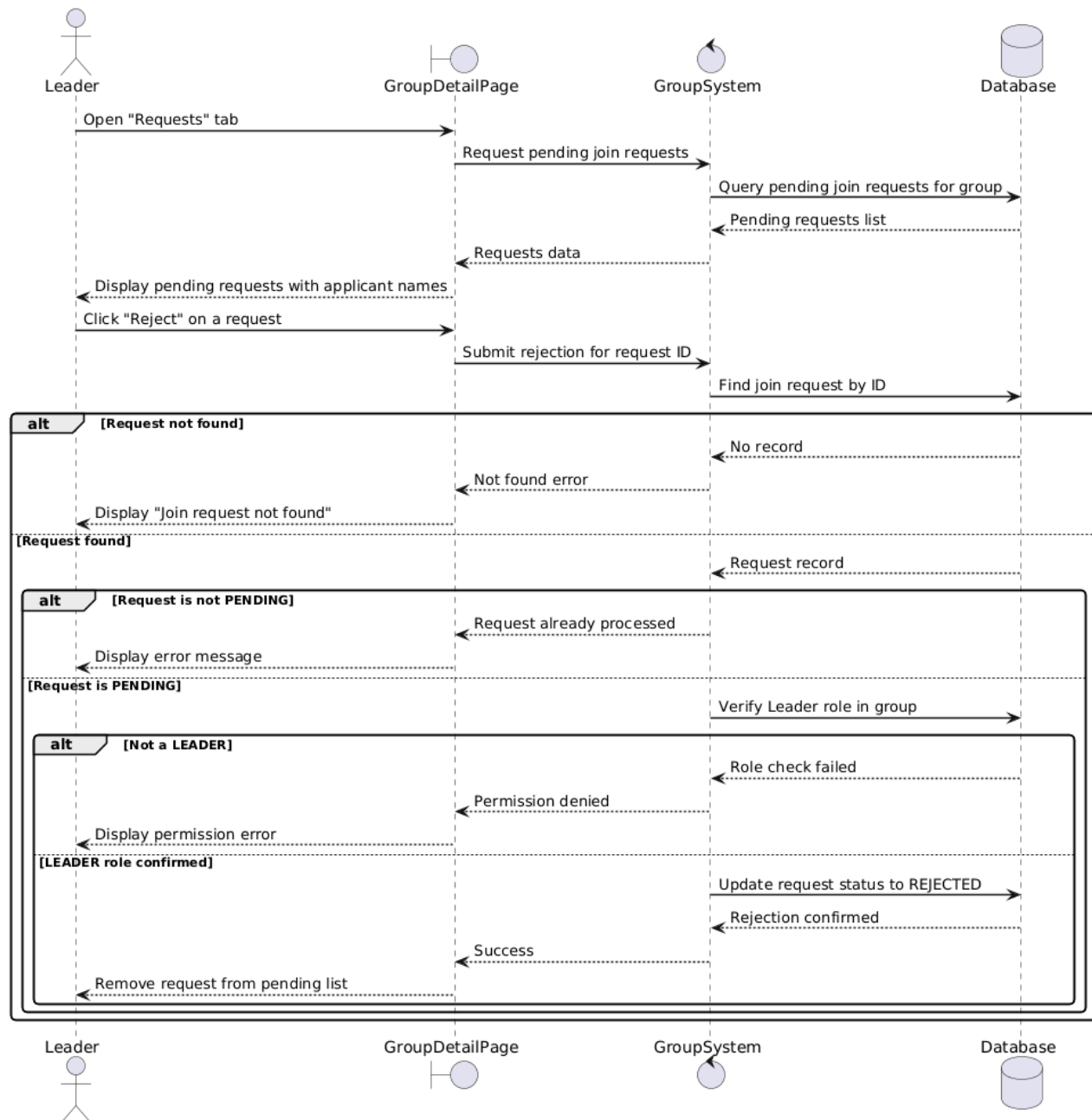
UC-L-03: Reject Join Request

Item	Description
Use Case ID	UC-L-03
Use Case Name	Reject Join Request

Description	Allows a Leader to reject a pending membership request from a user who has applied to join their study group, preventing that user from becoming a member.
Actor(s)	Leader
Priority	High
Trigger	Leader clicks "Reject" on a pending join request listed in the Requests tab of the group detail page.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader holds the LEADER role within the target group. 3. A pending join request from a non-member exists for the group.
Post-Conditions	1. The join request is marked as Rejected. 2. The requesting user is not added to the group. 3. The pending requests list is updated.
Basic Flow	1. Leader navigates to the group detail page and opens the Requests tab. 2. System displays the list of pending join requests with applicant names. 3. Leader reviews a request and clicks "Reject". 4. System verifies the Leader's role within the group. 5. System updates the join request status to Rejected.

	6. System removes the request from the pending list and confirms the action.
Alternative Flow	<i>No Pending Requests:</i> 1. Leader opens the Requests tab and there are no pending requests. 2. System displays a "No pending requests" message.
Exception Flow	<i>Unauthorized Attempt:</i> 1. A user who is not a LEADER of the group attempts to reject a request. 2. System denies the action and displays a permission error. <i>Request No Longer Pending:</i> 1. Leader attempts to reject a request that has already been processed. 2. System displays an error indicating the request is no longer pending. 3. No change is made. <i>Request Not Found:</i> 1. The target join request cannot be found (e.g., applicant withdrew). 2. System displays a "Join request not found" error. <i>System Error:</i> 1. System encounters an unexpected failure.

2. System displays an error message.
3. Leader may retry the action.



UC-L-04: Approve Join Request

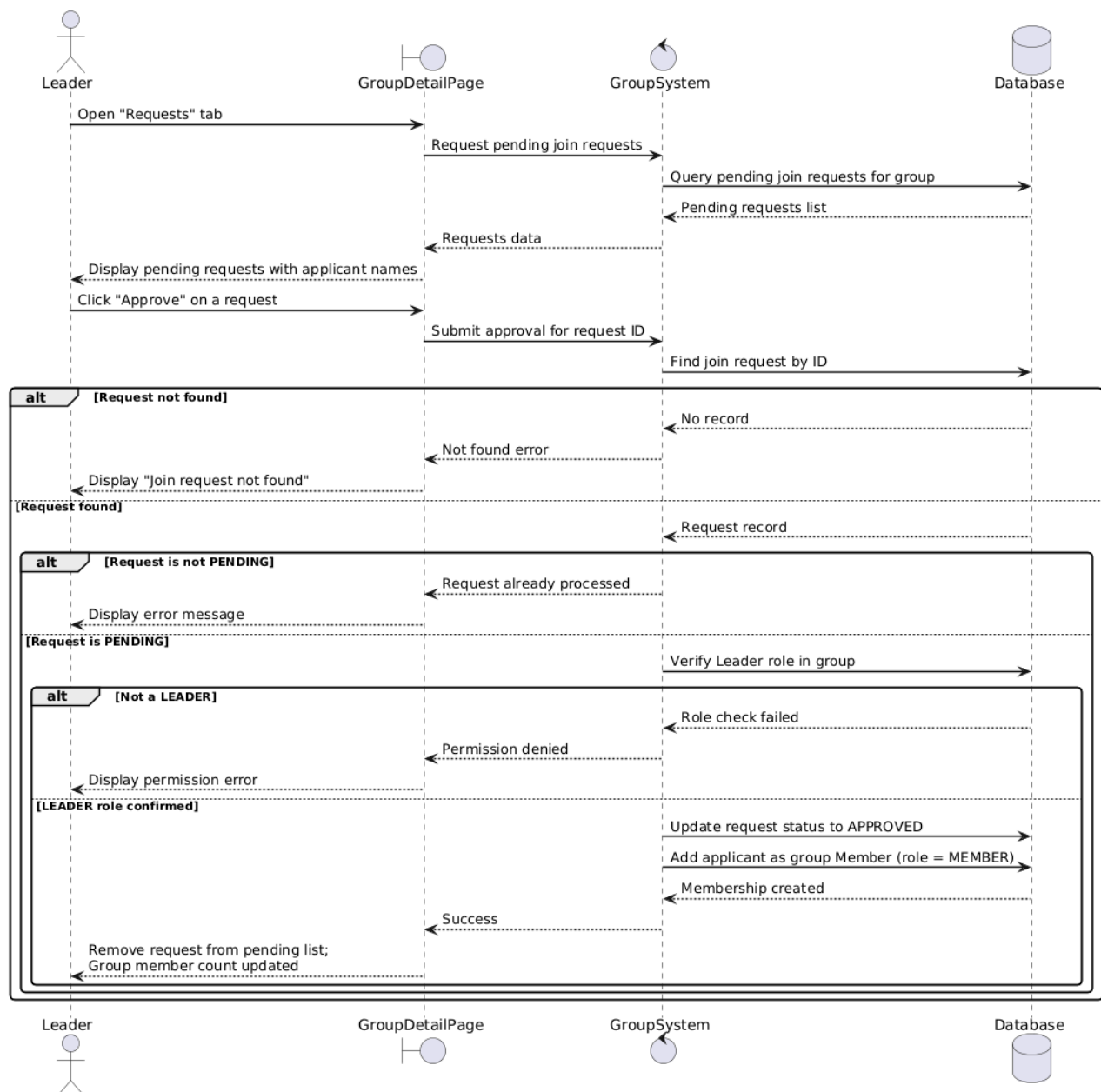
Item	Description
Use Case ID	UC-L-04
Use Case Name	Approve Join Request
Description	Allows a Leader to approve a pending membership request, granting the applicant full Member status within the study group.
Actor(s)	Leader
Priority	High
Trigger	Leader clicks "Approve" on a pending join request listed in the Requests tab of the group detail page.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader holds the LEADER role within the target group. 3. A pending join request from a non-member exists. 4. The group has not reached its maximum member capacity.
Post-Conditions	1. The join request is marked as Approved. 2. The applicant is added as a Member of the group. 3. The pending requests list is updated to remove the processed request.
Basic Flow	1. Leader navigates to the group detail page and opens the Requests tab. 2. System displays the list of pending join requests with

	applicant names. 3. Leader reviews a request and clicks "Approve". 4. System verifies the Leader's role within the group. 5. System updates the join request status to Approved. 6. System adds the applicant as a Member of the group. 7. System removes the request from the pending list and confirms the action.
Alternative Flow	<i>No Pending Requests:</i> 1. Leader opens the Requests tab and there are no pending requests. 2. System displays a "No pending requests" message.
Exception Flow	<i>Unauthorized Attempt:</i> 1. A user who is not a LEADER of the group attempts to approve a request. 2. System denies the action and displays a permission error. <i>Request No Longer Pending:</i> 1. Leader attempts to approve a request that has already been processed. 2. System displays an error indicating the request is no longer pending. 3. No change is made. <i>Request Not Found:</i> 1. The target join request cannot be found.

2. System displays a "Join request not found" error.

System Error:

1. System encounters an unexpected failure.
2. System displays an error message.
3. Leader may retry the action.

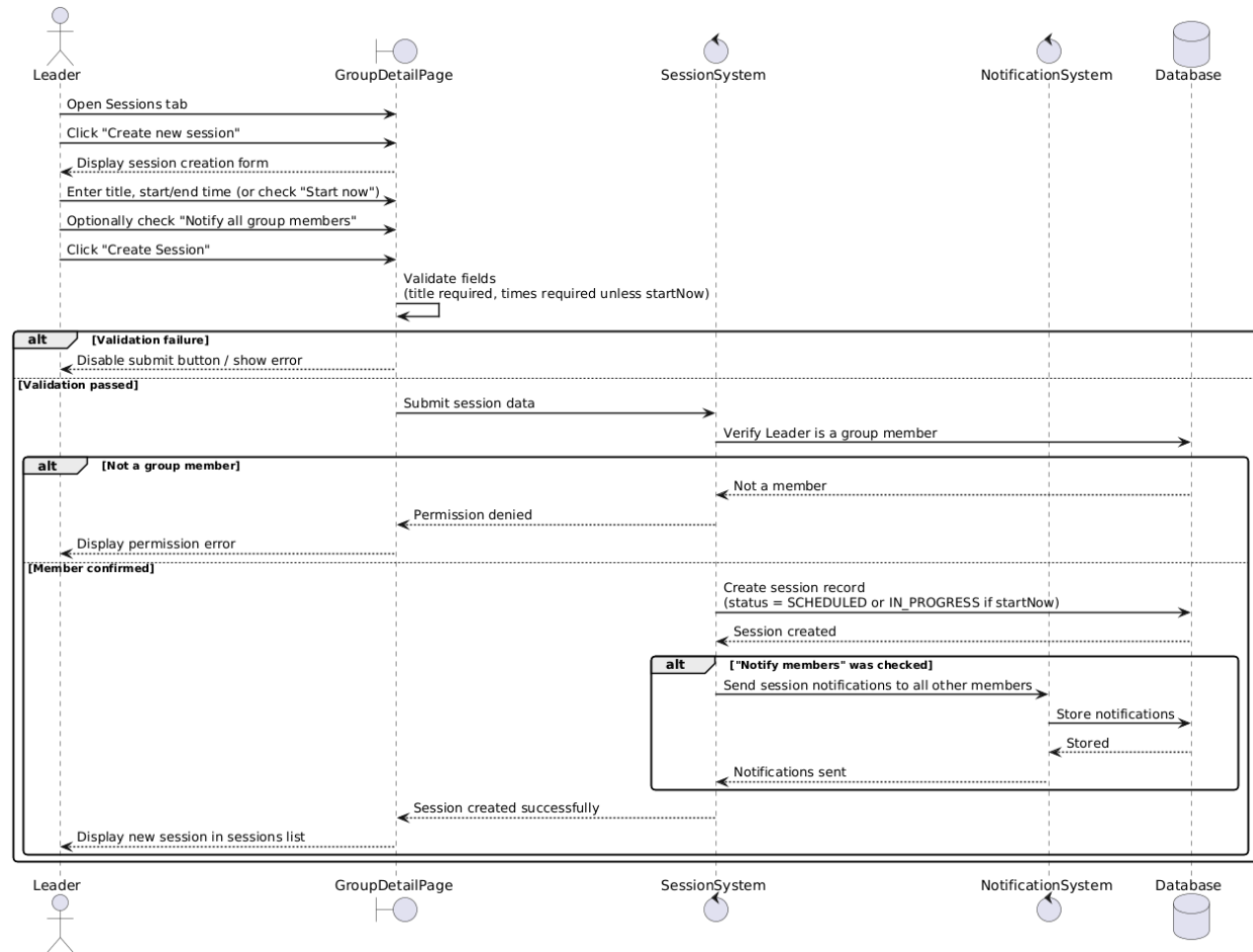


UC-L-05: Create Session

Item	Description
Use Case ID	UC-L-05
Use Case Name	Create Session
Description	Allows a Leader to schedule a new study session for their group, either as a future scheduled event with a defined start and end time, or as an immediately started live session. Optionally, group members can be notified.
Actor(s)	Leader
Priority	High
Trigger	Leader clicks "Create new session" in the Sessions tab of the group detail page and submits the session form.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader is an active member of the target group. 3. The target group exists and is active.
Post-Conditions	1. A new study session is created and associated with the group. 2. If "Start now" was selected, the session status is set to In Progress immediately. 3. If "Notify members" was checked, all group members receive a notification about the new session. 4. The session appears in the group's Sessions tab.

Basic Flow	1. Leader opens the Sessions tab on the group detail page. 2. Leader clicks "Create new session". 3. System displays the session creation form with title, start time, end time, and option fields. 4. Leader enters a session title. 5. Leader either selects a start date/time and end date/time, or checks "Start now". 6. Optionally, Leader checks "Notify all group members". 7. Leader clicks "Create Session". 8. System validates the entered information. 9. System verifies the Leader is a group member. 10. System saves the new session record. 11. If "Notify members" was checked, system sends notifications to all other group members. 12. System displays the new session in the Sessions list.
Alternative Flow	<i>Start Now:</i> 1. Leader checks "Start now". 2. System hides the start and end time fields. 3. Leader submits the form with only a title. 4. System creates the session with In Progress status and the current time as the start time. <i>Cancel:</i> 1. Leader clicks "Cancel".

	2. System hides the form. 3. No session is created.
Exception Flow	<i>Validation Failure — Title:</i> 1. Leader submits an empty or too-short title (under 3 characters). 2. System disables the submit button or displays a validation error. 3. Session is not created. <i>Validation Failure — Missing Times:</i> 1. Leader submits the form without "Start now" checked but without providing start or end time. 2. System displays an error: "End time is required for scheduled sessions". 3. Session is not created. <i>Not a Group Member:</i> 1. System determines the requester is not a member of the group. 2. System displays a permission error. 3. Session is not created. <i>System Error:</i> 1. System encounters an unexpected failure during creation. 2. System displays an error message. 3. Leader may retry.

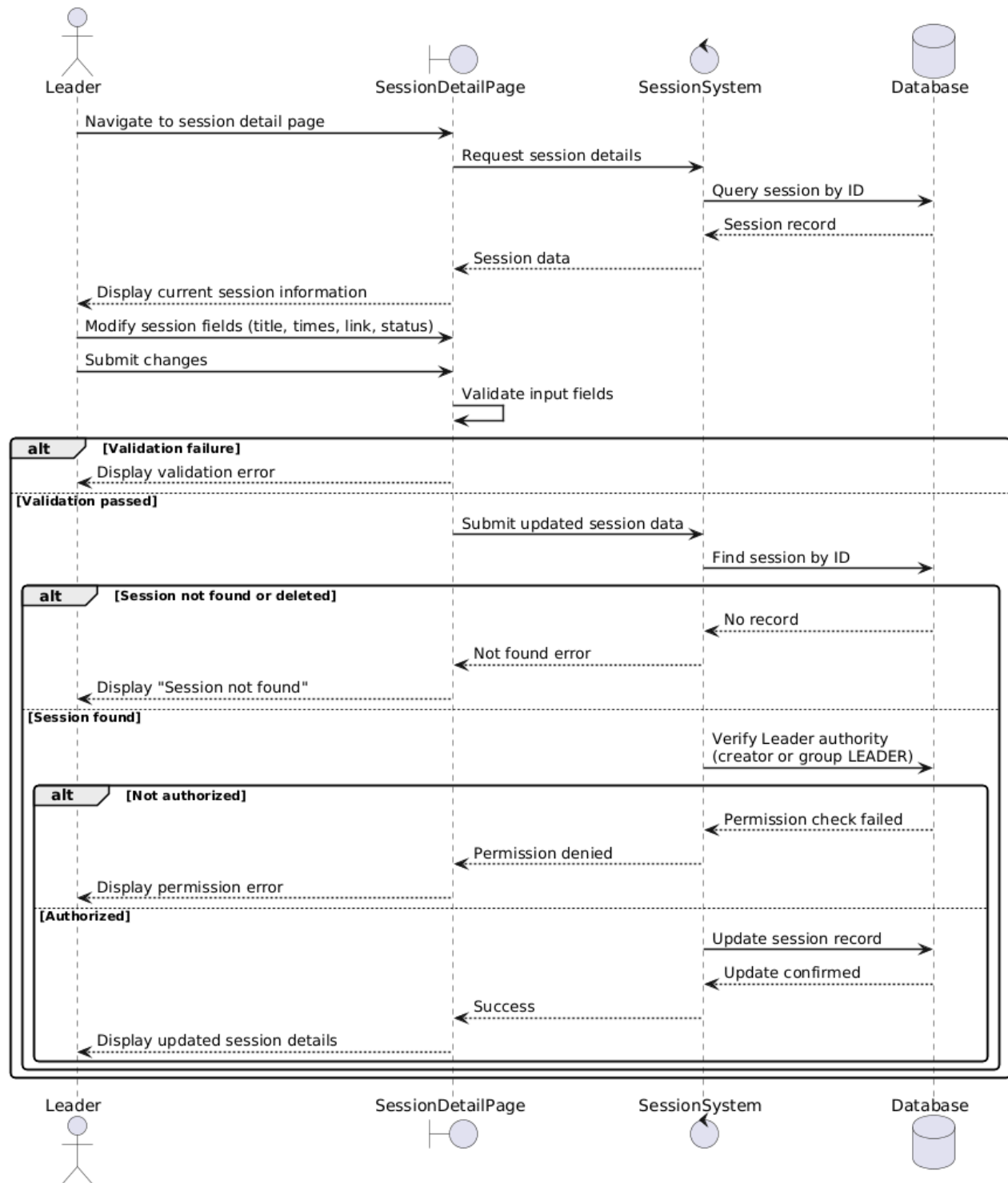


UC-L-06: Edit Session

Item	Description
Use Case ID	UC-L-06
Use Case Name	Edit Session
Description	Allows a Leader to update the details of an existing study session, such as the title, description, scheduled start/end time, meeting link, or status.

Actor(s)	Leader
Priority	High
Trigger	Leader submits updated session information through the session detail or session management interface.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader is the session creator or holds the LEADER role in the associated group. 3. The target session exists and has not been deleted.
Post-Conditions	1. The session's details are updated in the system. 2. The updated information is reflected on the session detail page.
Basic Flow	1. Leader navigates to the session detail page. 2. System displays the session's current information. 3. Leader accesses the edit function for the session. 4. Leader modifies one or more fields (title, description, start time, end time, meeting link, or status). 5. Leader submits the changes. 6. System validates the entered data. 7. System verifies the Leader's authority to manage the session. 8. System saves the updated session information. 9. System confirms the update and displays the refreshed session details.

Alternative Flow	<i>Cancel Edit:</i> 1. Leader decides not to save changes. 2. Leader dismisses the form. 3. System discards unsaved changes.
Exception Flow	<i>Unauthorized Attempt:</i> 1. A user who is neither the session creator nor a group Leader attempts to edit the session. 2. System denies the action and displays a permission error. <i>Session Not Found:</i> 1. Leader attempts to edit a session that does not exist or has been deleted. 2. System displays a "Session not found" error. <i>Validation Failure:</i> 1. Leader submits a title that is too short or an invalid date format. 2. System displays a validation error. 3. Changes are not saved. <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message.



UC-L-07: End Session

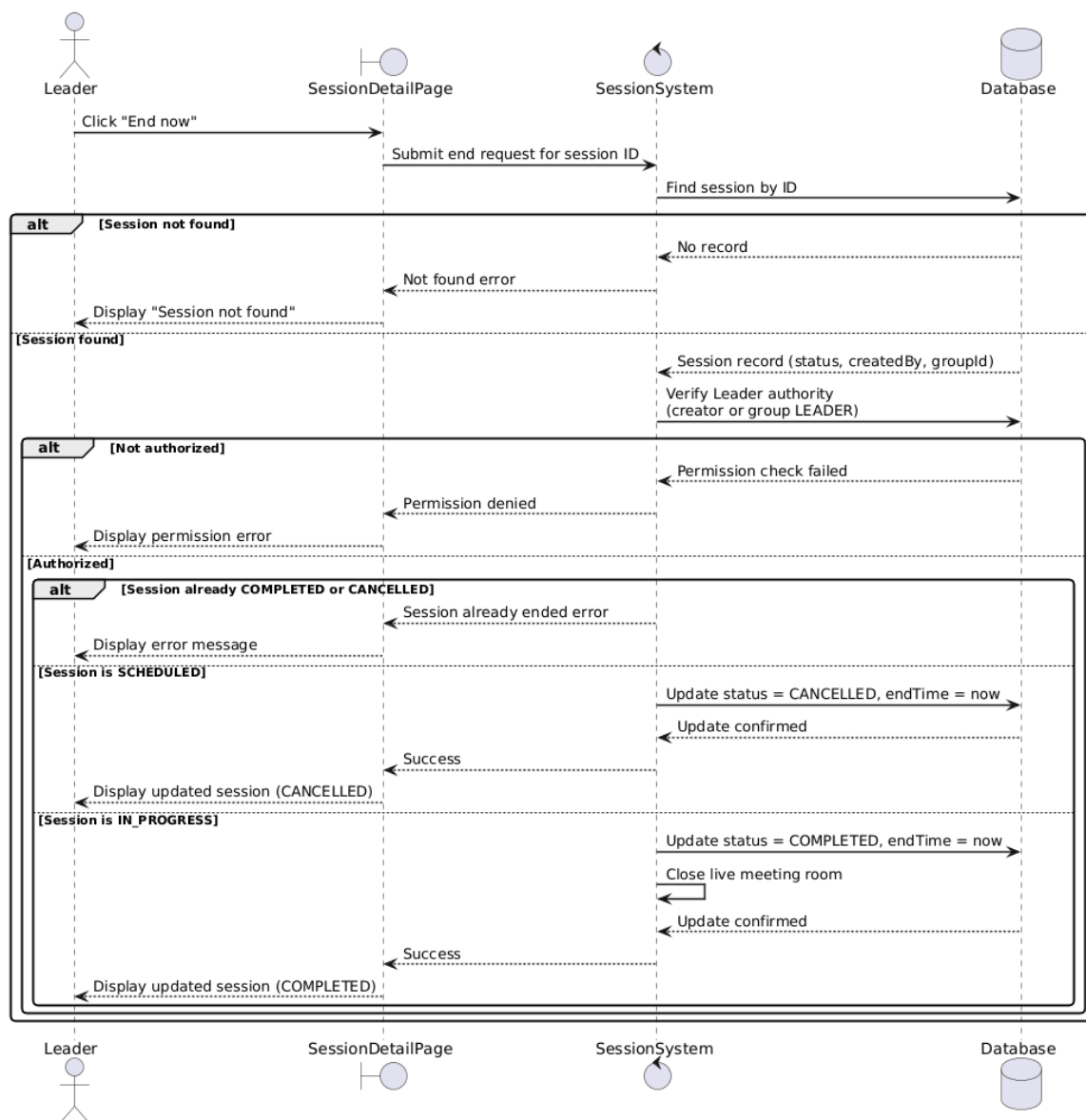
Item	Description
Use Case ID	UC-L-07
Use Case Name	End Session
Description	Allows a Leader to manually terminate an active or scheduled study session. If the session was In Progress, its status changes to Completed; if Scheduled, it is marked as Cancelled.
Actor(s)	Leader
Priority	High
Trigger	Leader clicks "End now" on an active or scheduled session card or on the session detail page.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader is the session creator or holds the LEADER role in the associated group. 3. The target session exists and has a status of Scheduled or In Progress.
Post-Conditions	1. The session status is changed to Completed (if it was In Progress) or Cancelled (if it was Scheduled). 2. The session's end time is recorded. 3. Any active meeting room associated with the session is closed.

Basic Flow	1. Leader navigates to the group's Sessions tab or the session detail page. 2. System displays the session with its current status and available actions. 3. Leader clicks "End now". 4. System verifies the Leader's authority to manage the session. 5. System determines the current session status. 6. System updates the session status to Completed (if In Progress) or Cancelled (if Scheduled). 7. System records the current time as the end time. 8. System closes the associated meeting room if the session was live. 9. System confirms the action and updates the display.
Alternative Flow	<i>Session Already Ended:</i> 1. Leader attempts to end a session that is already Completed or Cancelled. 2. System displays an error indicating the session has already ended. 3. No change is made.
Exception Flow	<i>Unauthorized Attempt:</i> 1. A user who is neither the session creator nor a group Leader attempts to end the session. 2. System denies the action and displays a permission error. <i>Session Not Found:</i>

1. Leader attempts to end a session that does not exist.
2. System displays a "Session not found" error.

System Error:

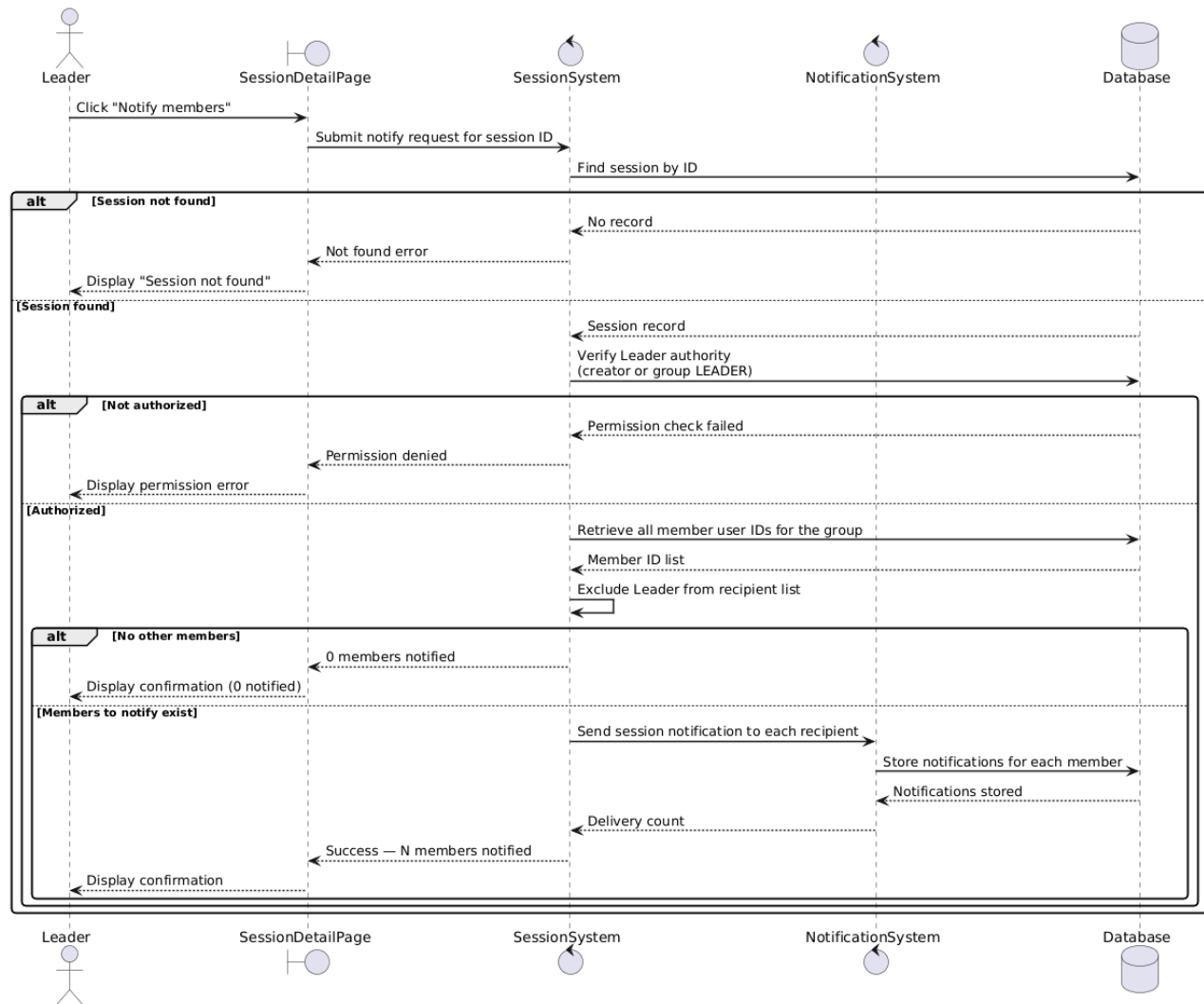
1. System encounters an unexpected failure.
2. System displays an error message.
3. Leader may retry.



UC-L-08: Notify Members

Item	Description
Use Case ID	UC-L-08
Use Case Name	Notify Members
Description	Allows a Leader to manually send a notification to all members of the group about a specific study session, reminding them of its details.
Actor(s)	Leader
Priority	Medium
Trigger	Leader clicks "Notify members" on a session card or on the session detail page.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader is the session creator or holds the LEADER role in the associated group. 3. The target session exists.
Post-Conditions	1. All group members except the Leader receive an in-app notification about the session. 2. The notification includes the session title, a message, and a link to the session detail page.
Basic Flow	1. Leader navigates to the Sessions tab or session detail page. 2. System displays session information and management actions.

	3. Leader clicks "Notify members". 4. System verifies the Leader's authority to manage the session. 5. System retrieves the list of all group members. 6. System sends a notification to each member (excluding the Leader themselves). 7. System confirms the number of members notified.
Alternative Flow	<i>No Other Members:</i> 1. The group has no members other than the Leader. 2. System sends no notifications. 3. System confirms with a count of 0 notified.
Exception Flow	<i>Unauthorized Attempt:</i> 1. A user who is not a LEADER of the group attempts to send notifications. 2. System denies the action and displays a permission error. <i>Session Not Found:</i> 1. Leader attempts to notify members for a session that does not exist. 2. System displays a "Session not found" error. <i>System Error:</i> 1. System encounters an unexpected failure during notification delivery. 2. System displays an error message. 3. Leader may retry.

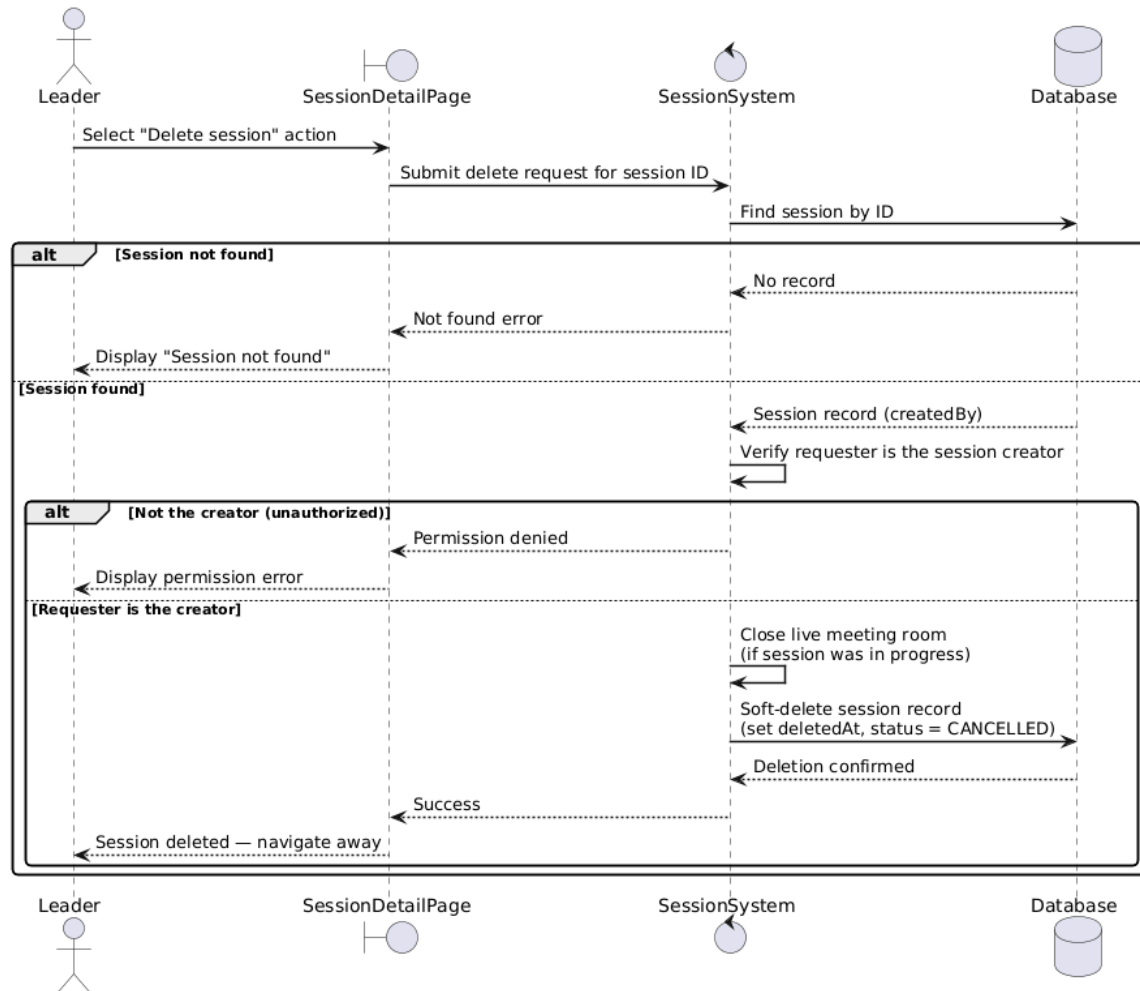


UC-L-09: Delete Session

Item	Description
Use Case ID	UC-L-09
Use Case Name	Delete Session
Description	Allows a Leader who created a session to permanently remove it from the system, closing any associated live meeting room.

Actor(s)	Leader
Priority	Medium
Trigger	Leader selects the option to delete a session from the session management interface.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader is the creator of the target session. 3. The target session exists and has not already been deleted.
Post-Conditions	1. The session is soft-deleted and no longer appears in any session list. 2. Any associated live meeting room is closed.
Basic Flow	1. Leader navigates to the session detail page or the group's Sessions tab. 2. System displays the session with management actions. 3. Leader selects the option to delete the session. 4. System verifies that the Leader is the session creator. 5. System closes the associated live meeting room (if applicable). 6. System marks the session as deleted. 7. System confirms the deletion.
Alternative Flow	<i>Cancel Deletion:</i> 1. Leader decides not to proceed with deletion. 2. Leader dismisses the action. 3. System takes no action and the session remains unchanged.

Exception Flow	<i>Unauthorized Attempt:</i> 1. A user who is not a LEADER of the group attempts to approve a request. 2. System denies the action and delete the session. <i>Session Not Found:</i> 1. Leader attempts to delete a session that does not exist. 2. System displays a "Session not found" error. <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message. 3. Leader may retry the action.
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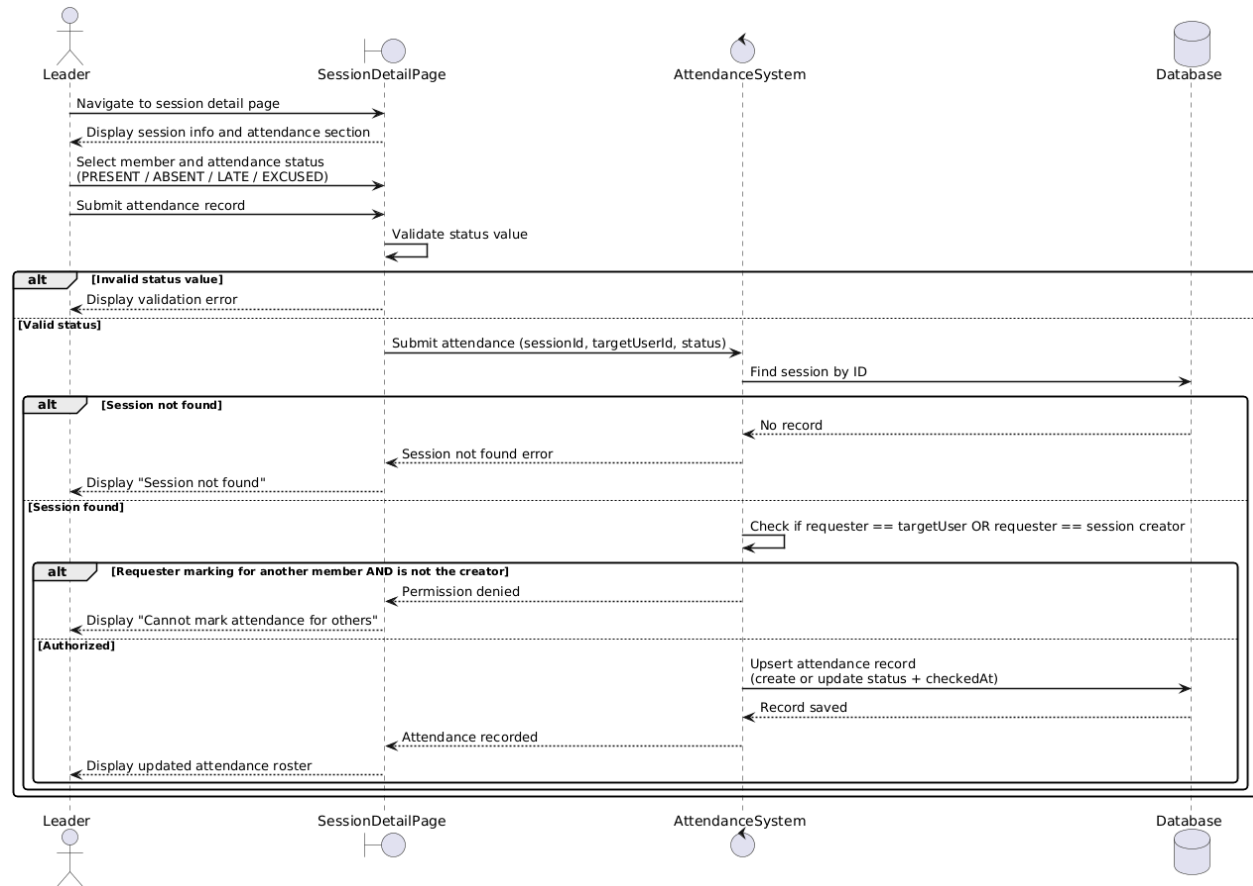


UC-L-10: Mark Attendance

Item	Description
Use Case ID	UC-L-10
Use Case Name	Mark Attendance
Description	Allows a Leader to record the attendance status of a group member for a specific study session, setting their status to Present, Absent, Late, or Excused.

Actor(s)	Leader
Priority	Medium
Trigger	Leader submits an attendance record for a specific member in a given session.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader is the session creator. 3. The target session exists. 4. The target member's user record exists.
Post-Conditions	1. The member's attendance record for the session is created or updated with the specified status and the current timestamp. 2. The updated attendance record is reflected in the attendance roster.
Basic Flow	1. Leader navigates to the session detail page or attendance management interface. 2. System displays the session details and available member list. 3. Leader selects a member and assigns an attendance status (Present, Absent, Late, or Excused). 4. Leader submits the attendance record. 5. System verifies that the Leader is the session creator. 6. System creates or updates the attendance record for the specified member. 7. System confirms the record was saved and displays the updated attendance roster.

Alternative Flow	<i>Self-Attendance (Member marking own attendance):</i> 1. A member marks their own attendance as Present. 2. System accepts the record without requiring session creator privileges. 3. System saves the attendance record.
Exception Flow	<i>Unauthorized Attempt:</i> 1. A user attempts to mark attendance for another member but is not the session creator. 2. System denies the action and displays a permission error. <i>Session Not Found:</i> 1. Leader attempts to mark attendance for a session that does not exist. 2. System displays a "Session not found" error. <i>Invalid Status:</i> 1. Leader submits an attendance status that is not one of Present, Absent, Late, or Excused. 2. System displays a validation error. <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message.



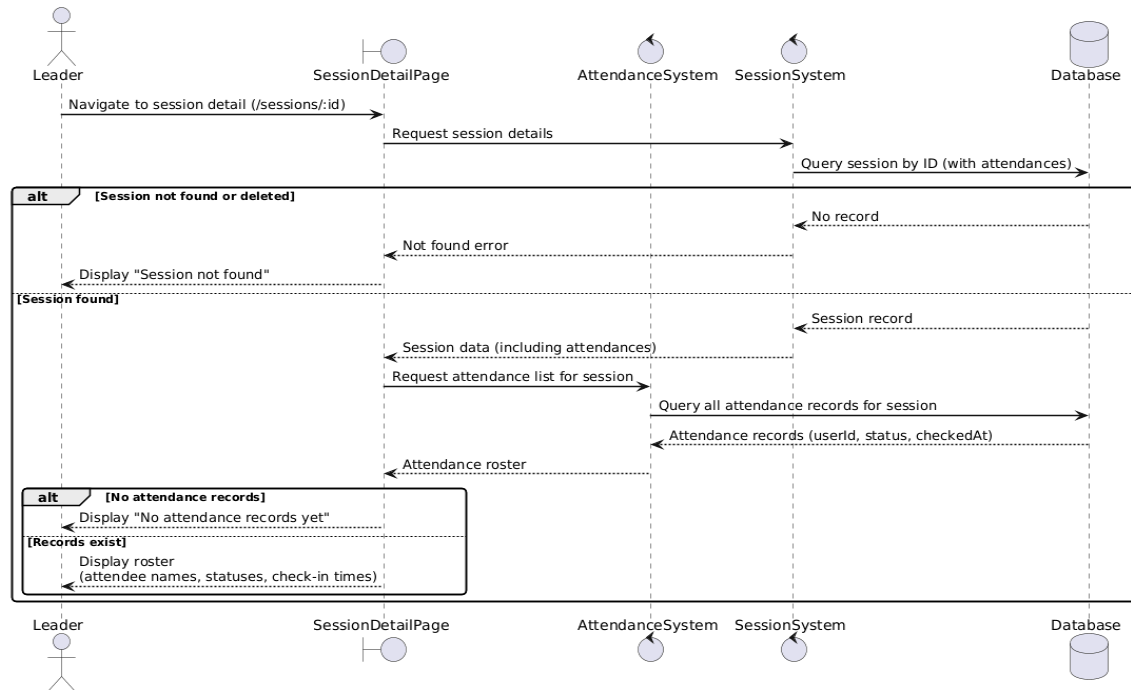
UC-L-11: View Attendance Roster

Item	Description
Use Case ID	UC-L-11
Use Case Name	View Attendance Roster
Description	Allows a Leader to view the complete list of attendance records for a specific study session, showing which members attended, their attendance statuses, and when they checked in.
Actor(s)	Leader

Priority	Medium
Trigger	Leader navigates to the session detail page, which displays the attendance section.
Pre-Conditions	1. The Leader is authenticated. 2. The target session exists. 3. The Leader is a member of the group associated with the session.
Post-Conditions	1. The attendance roster is displayed, listing all recorded attendees and their statuses. 2. No data is modified.
Basic Flow	1. Leader navigates to the session detail page. 2. System retrieves and displays the session information. 3. System retrieves all attendance records for the session. 4. System displays the attendance roster with each attendee's name and status. 5. Leader reviews the attendance list.
Alternative Flow	<i>No Attendance Records:</i> 1. No members have checked in or been marked yet. 2. System displays a "No attendance records yet" message.
Exception Flow	<i>Session Not Found:</i> 1. Leader navigates to a session that does not exist or has been deleted. 2. System displays a "Session not found" error.

System Error:

1. System fails to retrieve attendance data.
2. System displays an error message.
3. Leader may refresh the page.



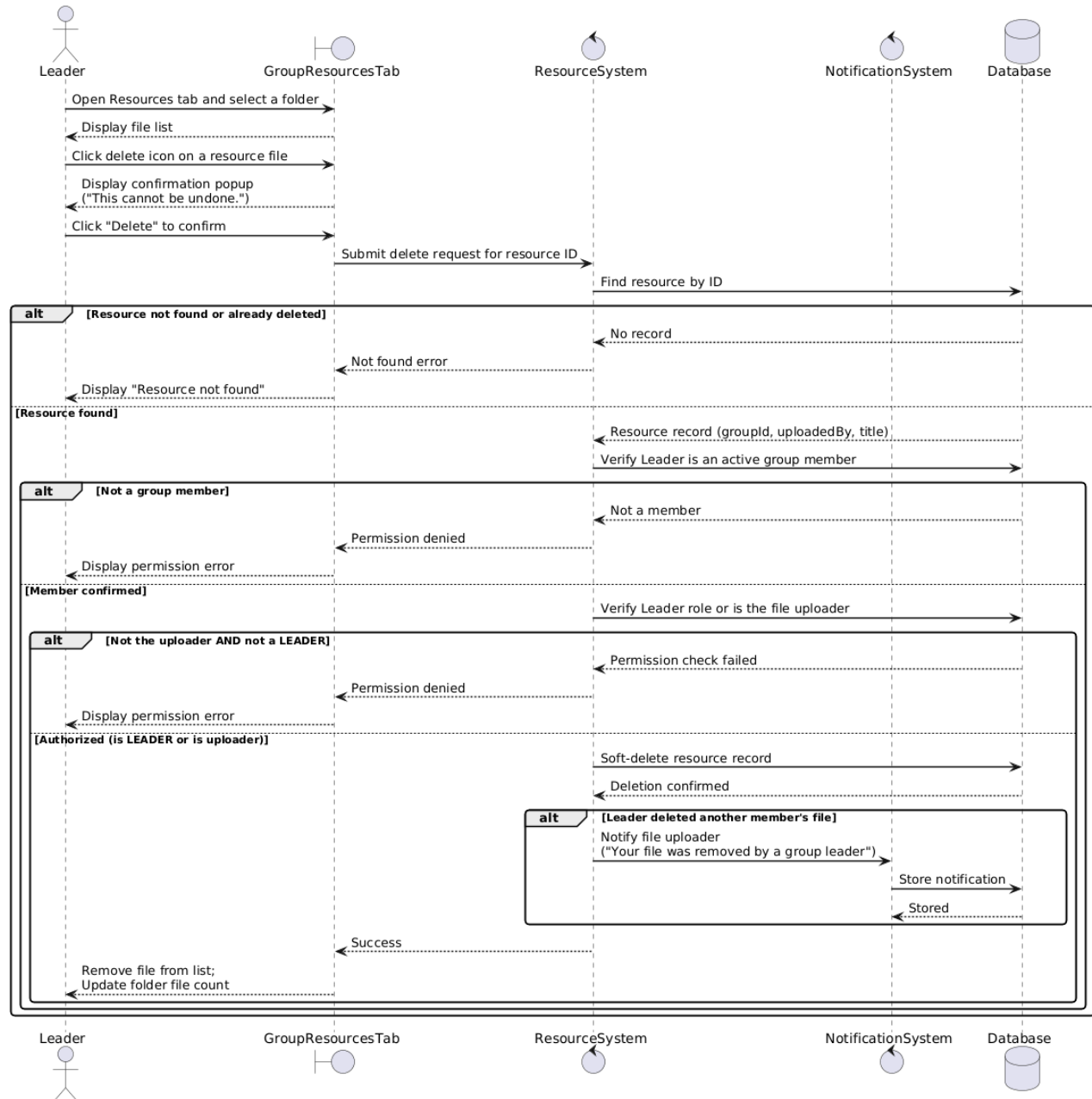
UC-L-12: Delete Resource

Item	Description
Use Case ID	UC-L-12
Use Case Name	Delete Resource

Description	Allows a Leader to remove any shared resource file from their group's resource folder. When the Leader deletes a file uploaded by another member, that member receives an in-app notification informing them of the removal.
Actor(s)	Leader
Priority	High
Trigger	Leader clicks the delete icon on a resource file row within the group's Resources tab and confirms the deletion.
Pre-Conditions	1. The Leader is authenticated. 2. The Leader holds the LEADER role or is the group creator. 3. The Leader is an active member of the group. 4. The target resource exists and has not already been deleted.
Post-Conditions	1. The resource is soft-deleted and removed from the folder's file list. 2. If the deleted resource was uploaded by another member, that member receives an in-app notification about the removal. 3. The folder's file count is updated.
Basic Flow	1. Leader opens the group detail page and navigates to the Resources tab. 2. System displays the resource folders for the group. 3. Leader opens a specific folder and views the file list. 4. Leader clicks the delete icon on a resource file. 5. System displays a confirmation popup.

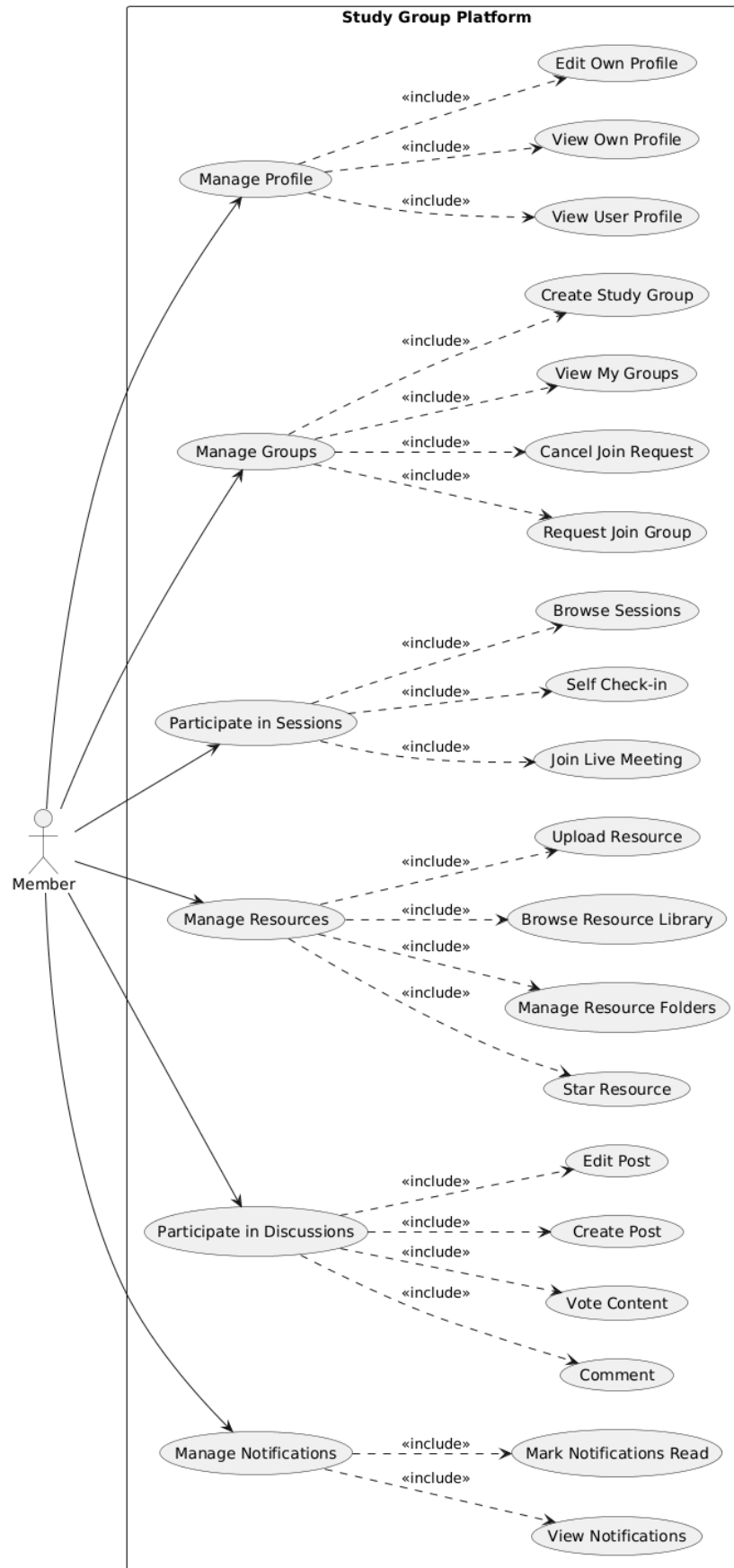
	6. Leader confirms the deletion. 7. System verifies the Leader's membership and role within the group. 8. System soft-deletes the resource file. 9. If the file was uploaded by a different member, system sends a notification to that member. 10. System updates the file list and confirms the deletion.
Alternative Flow	<i>Cancel Deletion:</i> 1. Leader clicks the delete icon but then clicks "Cancel" in the confirmation popup. 2. System dismisses the popup. 3. Resource remains unchanged. <i>Delete Own Resource:</i> 1. Leader deletes a file that they themselves uploaded. 2. System proceeds with deletion without sending a notification.
Exception Flow	<i>Unauthorized Attempt:</i> 1. A user who is not the file uploader and not a group Leader attempts to delete a resource. 2. System denies the action and displays a permission error. <i>Resource Not Found:</i> 1. Leader attempts to delete a resource that no longer exists. 2. System displays a "Resource not found" error.

	<i>Not a Group Member:</i> 1. Leader is no longer an active member of the group. 2. System displays a permission error. <i>System Error:</i> 1. System encounters an unexpected failure during deletion. 2. System displays an error message. 3. Leader may retry.
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3.3 Use Case for Members

Members are registered users who can manage their profiles, participate in study groups and sessions, share learning resources, and engage in discussions with other members.

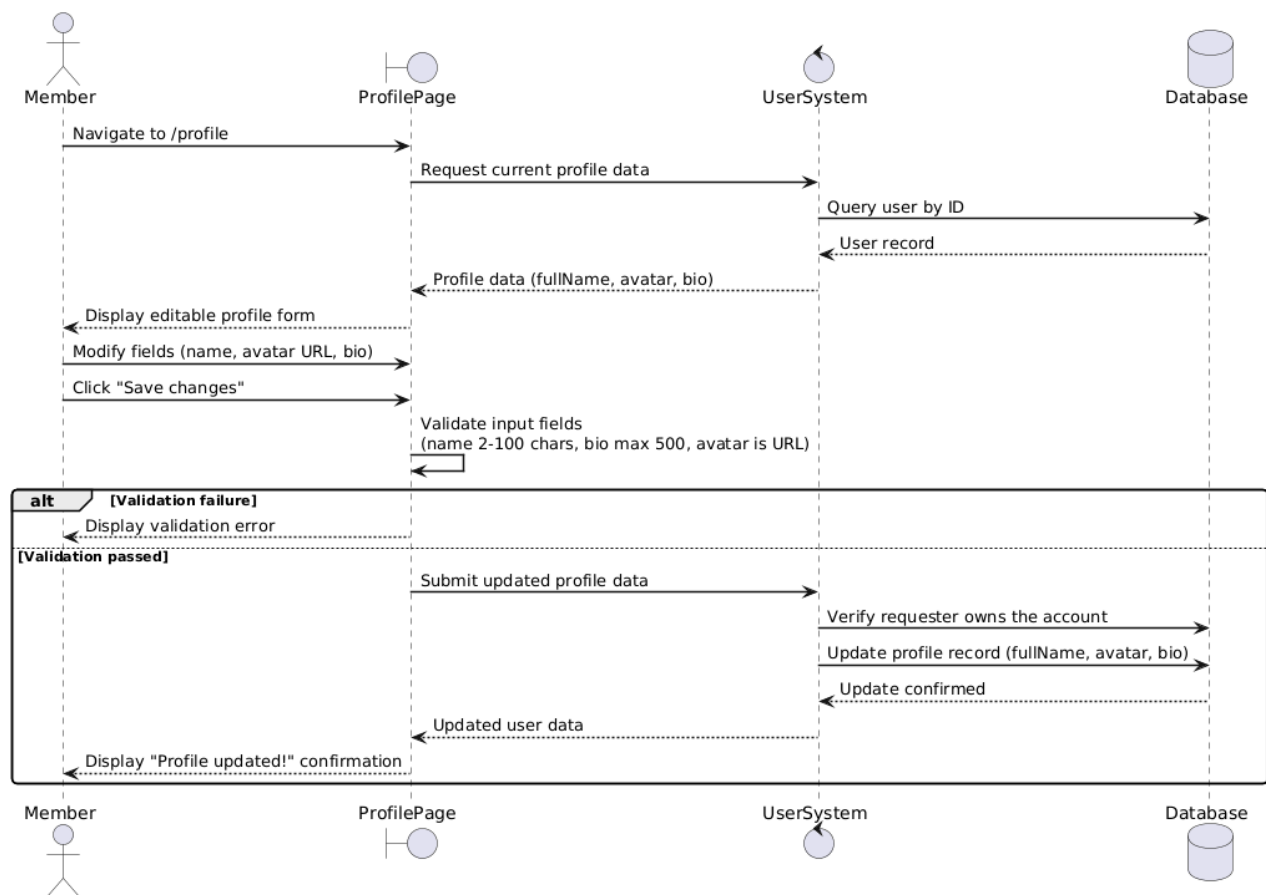


UC-M-01: Edit Own Profile

Item	Description
Use Case ID	UC-M-01
Use Case Name	Edit Own Profile
Description	Allows a Member to update their personal profile information, including their full name, avatar image URL, and biographical description.
Actor(s)	Member
Priority	Medium
Trigger	Member navigates to the Profile page, modifies their information, and submits the changes.
Pre-Conditions	1. The Member is authenticated. 2. The Member is editing their own profile.
Post-Conditions	1. The Member's profile information is updated in the system. 2. The updated details are immediately reflected across the platform.
Basic Flow	1. Member navigates to the Profile page. 2. System displays the current profile information in an editable form (full name, avatar URL, and bio). 3. Member modifies one or more fields.

	4. Member clicks "Save changes". 5. System validates the entered data. 6. System updates the profile record. 7. System displays a success confirmation message.
Alternative Flow	<i>No Changes Made:</i> 1. Member clicks "Save changes" without modifying any fields. 2. System saves the unchanged data and displays a success confirmation.
Exception Flow	<i>Validation Failure — Full Name:</i> 1. Member submits a name shorter than 2 characters or longer than 100 characters. 2. System displays a validation error. 3. Profile is not updated. <i>Validation Failure — Bio:</i> 1. Member submits a bio exceeding 500 characters. 2. System displays a validation error. 3. Profile is not updated. <i>Validation Failure — Avatar URL:</i> 1. Member enters a malformed URL for the avatar. 2. System displays a validation error. 3. Profile is not updated.

	<i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message. 3. Member may retry.
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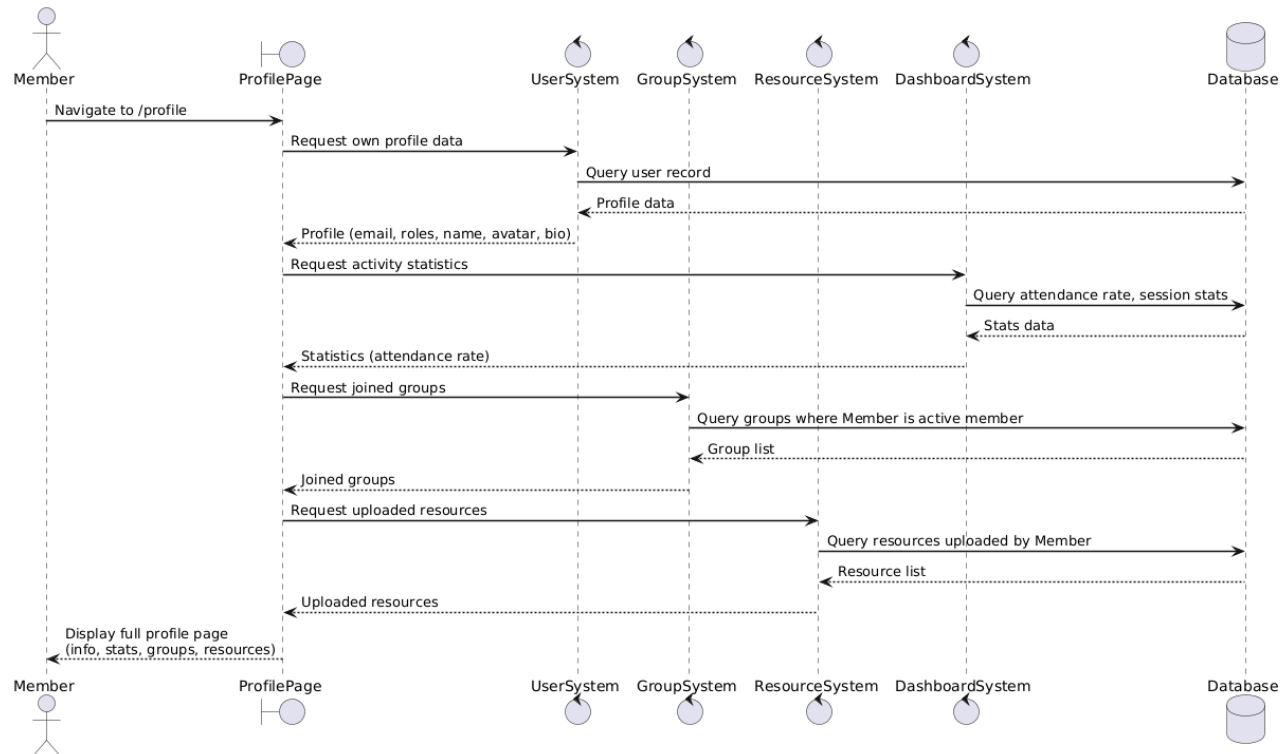


UC-M-02: View Own Profile

Item	Description
Use Case ID	UC-M-02
Use Case Name	View Own Profile

Description	Allows a Member to view their own profile page, which displays their personal information, assigned roles, activity statistics (group count, attendance rate, uploaded resources), a list of joined groups, and their uploaded resources.
Actor(s)	Member
Priority	Medium
Trigger	Member navigates to the Profile page.
Pre-Conditions	1. The Member is authenticated.
Post-Conditions	1. The Member's profile page is displayed with up-to-date personal information and activity statistics. 2. No data is modified.
Basic Flow	1. Member navigates to the Profile page. 2. System retrieves the Member's profile data, activity statistics, joined groups, and uploaded resources. 3. System displays the Member's email address, roles, full name, avatar, and bio. 4. System displays summary statistics: number of joined groups, attendance rate, and number of uploaded resources. 5. System displays a list of the Member's joined groups with links to each. 6. System displays a list of resources the Member has uploaded.

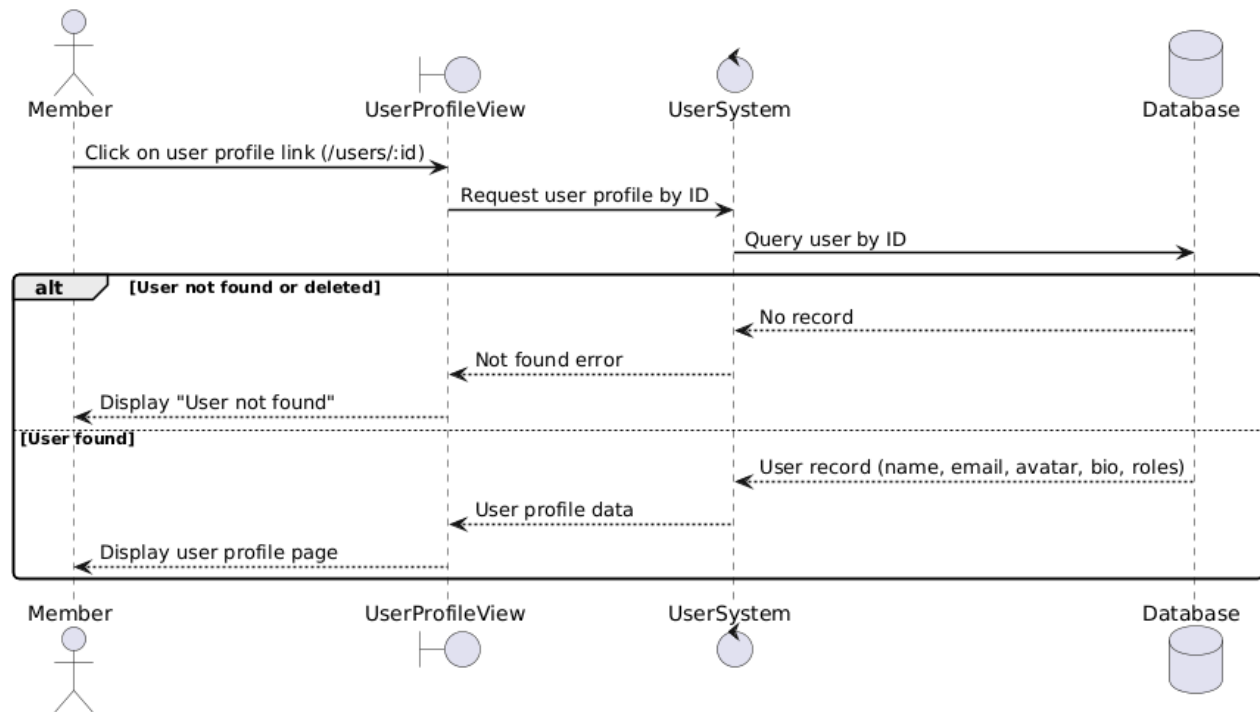
Alternative Flow	<i>No Joined Groups:</i> 1. The Member has not joined any groups. 2. System displays "No groups joined." in the groups section. <i>No Uploaded Resources:</i> 1. The Member has not uploaded any resources. 2. System displays "No uploads yet." in the resources section.
Exception Flow	<i>System Error:</i> 1. System fails to retrieve profile or statistics data. 2. System displays an error or empty state. 3. Member may refresh the page.



UC-M-03: View User Profile

Item	Description
Use Case ID	UC-M-03
Use Case Name	View User Profile
Description	Allows a Member to view the public profile of another registered user on the platform, including their name, avatar, bio, and role information.
Actor(s)	Member
Priority	Low

Trigger	Member navigates to another user's profile page by clicking their name or profile link.
Pre-Conditions	1. The Member is authenticated. 2. The target user exists in the system.
Post-Conditions	1. The target user's profile information is displayed. 2. No data is modified.
Basic Flow	1. Member clicks on another user's name or profile link. 2. System retrieves the target user's profile data. 3. System displays the user's full name, email, avatar, bio, roles, and account status.
Alternative Flow	None.
Exception Flow	<i>User Not Found:</i> 1. The target user does not exist or has been deleted. 2. System displays a "User not found" error. <i>System Error:</i> 1. System fails to retrieve user data. 2. System displays an error message.

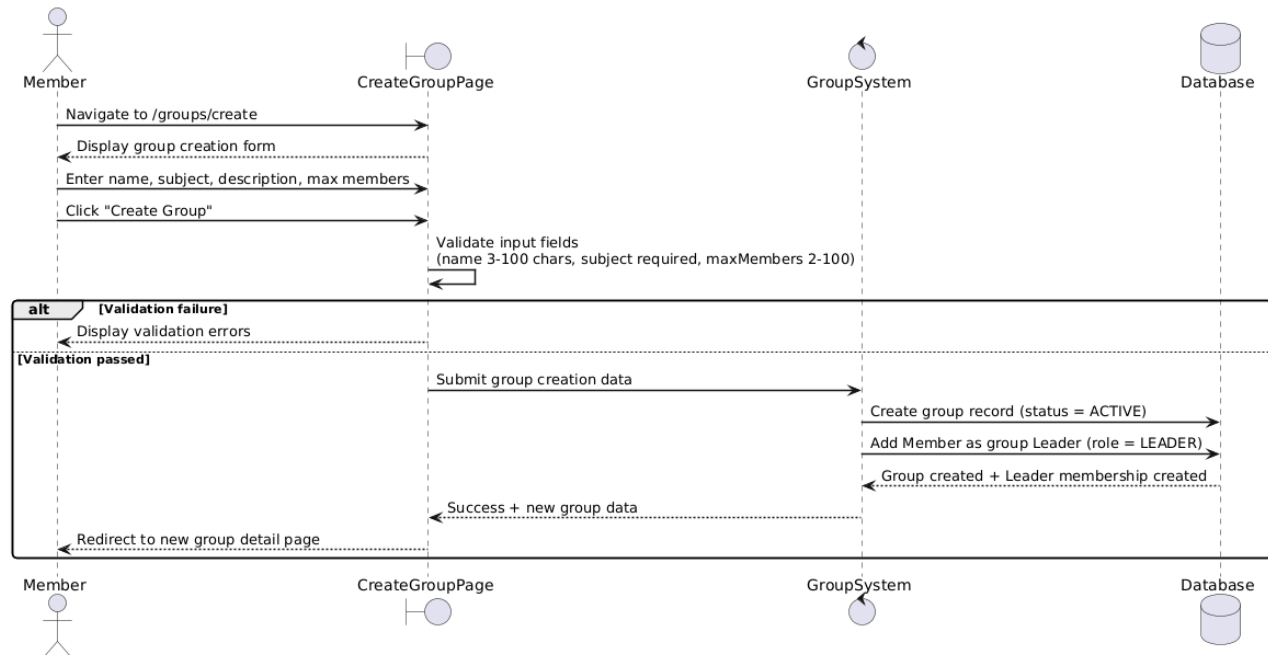


UC-M-04: Create Study Group

Item	Description
Use Case ID	UC-M-04
Use Case Name	Create Study Group
Description	Allows a Member to create a new study group on the platform. The Member who creates the group automatically becomes its Leader.
Actor(s)	Member
Priority	High

Trigger	Member clicks "Create group" and submits the group creation form.
Pre-Conditions	1. The Member is authenticated.
Post-Conditions	1. A new study group is created with Active status. 2. The Member is automatically enrolled as the group's Leader. 3. The new group is accessible in the study groups listing.
Basic Flow	1. Member navigates to the Study Groups page and clicks "Create group". 2. System displays the group creation form (Name, Subject, Description, Max Members). 3. Member enters the group name, subject, optional description, and maximum member count. 4. Member submits the form. 5. System validates the entered information. 6. System creates the new group and assigns the Member as its Leader. 7. System redirects the Member to the newly created group's detail page.
Alternative Flow	<i>Cancel:</i> 1. Member navigates away without submitting. 2. System discards the entered data. 3. No group is created.

Exception Flow	<i>Validation Failure — Name:</i> 1. Member submits a group name shorter than 3 characters or longer than 100 characters. 2. System displays a validation error. 3. Group is not created. <i>Validation Failure — Subject:</i> 1. Member submits an empty subject field. 2. System displays a validation error. 3. Group is not created. <i>Validation Failure — Max Members:</i> 1. Member enters a value outside the allowed range (2–100). 2. System displays a validation error. 3. Group is not created. <i>System Error:</i> 1. System encounters an unexpected failure during creation. 2. System displays an error message. 3. Member may retry.
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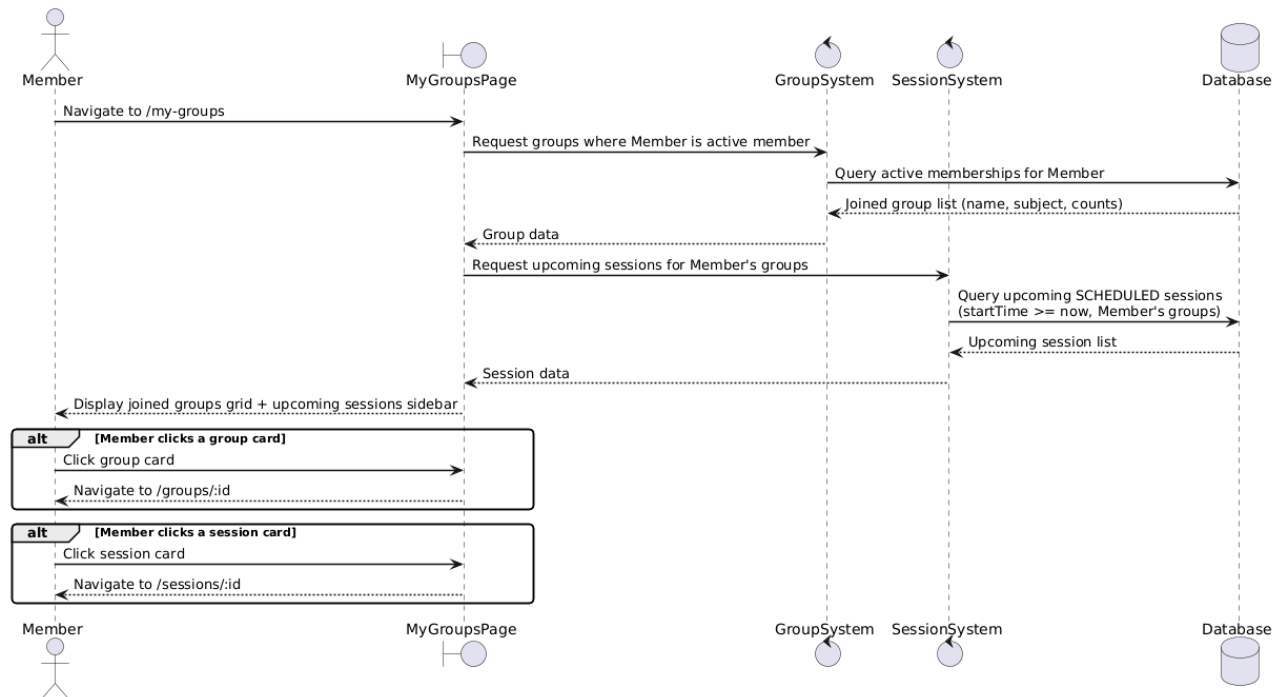


UC-M-05: View My Groups

Item	Description
Use Case ID	UC-M-05
Use Case Name	View My Groups
Description	Allows a Member to view a personalized dashboard of all study groups they have joined, along with a sidebar showing their upcoming scheduled sessions.
Actor(s)	Member
Priority	High
Trigger	Member navigates to the My Groups page.

Pre-Conditions	1. The Member is authenticated.
Post-Conditions	1. All groups the Member belongs to are displayed, along with their upcoming sessions. 2. No data is modified.
Basic Flow	1. Member navigates to the My Groups page. 2. System retrieves all groups the Member has joined and their upcoming scheduled sessions. 3. System displays joined groups in a grid, showing each group's name, subject, member count, and session count. 4. System displays a sidebar with upcoming sessions showing title, group name, and scheduled start time. 5. Member may click on a group card to navigate to the group's detail page. 6. Member may click on a session card to navigate to the session's detail page.
Alternative Flow	<i>No Joined Groups:</i> 1. Member has not joined any groups. 2. System displays a prompt to explore groups with a link to the Groups page. <i>No Upcoming Sessions:</i> 1. Member has no upcoming sessions. 2. System displays "No upcoming sessions." in the sidebar.

Exception Flow	<i>System Error:</i> 1. System fails to retrieve groups or session data. 2. System displays an error or loading state. 3. Member may refresh.
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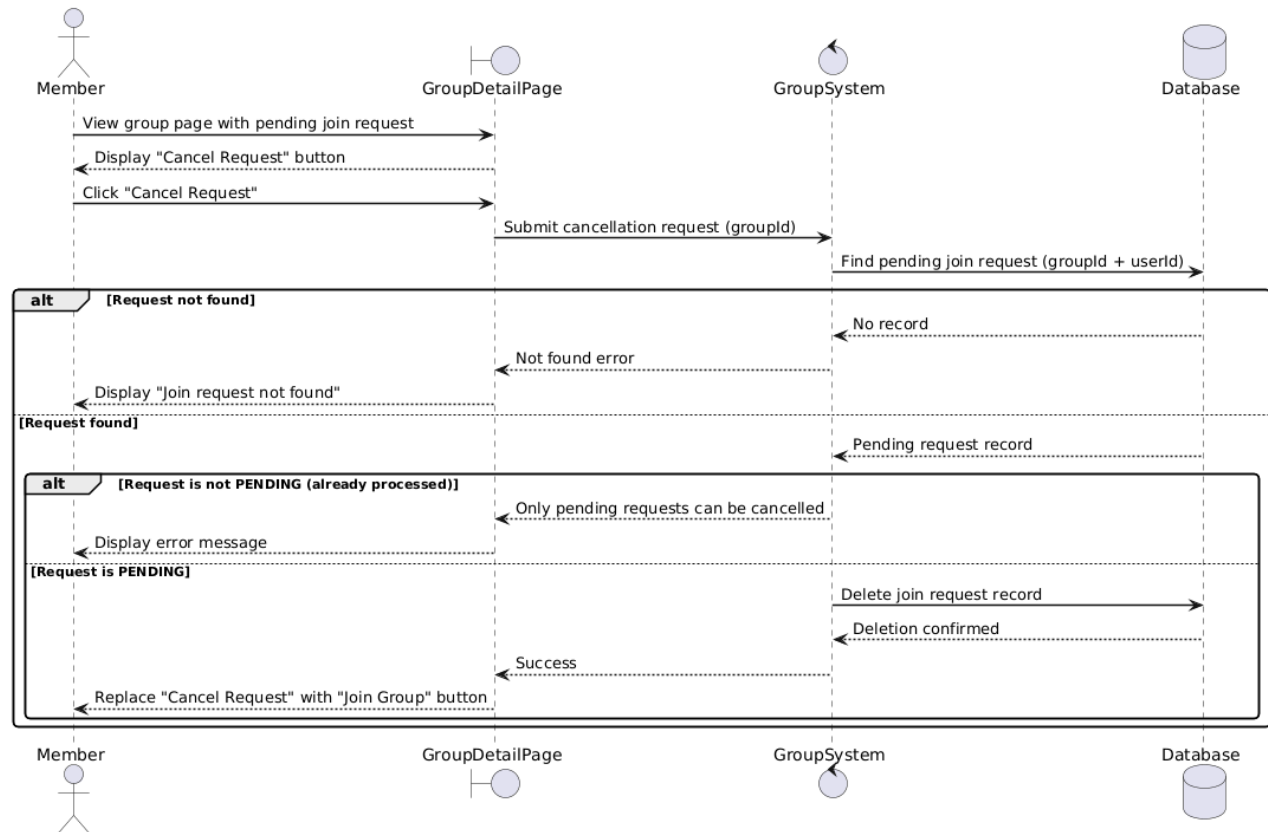
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UC-M-06: Cancel Join Request

Item	Description
Use Case ID	UC-M-06
Use Case Name	Cancel Join Request

Description	Allows a Member to withdraw a pending join request they submitted for a study group before the group Leader has acted on it.
Actor(s)	Member
Priority	Medium
Trigger	Member clicks "Cancel Request" on a group detail page where they have a pending join request.
Pre-Conditions	1. The Member is authenticated. 2. The Member has a pending join request for the target group. 3. The join request has not yet been approved or rejected.
Post-Conditions	1. The pending join request is removed from the system. 2. The Member is no longer listed as a pending applicant for the group.
Basic Flow	1. Member navigates to the group detail page where they have a pending request. 2. System displays a "Cancel Request" button indicating their pending request status. 3. Member clicks "Cancel Request". 4. System locates the Member's pending join request for the group. 5. System deletes the pending join request.

	6. System updates the page to show the "Join Group" button again.
Alternative Flow	None.
Exception Flow	<i>Request Not Found:</i> 1. The join request no longer exists (e.g., was processed by the Leader). 2. System displays a "Join request not found" error. <i>Request Already Processed:</i> 1. Member attempts to cancel a request that has already been approved or rejected. 2. System displays an error indicating only pending requests can be cancelled. <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message.



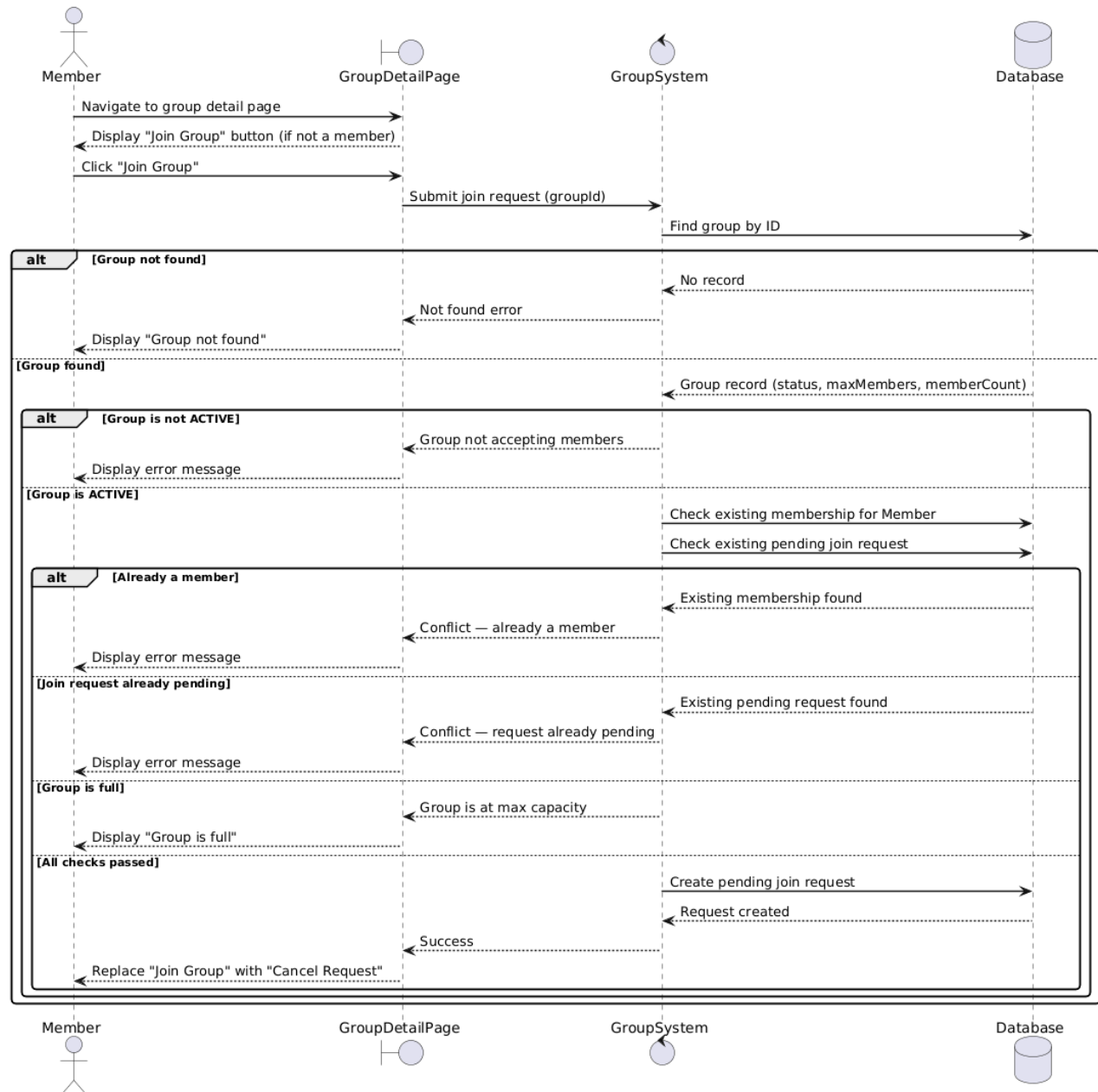
UC-M-07: Request Join Group

Item	Description
Use Case ID	UC-M-07
Use Case Name	Request Join Group
Description	Allows a Member to submit a request to join a study group. The request is placed in a pending state and must be approved by the group Leader before the Member gains access.

Actor(s)	Member
Priority	High
Trigger	Member clicks "Join Group" on a group's detail page or group card.
Pre-Conditions	1. The Member is authenticated. 2. The target group exists and has an Active status. 3. The Member is not already a member of the group. 4. The Member does not already have a pending join request for the group. 5. The group has not reached its maximum member capacity.
Post-Conditions	1. A pending join request is created for the Member in the target group. 2. The "Join Group" button is replaced with a "Cancel Request" button on the group page.
Basic Flow	1. Member navigates to the group detail page or group listing. 2. System displays a "Join Group" button for groups the Member has not joined. 3. Member clicks "Join Group". 4. System verifies the group is Active and the Member has no existing membership or pending request. 5. System checks that the group has not reached its

	maximum member capacity. 6. System creates a pending join request for the Member. 7. System updates the page to show a "Cancel Request" button.
Alternative Flow	<i>Click Join from Group Listing:</i> 1. Member clicks "Join" on a group card in the groups listing. 2. System navigates the Member to the group detail page. 3. The join request flow proceeds from the group detail page.
Exception Flow	<i>Already a Member:</i> 1. Member attempts to join a group they already belong to. 2. System displays a conflict error: "Already a member". <i>Request Already Pending:</i> 1. Member attempts to submit a duplicate join request. 2. System displays a conflict error: "Join request already pending". <i>Group Not Active:</i> 1. Member attempts to join a group that is not in Active status. 2. System displays an error: "This group is not accepting members".

	<i>Group is Full:</i> 1. The group has reached its maximum member capacity. 2. System displays an error: "Group is full". <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message.
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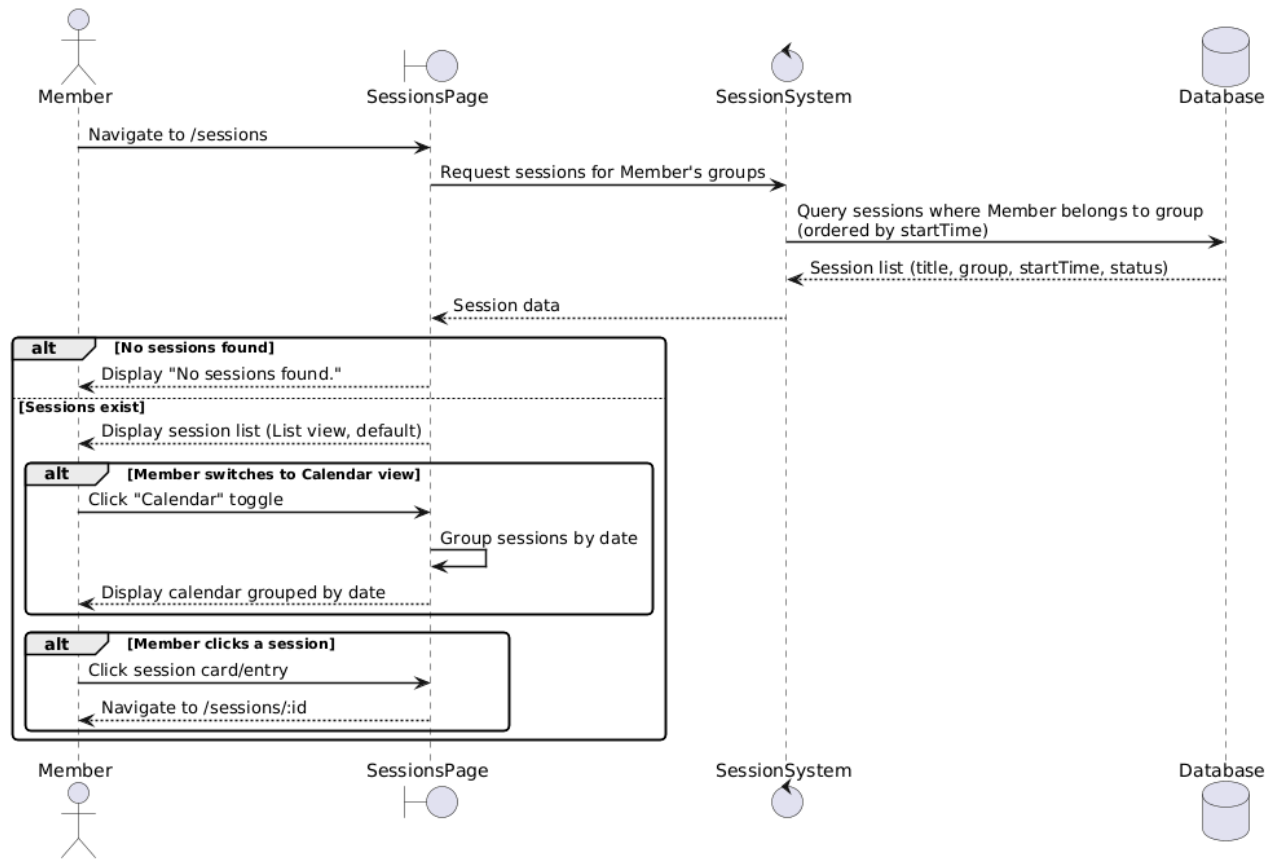


UC-M-08: Browse Sessions

Item	Description
Use Case ID	UC-M-08

Use Case Name	Browse Sessions
Description	Allows a Member to view a list of study sessions associated with the groups they have joined, including upcoming scheduled sessions and session history. Sessions can be viewed in list or calendar view.
Actor(s)	Member
Priority	High
Trigger	Member navigates to the Study Sessions page.
Pre-Conditions	1. The Member is authenticated. 2. The Member belongs to at least one study group with scheduled sessions.
Post-Conditions	1. A list of the Member's relevant study sessions is displayed. 2. No data is modified.
Basic Flow	1. Member navigates to the Study Sessions page. 2. System retrieves all sessions associated with the Member's groups. 3. System displays each session's title, group name, start time, and status (Scheduled, In Progress, Completed, or Cancelled). 4. Member may toggle between List view and Calendar

	view (grouped by date). 5. Member may click a session to navigate to its detail page.
Alternative Flow	<i>Calendar View:</i> 1. Member clicks the "Calendar" toggle. 2. System groups and displays sessions by their scheduled date. <i>No Sessions:</i> 1. The Member has no sessions in their groups. 2. System displays "No sessions found."
Exception Flow	<i>System Error:</i> 1. System fails to retrieve sessions. 2. System displays an error state. 3. Member may refresh.

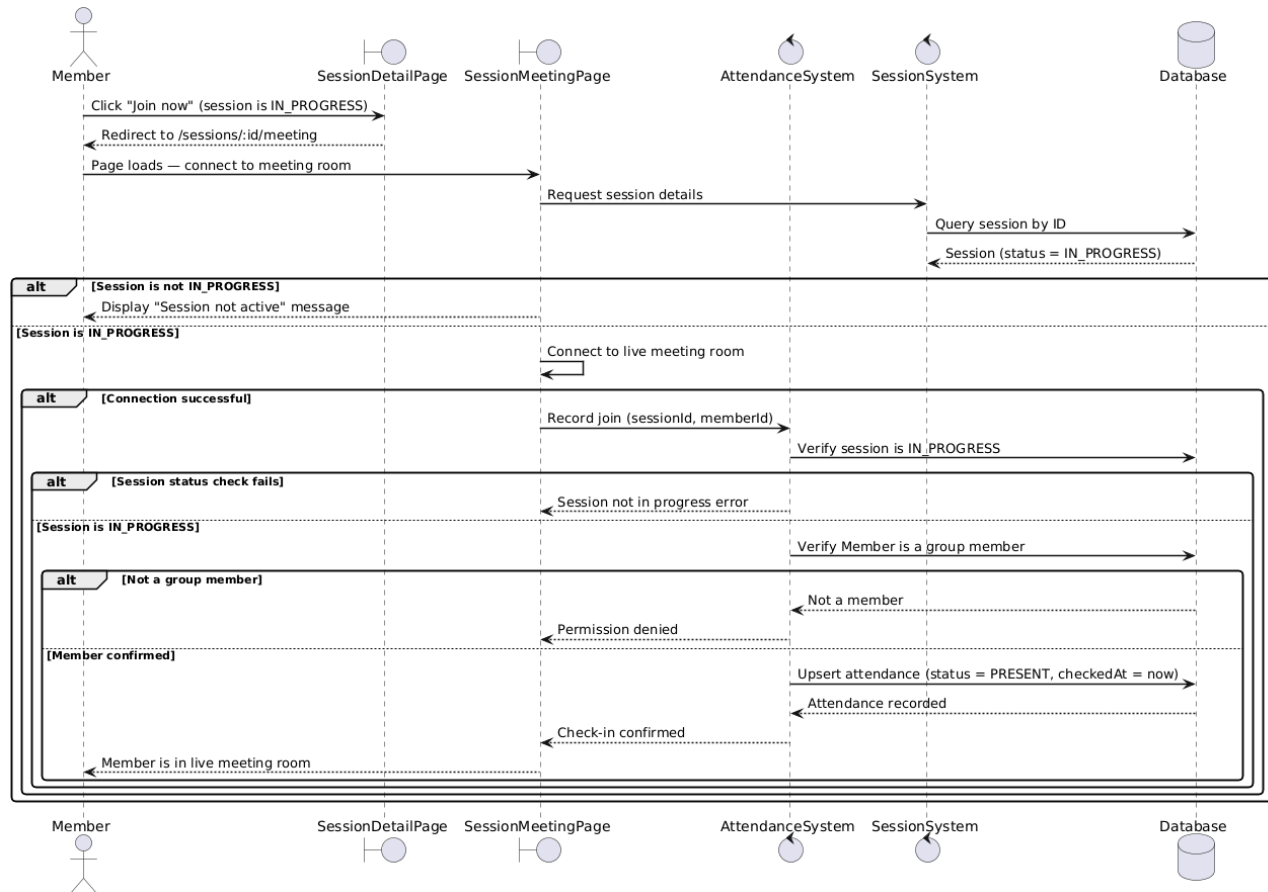


UC-M-09: Self Check-in

Item	Description
Use Case ID	UC-M-09
Use Case Name	Self Check-in
Description	Allows a Member to mark themselves as Present for an In Progress study session by joining the live meeting room. The system automatically records their attendance upon successful connection.

Actor(s)	Member
Priority	High
Trigger	Member joins a live meeting room for an In Progress session, triggering an automatic attendance record.
Pre-Conditions	1. The Member is authenticated. 2. The target session exists and has a status of In Progress. 3. The Member is an active member of the group associated with the session.
Post-Conditions	1. An attendance record with status "Present" is created or updated for the Member in the session. 2. The Member's attendance is reflected in the session's attendance roster.
Basic Flow	1. Member navigates to the session detail page and clicks "Join now". 2. System verifies the session is In Progress and the Member belongs to the group. 3. System redirects the Member to the live meeting room page. 4. Upon successful connection to the meeting room, the system automatically records the Member's attendance as Present. 5. System updates the attendance roster for the session.
Alternative Flow	None.

Exception Flow	<i>Session Not In Progress:</i> 1. Member attempts to check in to a session that is not In Progress. 2. System displays an error: "Session is not in progress". 3. No attendance record is created. <i>Not a Group Member:</i> 1. Member attempts to check in to a session for a group they do not belong to. 2. System denies access and displays a permission error. <i>System Error:</i> 1. System encounters an unexpected failure during attendance recording. 2. System displays an error message.
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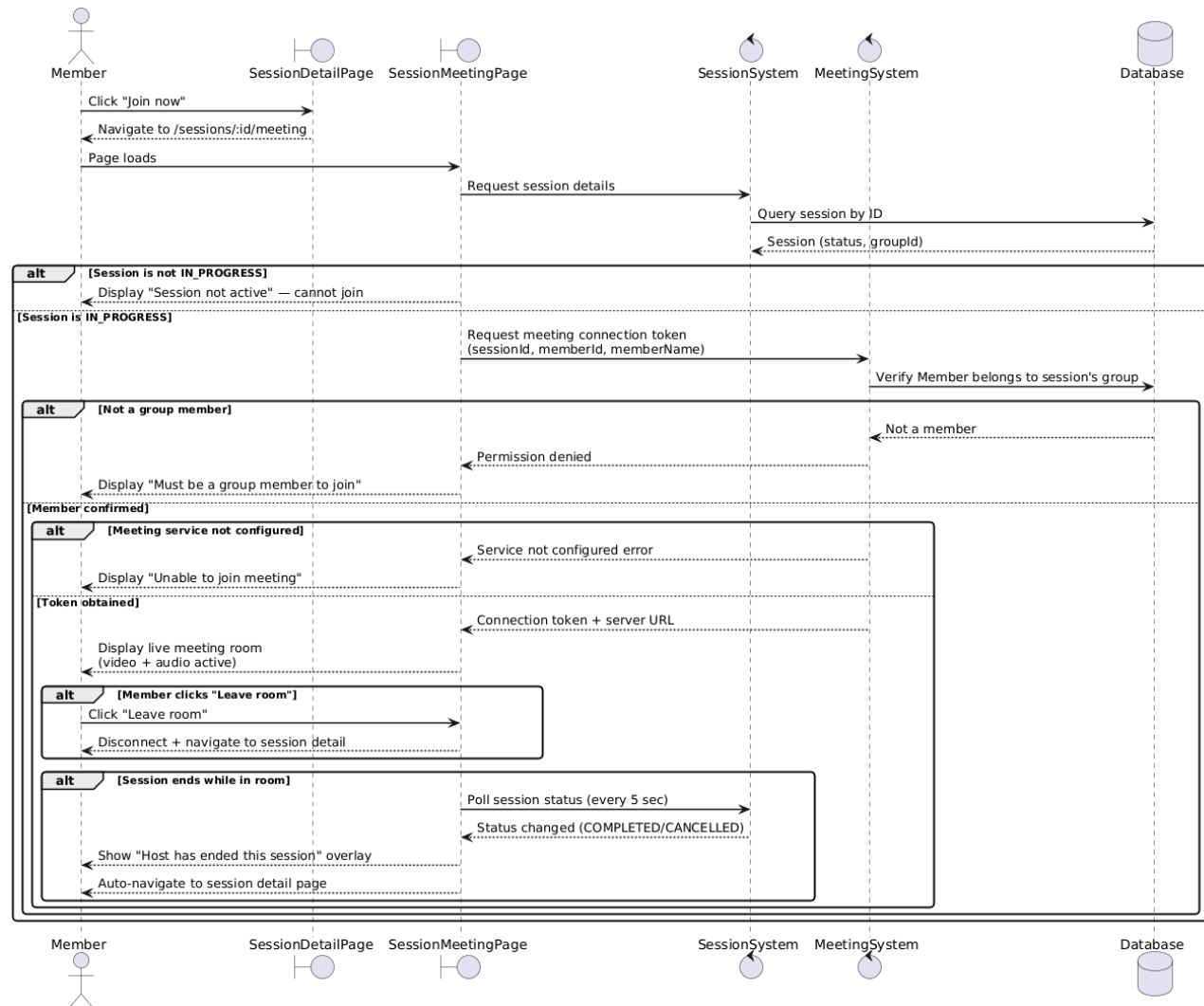
UC-M-10: Join Live Meeting

Item	Description
Use Case ID	UC-M-10
Use Case Name	Join Live Meeting
Description	Allows a Member to join the live video meeting room for an In Progress study session, enabling real-time audio and video participation with other group members.

Actor(s)	Member
Priority	High
Trigger	Member clicks "Join now" on a session detail page or session card when the session is In Progress.
Pre-Conditions	1. The Member is authenticated. 2. The target session exists and has a status of In Progress. 3. The Member is an active member of the group associated with the session. 4. The live meeting service is properly configured on the platform.
Post-Conditions	1. The Member is connected to the live meeting room. 2. The Member can participate in audio and video communication with other attendees. 3. The Member's attendance is automatically recorded as Present.
Basic Flow	1. Member navigates to a session detail page showing an In Progress session. 2. System displays a "Join now" button. 3. Member clicks "Join now". 4. System validates that the session is still In Progress and the Member belongs to the group. 5. System obtains a connection token for the live meeting room.

	6. System redirects the Member to the meeting room page with live audio and video. 7. The meeting room displays the session title and group name in the header. 8. Member can participate in the live session. 9. Member may click "Leave room" to exit the meeting without ending the session.
Alternative Flow	<i>Leave Room:</i> 1. Member clicks "Leave room". 2. System disconnects the Member from the meeting room. 3. System returns the Member to the session detail page. <i>Session Ends While in Room:</i> 1. The session Leader ends the session while the Member is in the room. 2. System displays a "The host has ended this session" overlay. 3. System automatically navigates the Member back to the session detail page.
Exception Flow	<i>Session No Longer In Progress:</i> 1. Member attempts to join but the session has since ended or been cancelled. 2. System displays a "Session not active" message. 3. Member is not connected to the meeting room.

	<i>Not a Group Member:</i> 1. Member attempts to join a meeting for a group they do not belong to. 2. System denies access and displays a permission error. <i>Meeting Service Not Configured:</i> 1. The live meeting service is not configured on the server. 2. System displays "Unable to join meeting — LiveKit may not be configured." 3. Member cannot connect. <i>System Error:</i> 1. System encounters an unexpected failure obtaining the connection token. 2. System displays an error message.
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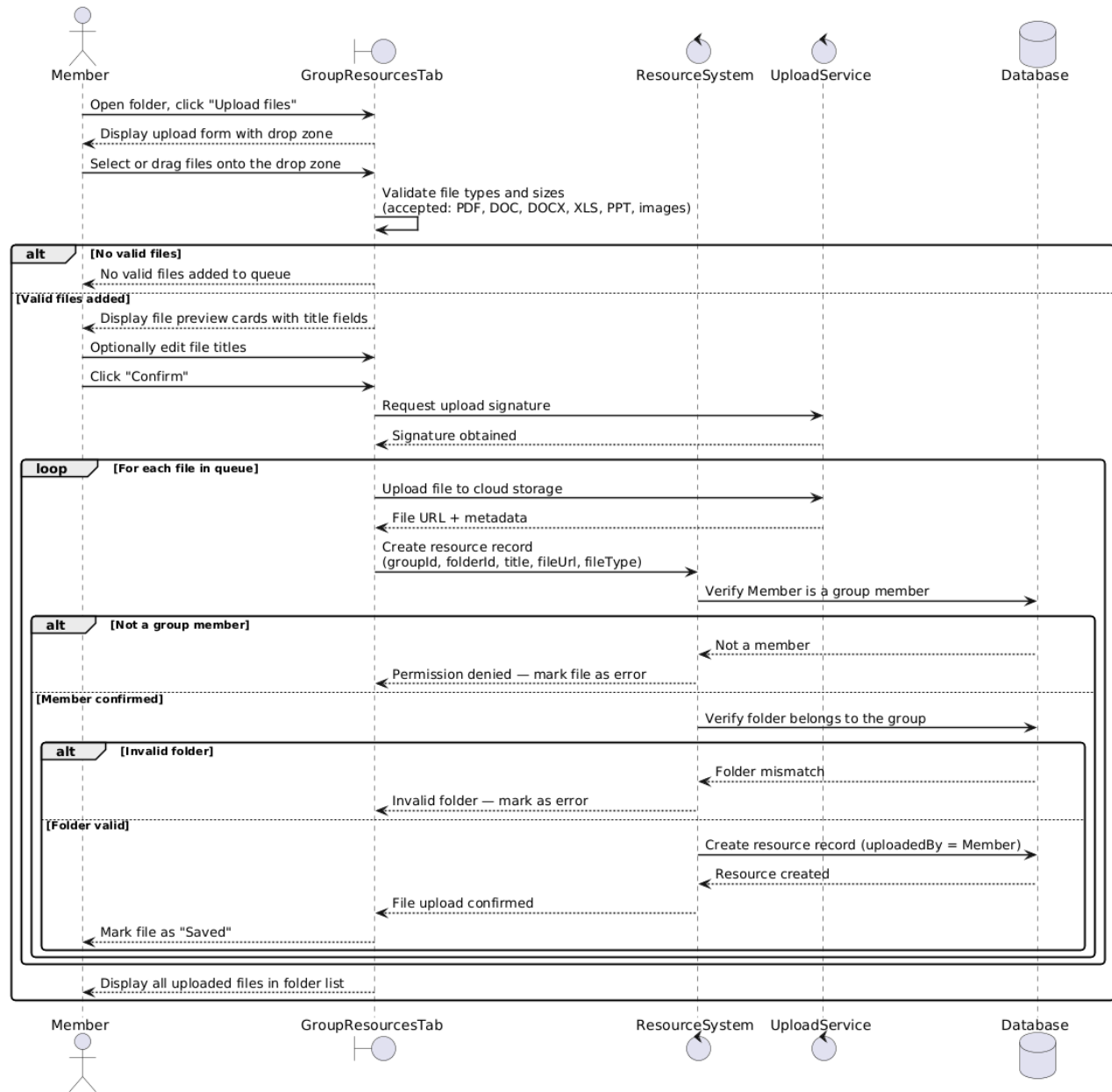
UC-M-11: Upload Resource

Item	Description
Use Case ID	UC-M-11
Use Case Name	Upload Resource

Description	Allows a Member to upload one or more study files (documents, images, PDFs, spreadsheets, presentations) to a specific resource folder within their group.
Actor(s)	Member
Priority	High
Trigger	Member clicks "Upload files" within a resource folder and confirms the file upload.
Pre-Conditions	1. The Member is authenticated. 2. The Member is an active member of the target group. 3. A resource folder exists within the group.
Post-Conditions	1. The uploaded files are stored in the selected resource folder. 2. Each resource record is created with the file title, URL, and type. 3. The resource appears in the folder's file list and the Member is recorded as the uploader.
Basic Flow	1. Member navigates to the group's Resources tab and opens a resource folder. 2. Member clicks "Upload files". 3. System displays the file upload form with a drag-and-drop area. 4. Member selects one or more files by clicking or dragging them onto the upload area.

	5. System validates each file (accepted types: PDF, DOC, DOCX, XLS, PPT, images; maximum file size applies). 6. System displays a preview card for each valid file with an editable title field. 7. Member may adjust the title for each file. 8. Member clicks "Confirm". 9. System uploads each file to the cloud storage service. 10. System creates a resource record for each successfully uploaded file, linking it to the folder and group. 11. System displays the uploaded files in the folder's file list.
Alternative Flow	<i>Drag and Drop:</i> 1. Member drags files directly onto the upload drop zone. 2. System validates and adds them to the upload queue. 3. The flow continues from step 7. <i>Cancel Before Upload:</i> 1. Member clicks "Cancel" before confirming. 2. System discards the selected files and hides the form. <i>Partial Success:</i> 1. Some files fail to upload while others succeed. 2. System marks failed items with an error indicator. 3. Successfully uploaded files appear in the folder. 4. Member may retry failed items.

Exception Flow	<i>Invalid File Type:</i> 1. Member selects a file with an unsupported type. 2. System silently excludes the invalid file from the upload queue. <i>File Too Large:</i> 1. Member selects a file exceeding the maximum allowed size. 2. System excludes the oversized file from the upload queue. <i>Not a Group Member:</i> 1. Member is no longer an active member of the group. 2. System denies the upload and displays a permission error. <i>Invalid Folder:</i> 1. The selected folder does not belong to the target group. 2. System displays an error: "Invalid folder for this group". <i>Upload Service Unavailable:</i> 1. The file hosting service is not configured. 2. System displays a configuration error. 3. Member cannot upload files.
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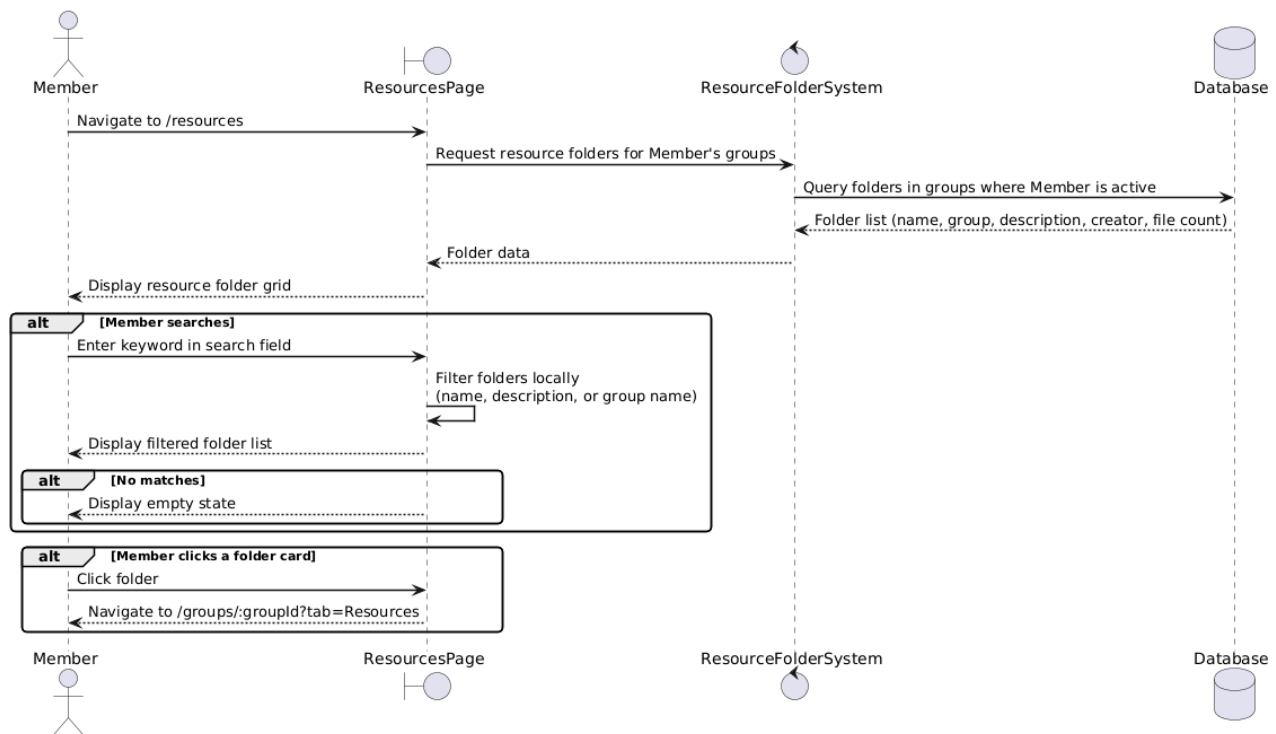


UC-M-12: Browse Resource Library

Item	Description
Use Case ID	UC-M-12

Use Case Name	Browse Resource Library
Description	Allows a Member to view and search through the resource folders and files shared across all groups they have joined. Members can search by folder name, description, or group name.
Actor(s)	Member
Priority	Medium
Trigger	Member navigates to the Resources page.
Pre-Conditions	1. The Member is authenticated. 2. The Member belongs to at least one study group.
Post-Conditions	1. Resource folders from all of the Member's groups are displayed. 2. No data is modified.
Basic Flow	1. Member navigates to the Resources page. 2. System retrieves all resource folders from groups the Member belongs to. 3. System displays each folder with its name, associated group, description, creator, file count, and creation date. 4. Member may enter a keyword in the search field to filter folders by name, description, or group name. 5. System updates the displayed list to show only matching folders.

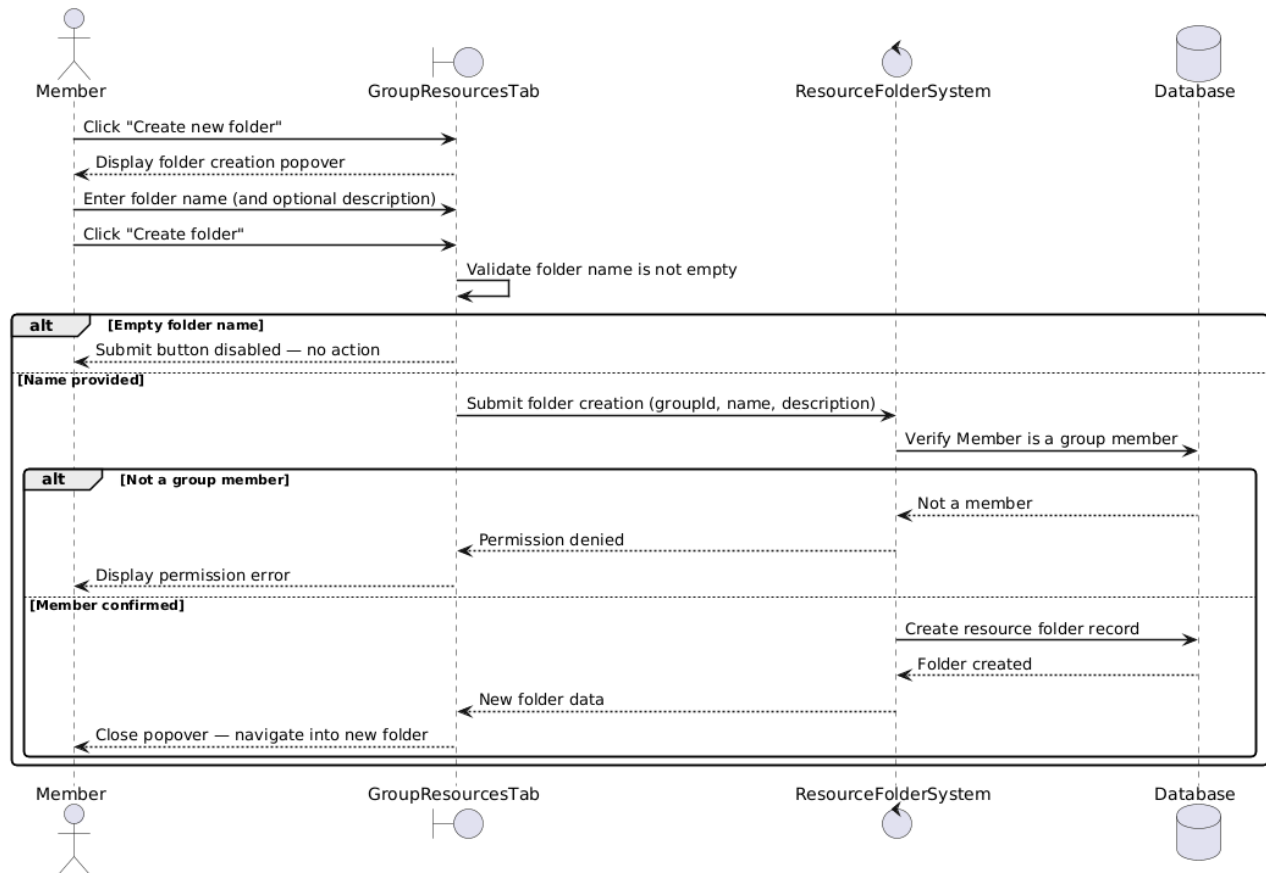
	6. Member clicks a folder card to navigate to the folder within the group's Resources tab.
Alternative Flow	<i>No Folders:</i> 1. Member's groups have no resource folders. 2. System displays "No resource folders yet." <i>Search Returns No Results:</i> 1. Member enters a keyword that matches no folders. 2. System shows an empty state.
Exception Flow	<i>System Error:</i> 1. System fails to retrieve resource folders. 2. System displays an error or loading state. 3. Member may refresh.



UC-M-13: Manage Resource Folders

Item	Description
Use Case ID	UC-M-13
Use Case Name	Manage Resource Folders
Description	Allows a Member (with Leader privileges) to create new resource folders within their group and edit existing folder names and descriptions to organize shared study materials.
Actor(s)	Member
Priority	Medium
Trigger	Member clicks "Create new folder" or edits an existing folder within the group's Resources tab.
Pre-Conditions	1. The Member is authenticated. 2. The Member is an active member of the target group.
Post-Conditions	A new resource folder is created and immediately available in the group's Resources tab.
Basic Flow	<i>Create Folder:</i> 1. Member navigates to the group's Resources tab. 2. Member clicks "Create new folder". 3. System displays a popover form with a folder name field and optional description. 4. Member enters a folder name and optional description.

	5. Member clicks "Create folder". 6. System validates the folder name is not empty. 7. System creates the new folder associated with the group. 8. System updates the folder list and navigates the Member into the new folder.
Alternative Flow	<i>Cancel Folder Creation:</i> 1. Member closes the popover without submitting. 2. No folder is created.
Exception Flow	<i>Validation Failure — Folder Name:</i> 1. Member submits an empty folder name. 2. System disables the submit button. 3. Folder is not created. <i>Not a Group Member:</i> 1. Member is not an active member of the group. 2. System displays a permission error. <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message.

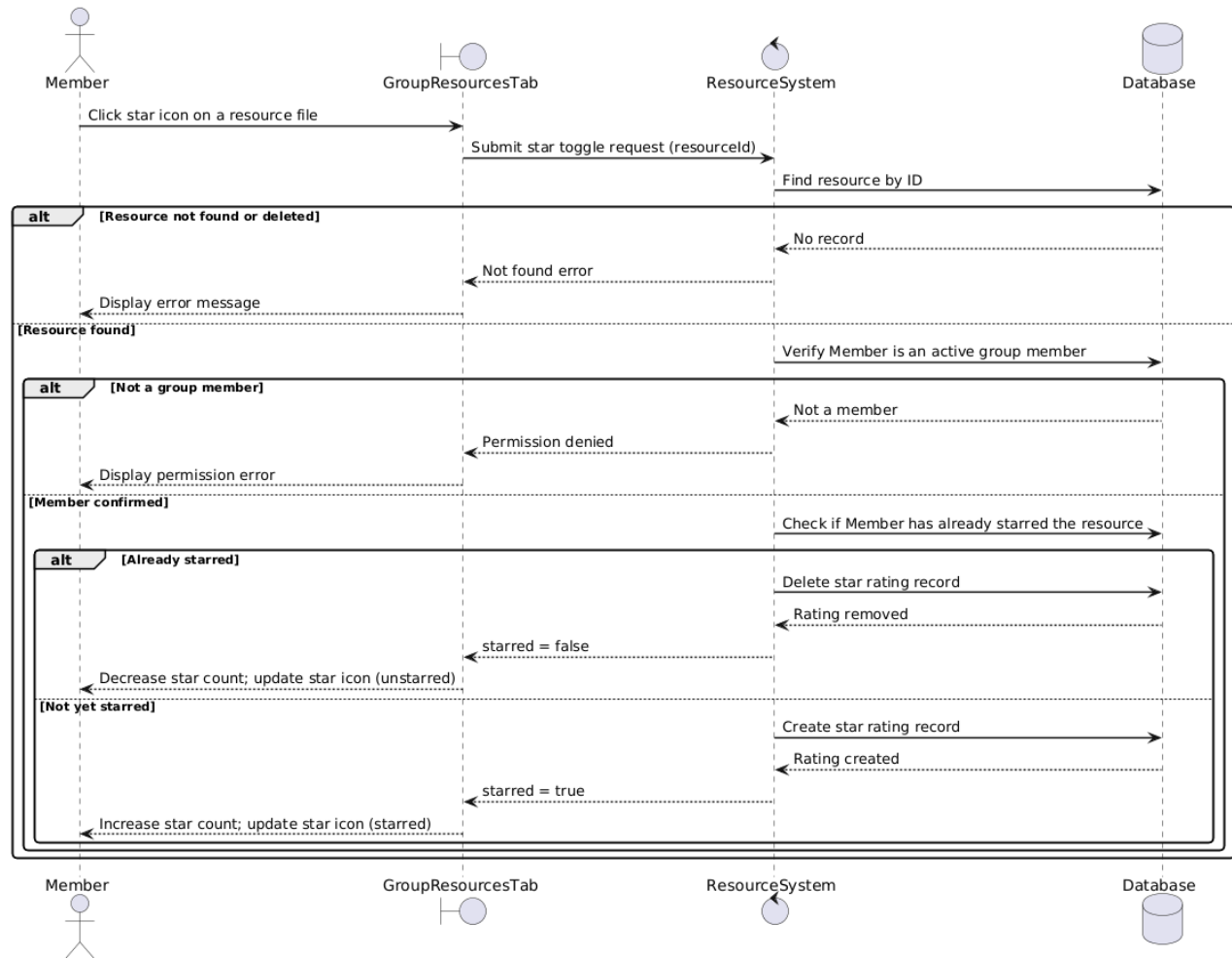


UC-M-14: Star Resource

Item	Description
Use Case ID	UC-M-14
Use Case Name	Star Resource
Description	Allows a Member to toggle a star rating on a shared resource file. Starring a resource marks it as a favourite; starring it again removes the rating. The star count is visible to all group members.

Actor(s)	Member
Priority	Low
Trigger	Member clicks the star icon on a resource file row within the group's Resources tab.
Pre-Conditions	1. The Member is authenticated. 2. The Member is an active member of the group that owns the resource. 3. The target resource exists.
Post-Conditions	1. If the resource was not starred: a star rating is created and the star count increments by one. 2. If the resource was already starred: the star rating is removed and the star count decrements by one.
Basic Flow	1. Member navigates to a resource folder within the group's Resources tab. 2. System displays the file list with a star icon and star count for each resource. 3. Member clicks the star icon on a resource. 4. System verifies the Member is an active group member. 5. System checks whether the Member has already starred the resource. 6. If not starred: system creates a star rating record; displayed count increases. 7. If already starred: system removes the star rating record;

	displayed count decreases. 8. System updates the star count displayed on the resource row.
Alternative Flow	None.
Exception Flow	<i>Resource Not Found:</i> 1. Member attempts to star a resource that has been deleted. 2. System displays a "Resource not found" error. <i>Not a Group Member:</i> 1. Member is no longer an active member of the resource's group. 2. System displays a permission error. <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message.



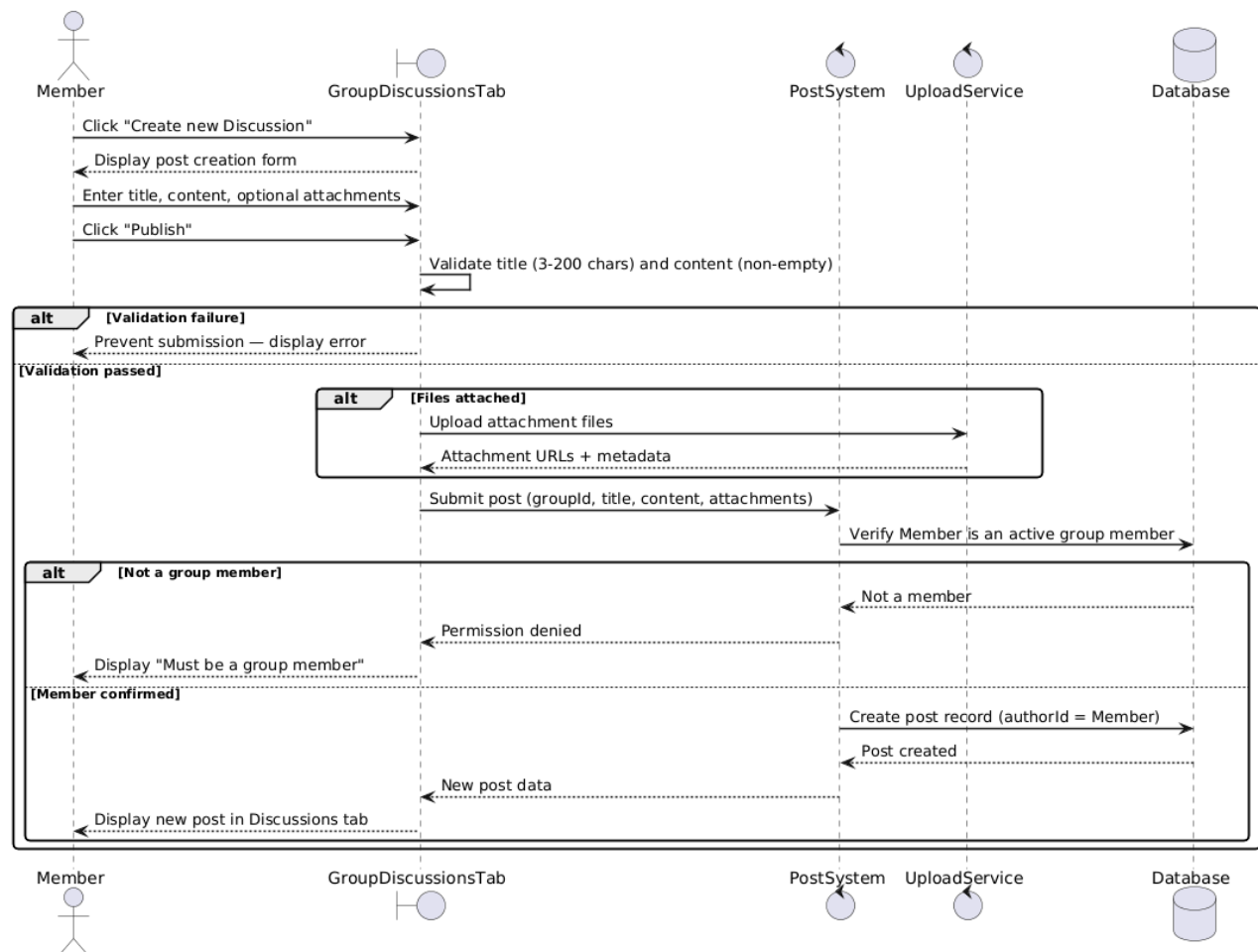
UC-M-15: Create Post

Item	Description
Use Case ID	UC-M-15
Use Case Name	Create Post
Description	Allows a Member to publish a new discussion post in a study group they belong to, optionally attaching files. The

	post appears in the group's Discussions tab and in the Member's Discussions feed.
Actor(s)	Member
Priority	High
Trigger	Member clicks "Create new Discussion" in the Discussions tab and submits the post form.
Pre-Conditions	1. The Member is authenticated. 2. The Member is an active member of the target group.
Post-Conditions	1. The discussion post is created and appears in the group's Discussions tab. 2. The post displays the author's name, timestamp, title, content, and any attachments.
Basic Flow	1. Member navigates to the group's Discussions tab. 2. Member clicks "Create new Discussion". 3. System displays the post creation form (Title, Content, optional file attachments). 4. Member enters a title and content for the post. 5. Optionally, Member attaches one or more files. 6. Member clicks "Publish". 7. If files are attached, system uploads them first and shows upload progress. 8. System validates the title (minimum 3 characters, maximum 200 characters) and content (non-empty).

	9. System verifies the Member belongs to the group. 10. System creates the discussion post with any uploaded attachments. 11. System displays the new post in the Discussions tab.
Alternative Flow	<i>Cancel:</i> 1. Member clicks "Cancel". 2. System hides the form. 3. No post is created.
Exception Flow	<i>Validation Failure — Title:</i> 1. Member submits a title shorter than 3 characters or longer than 200 characters. 2. System prevents submission. 3. Post is not created. <i>Validation Failure — Content:</i> 1. Member submits an empty content field. 2. System prevents submission. 3. Post is not created. <i>Not a Group Member:</i> 1. Member is not an active member of the group. 2. System displays a permission error: "Must be a group member". <i>File Upload Failure:</i>

	1. One or more attachments fail to upload. 2. System displays an error. 3. Member may retry.
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UC-M-16: Edit Post

Item	Description
Use Case ID	UC-M-16

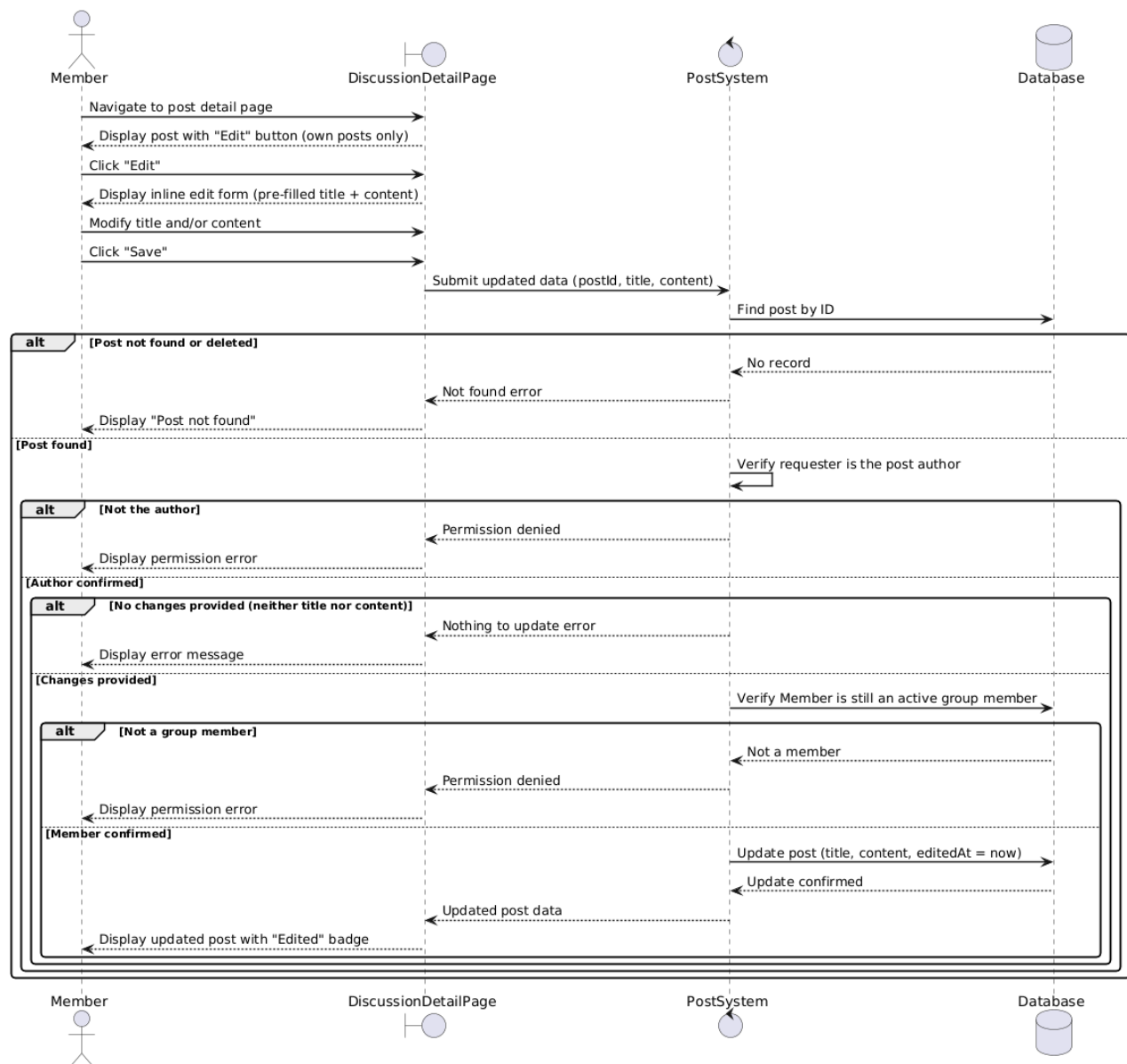
Use Case Name	Edit Post
Description	Allows a Member to edit the title or content of a discussion post they have authored. Edited posts are marked with an "Edited" badge to indicate they have been modified.
Actor(s)	Member
Priority	Medium
Trigger	Member clicks the "Edit" button on their own discussion post on the post detail page.
Pre-Conditions	1. The Member is authenticated. 2. The Member is the author of the post to be edited. 3. The Member is still an active member of the group. 4. The target post exists and has not been deleted.
Post-Conditions	1. The post's title and/or content is updated. 2. The post is marked with an "Edited" label indicating it has been modified.
Basic Flow	1. Member navigates to the discussion post detail page. 2. System displays the post with an "Edit" button visible only to the post author. 3. Member clicks "Edit". 4. System replaces the post view with an inline edit form pre-filled with the current title and content. 5. Member modifies the title and/or content.

	6. Member clicks "Save". 7. System validates the updated data. 8. System verifies the Member is the post author and an active group member. 9. System saves the updated post and marks it as edited. 10. System displays the updated post content with an "Edited" badge.
Alternative Flow	<i>Cancel Edit:</i> 1. Member clicks "Cancel". 2. System discards the changes. 3. The original post content is restored.
Exception Flow	<i>Not the Author:</i> 1. A Member who is not the post author attempts to edit. 2. System denies the action and displays a permission error. <i>Post Not Found:</i> 1. The post has been deleted before the edit is submitted. 2. System displays a "Post not found" error. <i>Nothing to Update:</i> 1. Member submits the form without changing the title or content. 2. System displays an error: "Nothing to update". <i>Not a Group Member:</i>

1. Member is no longer an active group member.
2. System displays a permission error.

Validation Failure:

1. Member submits a title shorter than 3 characters.
2. System displays a validation error.

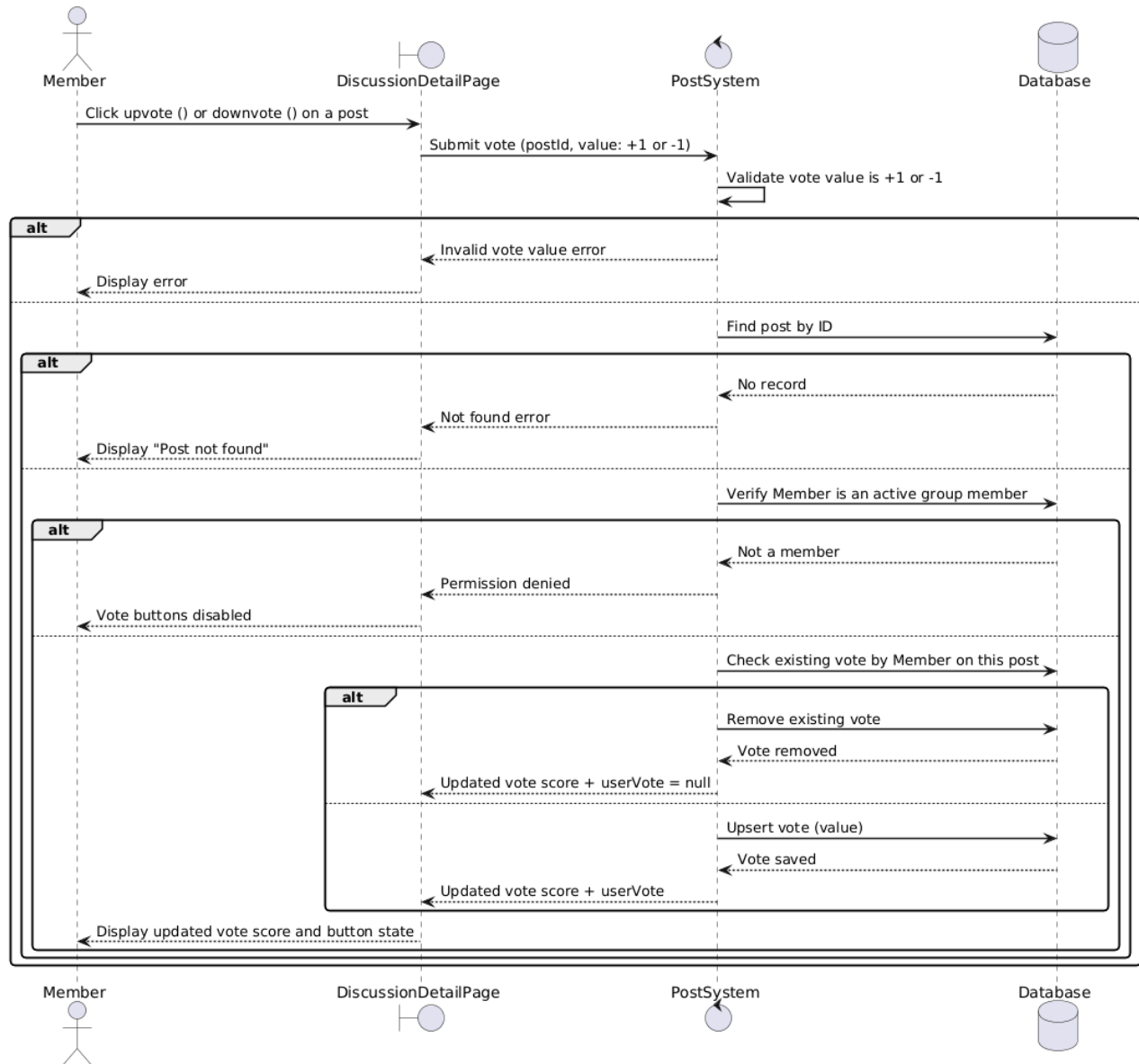


UC-M-17: Vote Content

Item	Description
Use Case ID	UC-M-17
Use Case Name	Vote Content
Description	Allows a Member to upvote or downvote a discussion post or comment. Voting the same value a second time removes the vote (toggle). The vote score influences the ranking of posts and comments in the Discussions tab.
Actor(s)	Member
Priority	Medium
Trigger	Member clicks the upvote (▲) or downvote (▼) button on a discussion post or comment.
Pre-Conditions	1. The Member is authenticated. 2. The Member is an active member of the group where the post or comment resides. 3. The target post or comment exists.
Post-Conditions	1. If voting a new value: the Member's vote is recorded and the vote score is updated. 2. If voting the same value as an existing vote: the vote is removed (toggled off) and the score is updated.

Basic Flow	1. Member navigates to the discussion post detail page. 2. System displays vote buttons (upvote and downvote) with the current vote score. 3. Member clicks an upvote or downvote button on a post or comment. 4. System verifies the vote value is valid (+1 or -1). 5. System verifies the Member is an active group member. 6. System checks if the Member has an existing vote on the item. 7. If no existing vote or a different vote value: system records or updates the vote. 8. If the same vote value: system removes the existing vote. 9. System updates the displayed vote score.
Alternative Flow	<i>Vote on Comment:</i> 1. Member clicks a vote button on a comment within a post. 2. System processes the vote for the comment rather than the post. 3. The comment's vote score is updated accordingly. <i>Switch Vote:</i> 1. Member previously voted and clicks the opposite vote button. 2. System replaces the existing vote with the new value. 3. Vote score is updated.

Exception Flow	<i>Post/Comment Not Found:</i> 1. The target post or comment does not exist. 2. System displays a "Post not found" or "Comment not found" error. <i>Not a Group Member:</i> 1. Member is not an active member of the group. 2. System disables voting and displays a permission-related message. <i>Invalid Vote Value:</i> 1. An invalid vote value is submitted. 2. System displays an error: "Invalid vote value".
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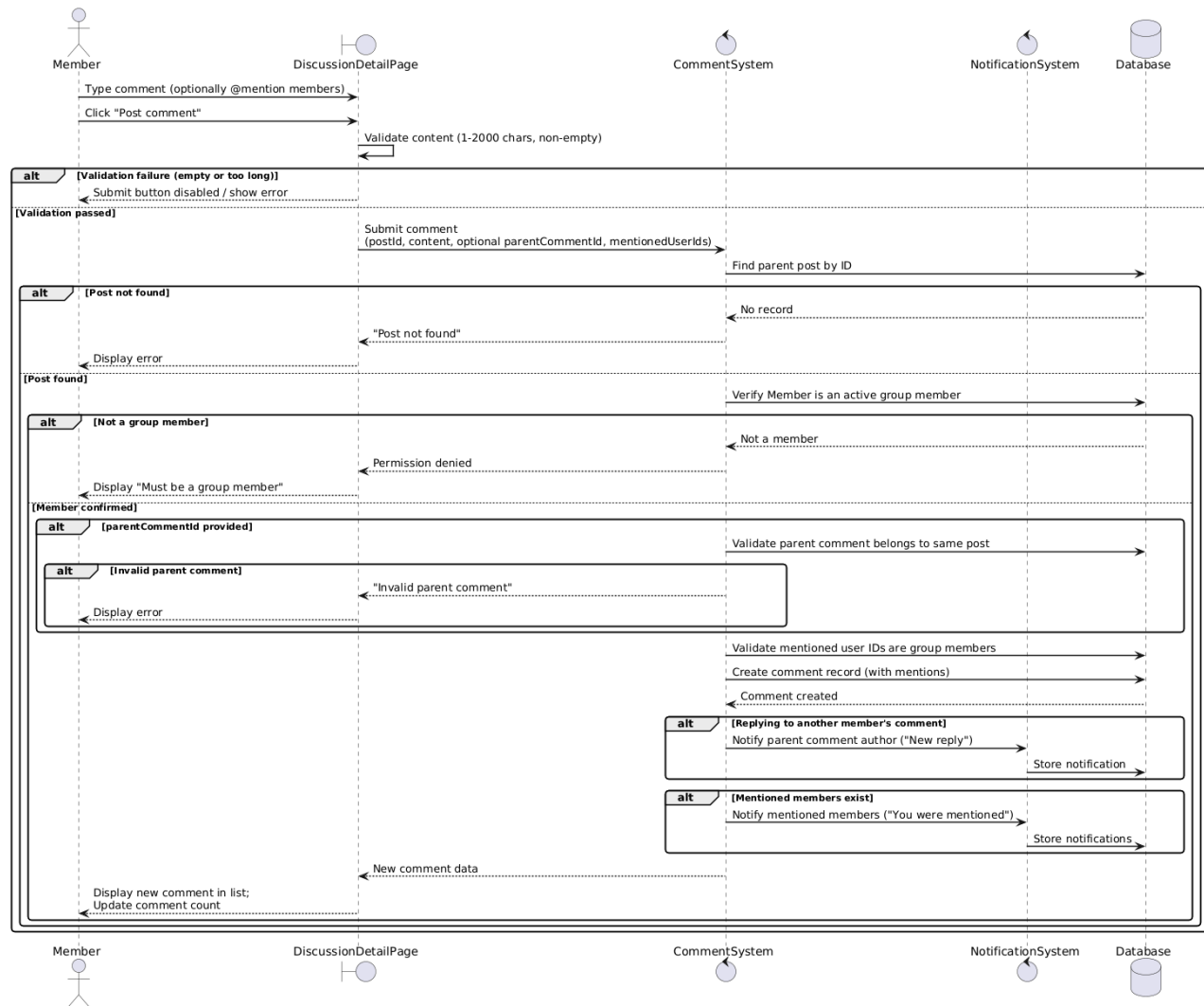
UC-M-18: Comment

Item	Description
Use Case ID	UC-M-18
Use Case Name	Comment

Description	Allows a Member to post a comment on a discussion post, reply to another comment, and mention other group members using the @ symbol. Mentioned users and the parent comment's author receive in-app notifications.
Actor(s)	Member
Priority	High
Trigger	Member types a comment in the comment input and clicks "Post comment".
Pre-Conditions	1. The Member is authenticated. 2. The Member is an active member of the group where the discussion post resides. 3. The target post exists.
Post-Conditions	1. The comment is created and appears in the comment list for the post. 2. If replying: the comment is nested under the parent comment. 3. Mentioned group members and the parent comment's author (if replying) receive in-app notifications. 4. The post's comment count is updated.
Basic Flow	1. Member navigates to the discussion post detail page. 2. System displays the comment input area with a text field. 3. Member types a comment. Optionally, Member uses the @ symbol to mention other group members.

	4. Member clicks "Post comment". 5. System validates the comment content is non-empty and within 2000 characters. 6. System verifies the Member is an active group member. 7. System creates the comment, linking any valid mentions to group members. 8. System sends notifications to mentioned members and the parent comment's author (if replying). 9. System adds the comment to the list and updates the comment count.
Alternative Flow	<i>Reply to a Comment:</i> 1. Member clicks "Reply" on an existing comment. 2. System populates the reply form referencing the parent comment. 3. Member types a reply and clicks "Post comment". 4. The reply is nested under the parent comment. 5. The parent comment's author receives a "New reply" notification. <i>Mention a Member (@):</i> 1. Member types @ followed by a name. 2. System suggests matching group members. 3. Member selects a suggested member. 4. The mentioned member receives a "You were mentioned" notification after the comment is posted.

	<i>Sort Comments:</i> 1. Member clicks "Newest" or "Top" sort toggle. 2. System re-orders the comment list by creation time or vote score.
Exception Flow	<i>Empty Comment:</i> 1. Member submits an empty comment. 2. System disables the submit button. 3. Comment is not posted. <i>Comment Too Long:</i> 1. Member submits a comment exceeding 2000 characters. 2. System displays a validation error. 3. Comment is not posted. <i>Post Not Found:</i> 1. The parent post does not exist. 2. System displays a "Post not found" error. <i>Invalid Parent Comment:</i> 1. Member attempts to reply to a comment from a different post. 2. System displays an error: "Invalid parent comment". <i>Not a Group Member:</i> 1. Member is not an active group member. 2. System displays a permission error.

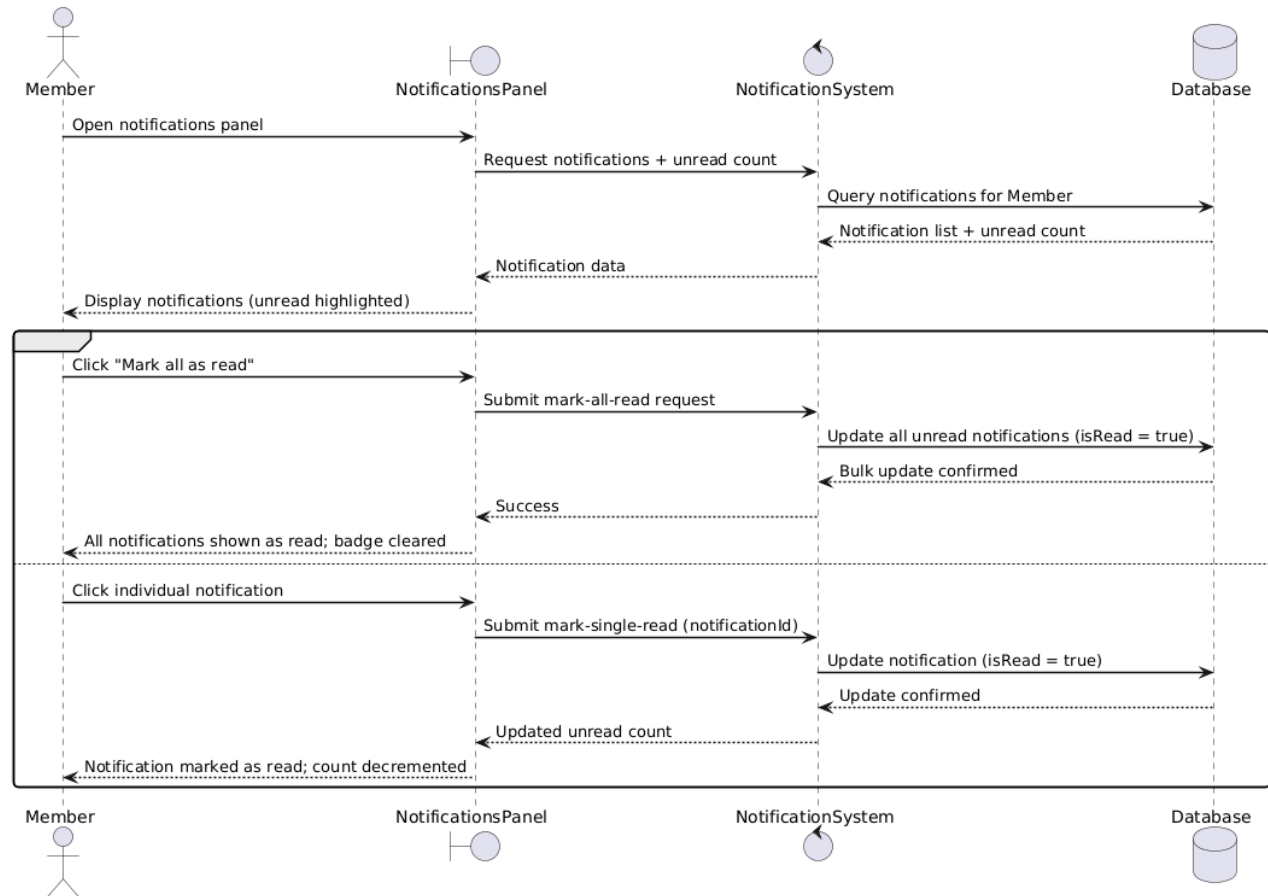


UC-M-19: Mark Notifications Read

Item	Description
Use Case ID	UC-M-19
Use Case Name	Mark Notifications Read

Description	Allows a Member to mark one or all of their in-app notifications as read, clearing the unread notification count indicator in the navigation.
Actor(s)	Member
Priority	Low
Trigger	Member clicks "Mark all as read" or interacts with an individual notification to mark it as read.
Pre-Conditions	1. The Member is authenticated. 2. The Member has at least one unread notification.
Post-Conditions	1. The selected notification(s) are marked as read. 2. The unread notification count badge in the navigation is updated to reflect the change.
Basic Flow	1. Member opens the notifications panel. 2. System displays a list of notifications with unread items highlighted. 3. Member clicks "Mark all as read". 4. System marks all of the Member's unread notifications as read. 5. System updates the notification list to show all items as read. 6. The unread count badge in the navigation clears.

Alternative Flow	<i>Mark Single Notification Read:</i> 1. Member clicks on an individual notification. 2. System marks only that specific notification as read. 3. The unread count decrements by one.
Exception Flow	<i>No Unread Notifications:</i> 1. Member attempts to mark all as read when there are no unread notifications. 2. System processes the request silently; no change is made. <i>System Error:</i> 1. System fails to update the notification status. 2. System displays an error message.

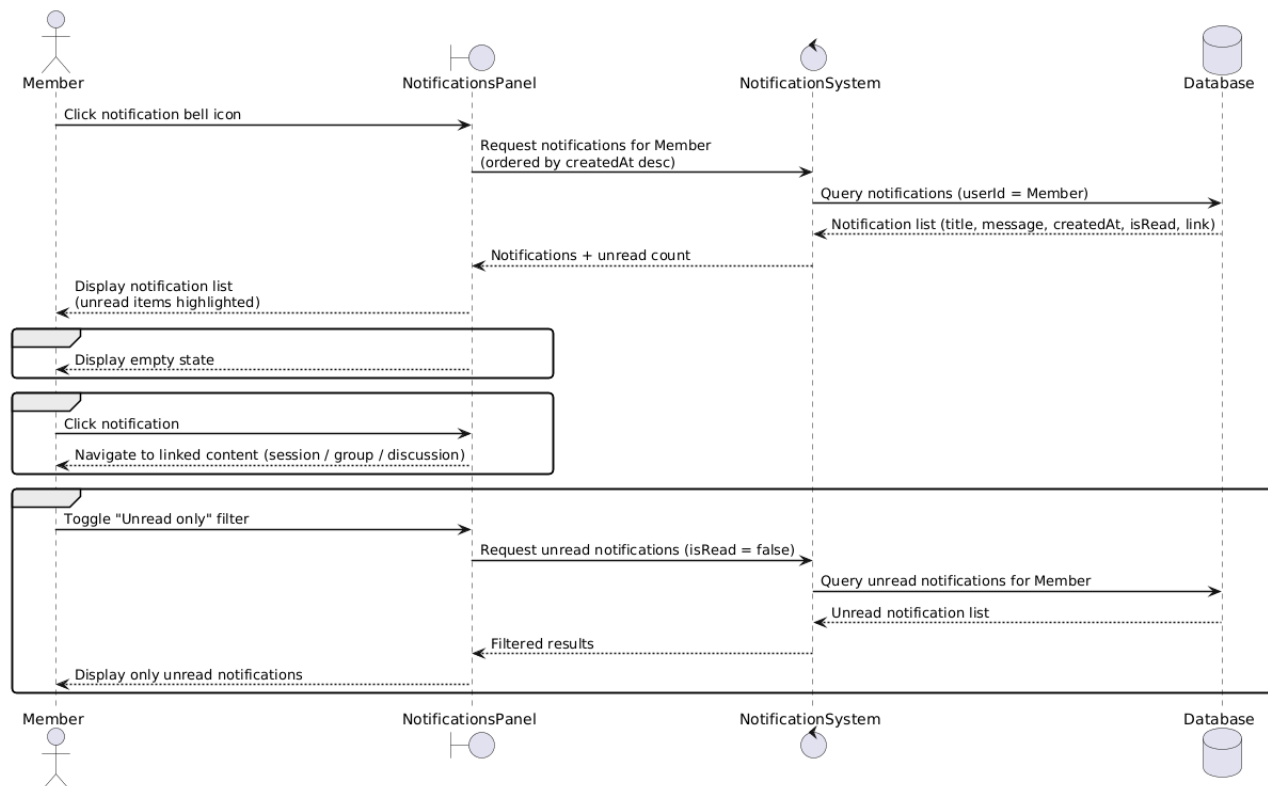


UC-M-20: View Notifications

Item	Description
Use Case ID	UC-M-20
Use Case Name	View Notifications
Description	Allows a Member to view a chronological list of their in-app notifications, including session reminders, join request updates, resource deletion notices, group status changes, mentions, and comment replies.

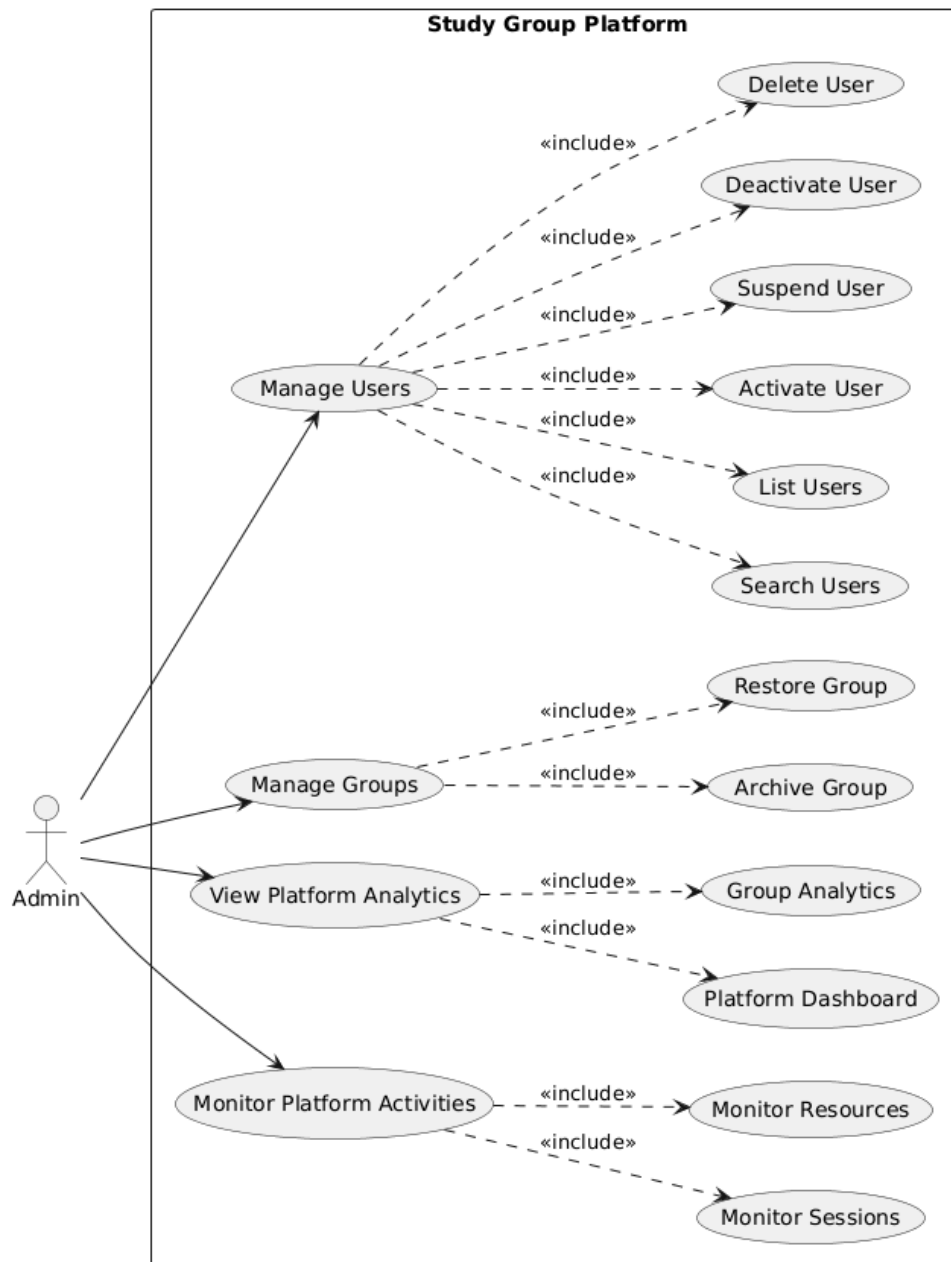
Actor(s)	Member
Priority	Medium
Trigger	Member opens the notifications panel or navigates to the notifications section.
Pre-Conditions	1. The Member is authenticated.
Post-Conditions	1. The Member's notifications are displayed in reverse-chronological order. 2. No data is modified.
Basic Flow	1. Member clicks the notification bell icon in the navigation. 2. System retrieves the Member's notifications, ordered by most recent first. 3. System displays a list of notifications, each showing the title, message content, and timestamp. 4. Unread notifications are visually distinguished from read ones. 5. Member may click a notification link to navigate to the relevant content (e.g., a session, group, or discussion). 6. Member may choose to filter to show only unread notifications.
Alternative Flow	<i>No Notifications:</i> 1. Member has no notifications. 2. System displays an empty state.

	<i>Filter Unread Only:</i> 1. Member toggles the "unread only" filter. 2. System retrieves only unread notifications for the Member. 3. System updates the displayed list.
Exception Flow	<i>System Error:</i> 1. System fails to retrieve notifications. 2. System displays an error or loading state. 3. Member may retry.



3.4 Use Case for Admin

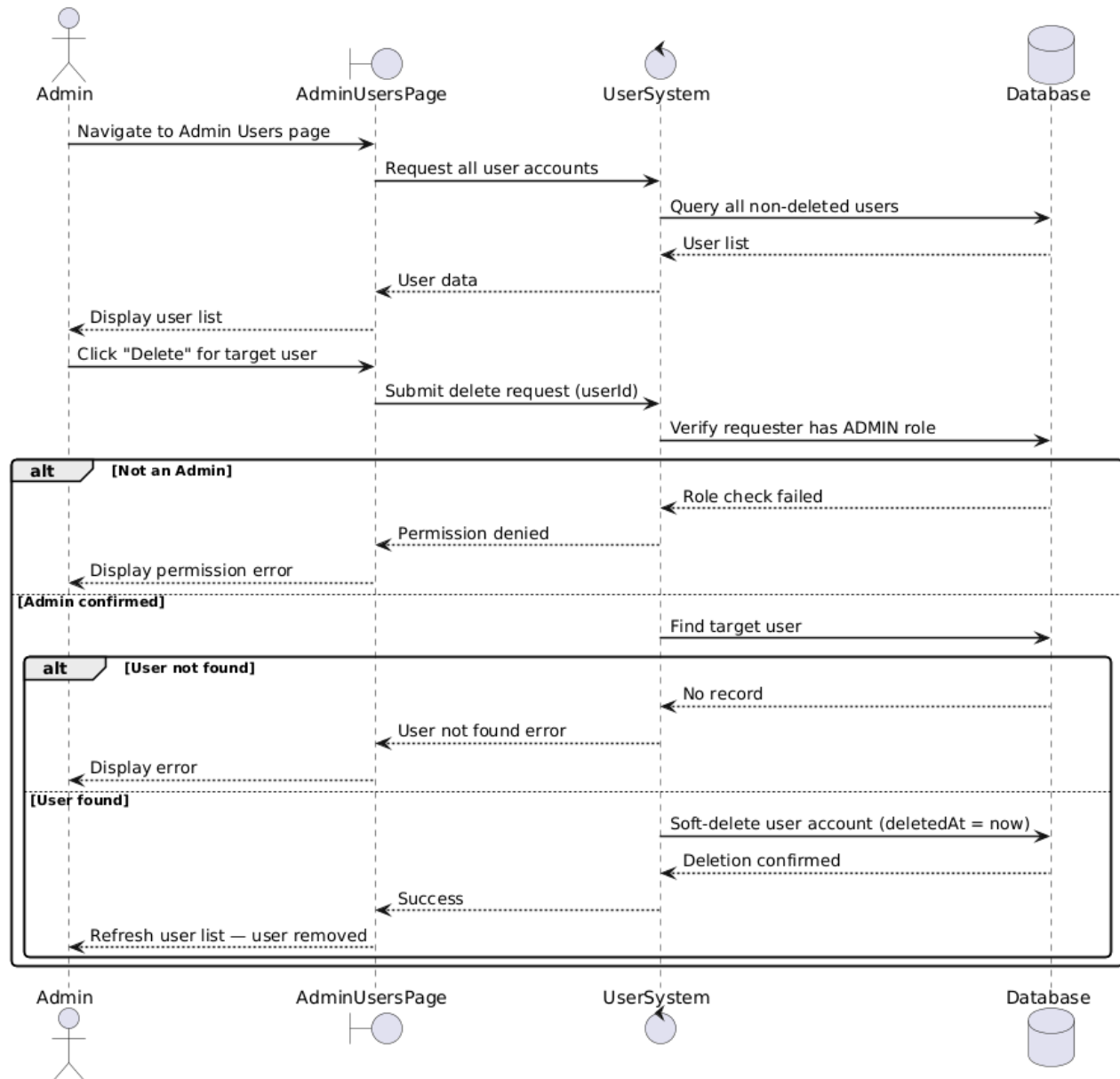
Administrators are platform managers who oversee users, groups, reports, and platform activities to ensure the system operates effectively and securely.



UC-A-01: Delete User

Item	Description
Use Case ID	UC-A-01
Use Case Name	Delete User
Description	Allows an Administrator to permanently remove a user account from the platform. The account is soft-deleted and can no longer be used to log in or access any platform feature.
Actor(s)	Admin
Priority	Medium
Trigger	Admin clicks the delete action for a specific user account on the Admin Users page.
Pre-Conditions	1. The Admin is authenticated. 2. The target user account exists on the platform.
Post-Conditions	1. The user account is marked as deleted and is no longer accessible. 2. The deleted user cannot log in or use the platform.
Basic Flow	1. Admin navigates to the Admin Users page. 2. System displays a list of all registered user accounts. 3. Admin locates the target user account. 4. Admin clicks the delete action for the target user.

	5. System removes the user account from the platform. 6. System confirms the deletion and refreshes the user list.
Alternative Flow	None.
Exception Flow	<i>User Not Found:</i> 1. The target account no longer exists at the time of the action. 2. System displays a "User not found" error. <i>Target is an Admin:</i> 1. Admin attempts to delete another Administrator account. 2. System displays a permission error. <i>Self-Deletion:</i> 1. Admin attempts to delete their own account. 2. System displays an error: "Cannot change your own account status." <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message. 3. Admin may retry.

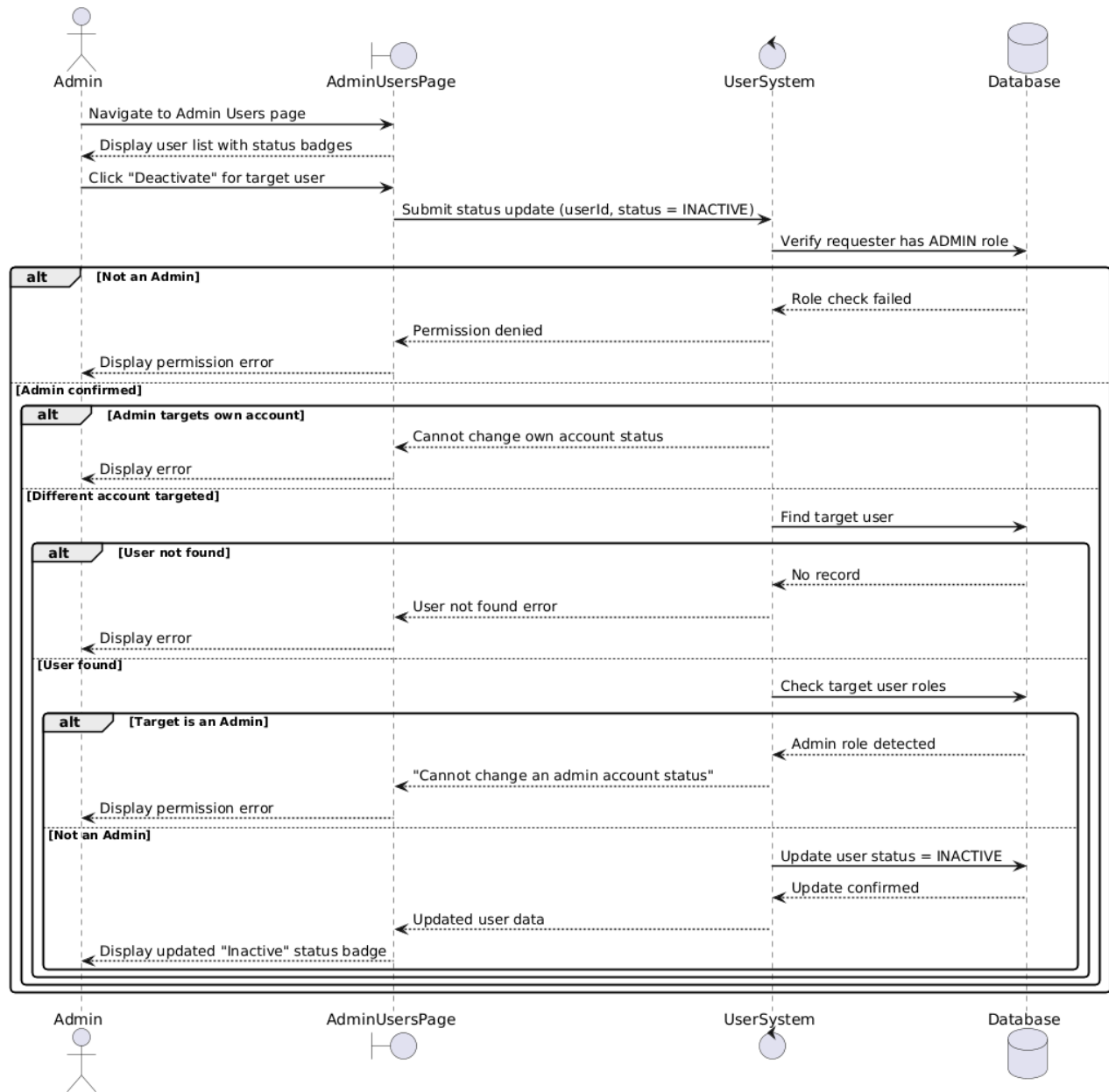


UC-A-02: Deactivate User

Item	Description
Use Case ID	UC-A-02

Use Case Name	Deactivate User
Description	Allows an Administrator to set a user account to an Inactive status, preventing the user from logging in without permanently removing their account or their data.
Actor(s)	Admin
Priority	Medium
Trigger	Admin sets a user's status to Inactive on the Admin Users page.
Pre-Conditions	1. The Admin is authenticated. 2. The target user account exists and is currently Active or Suspended. 3. The target user does not hold the Admin role.
Post-Conditions	1. The user's account status is updated to Inactive. 2. The user can no longer log in until the status is changed.
Basic Flow	1. Admin navigates to the Admin Users page. 2. System displays a list of all registered user accounts with their current status. 3. Admin locates the target user and selects the Deactivate action. 4. System verifies the Admin's authority and checks the target user is not an Administrator. 5. System updates the user's account status to Inactive.

	6. System confirms the change and updates the status badge displayed for that user.
Alternative Flow	None.
Exception Flow	<i>User Not Found:</i> 1. The target account no longer exists at the time of the action. 2. System displays a "User not found" error. <i>Target is an Admin:</i> 1. Admin attempts to deactivate another Administrator account. 2. System displays a permission error. <i>Self-Deletion:</i> 1. Admin attempts to delete their own account. 2. System displays an error: "Cannot change your own account status." <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message. 3. Admin may retry.

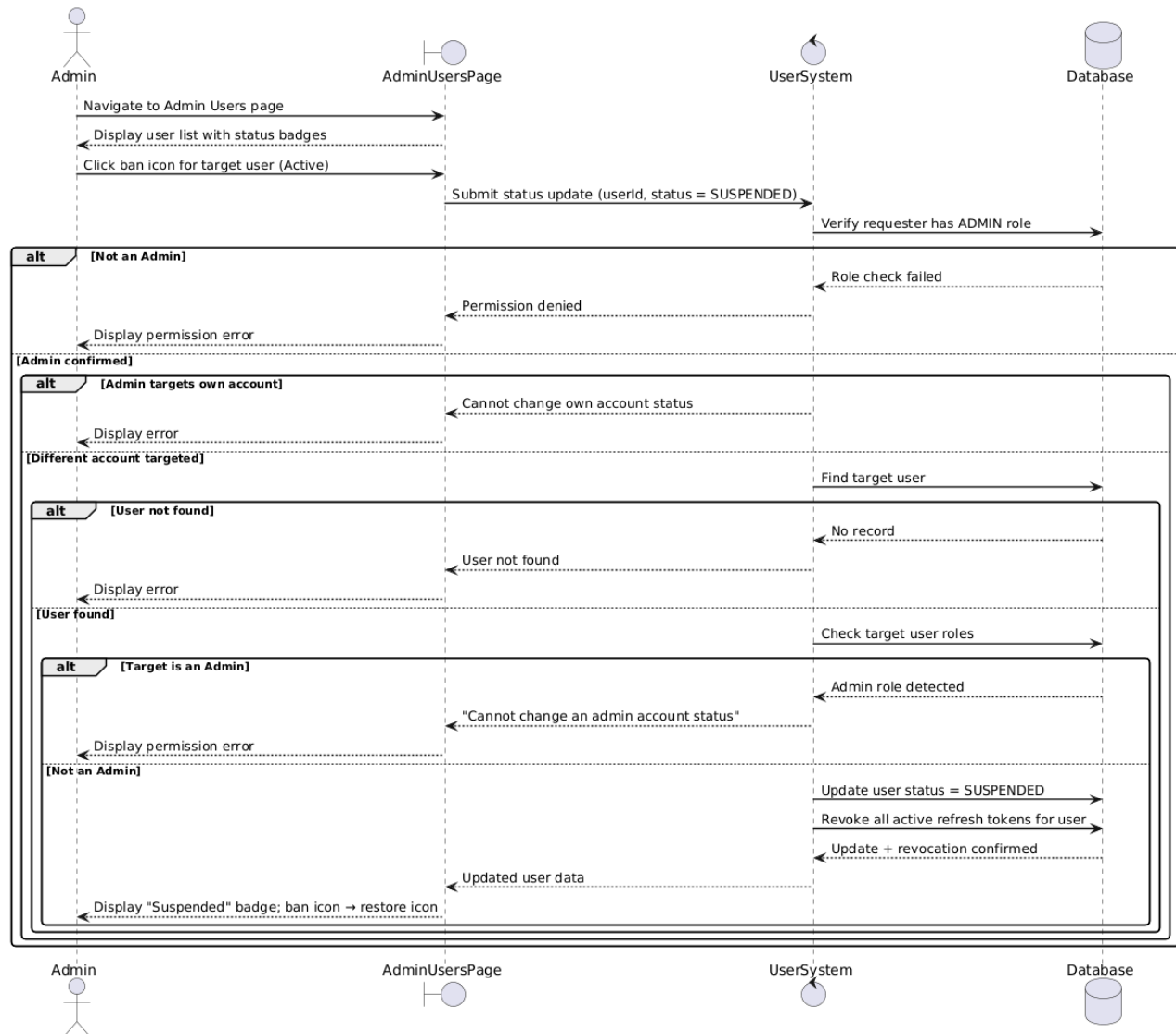


UC-A-03: Suspend User

Item	Description
Use Case ID	UC-A-03
Use Case Name	Suspend User

Description	Allows an Administrator to suspend a user account, immediately revoking the user's active session and blocking future logins until the account is restored. This is used for accounts that have violated platform policies.
Actor(s)	Admin
Priority	High
Trigger	Admin clicks the suspend (ban) action for a user on the Admin Users page.
Pre-Conditions	1. The Admin is authenticated. 2. The target user account exists and is currently Active. 3. The target user does not hold the Admin role.
Post-Conditions	1. The user's account status is updated to Suspended. 2. All of the user's active login sessions are immediately invalidated. 3. The suspended user cannot log in until the account is restored.
Basic Flow	1. Admin navigates to the Admin Users page. 2. System displays all user accounts with their current status. 3. Admin locates the target user and clicks the ban (suspend) icon. 4. System verifies the Admin's authority and confirms the target is not an Administrator.

	5. System updates the user's status to Suspended. 6. System immediately invalidates all of the user's active sessions. 7. System confirms the change and displays a "Suspended" status badge for that user.
Alternative Flow	None.
Exception Flow	<i>User Not Found:</i> 1. The target account no longer exists at the time of the action. 2. System displays a "User not found" error. <i>Target is an Admin:</i> 1. Admin attempts to delete another Administrator account. 2. System displays a permission error. <i>Self-Deletion:</i> 1. Admin attempts to delete their own account. 2. System displays an error: "Cannot change your own account status." <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message. 3. Admin may retry.

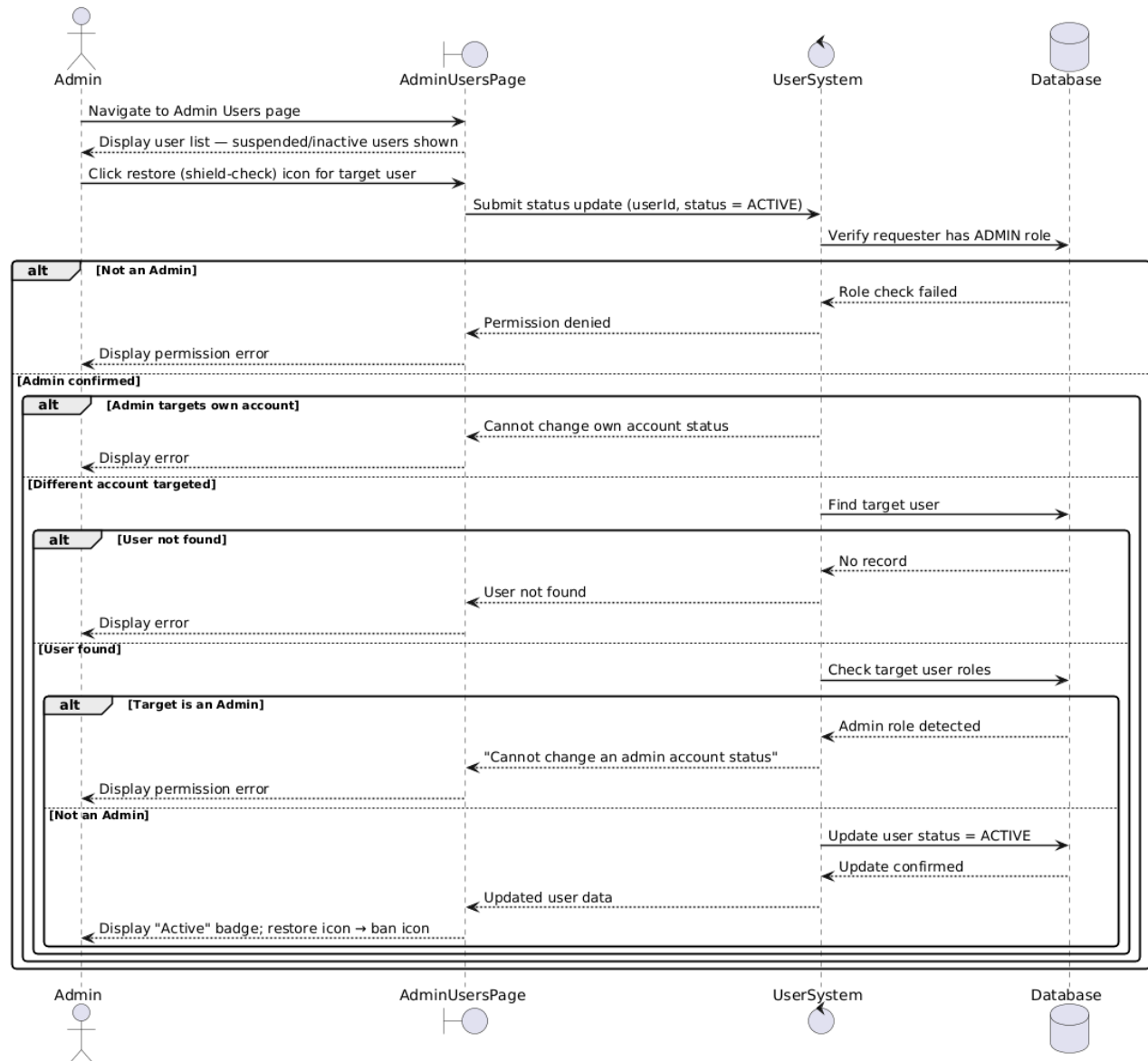


UC-A-04: Activate User

Item	Description
Use Case ID	UC-A-04
Use Case Name	Activate User

Description	Allows an Administrator to restore a Suspended or Inactive user account to Active status, re-enabling the user's ability to log in and use the platform.
Actor(s)	Admin
Priority	Medium
Trigger	Admin clicks the restore (unban) action for a suspended or inactive user on the Admin Users page.
Pre-Conditions	1. The Admin is authenticated. 2. The target user account exists on the platform.
Post-Conditions	1. The user's account status is updated to Active. 2. The user can log in and use the platform normally again.
Basic Flow	1. Admin navigates to the Admin Users page. 2. System displays a list of all registered user accounts. 3. Admin locates the target Suspended or Inactive user and clicks the restore (shield-check) icon. 4. System verifies the Admin's authority. 5. System updates the user's account status to Active. 6. System confirms the change and displays an "Active" status badge for that user.
Alternative Flow	None.
Exception Flow	<i>User Not Found:</i> 1. The target account no longer exists at the time of the

	action. 2. System displays a "User not found" error. <i>Target is an Admin:</i> 1. Admin attempts to delete another Administrator account. 2. System displays a permission error. <i>Self-Deletion:</i> 1. Admin attempts to delete their own account. 2. System displays an error: "Cannot change your own account status." <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message. 3. Admin may retry.
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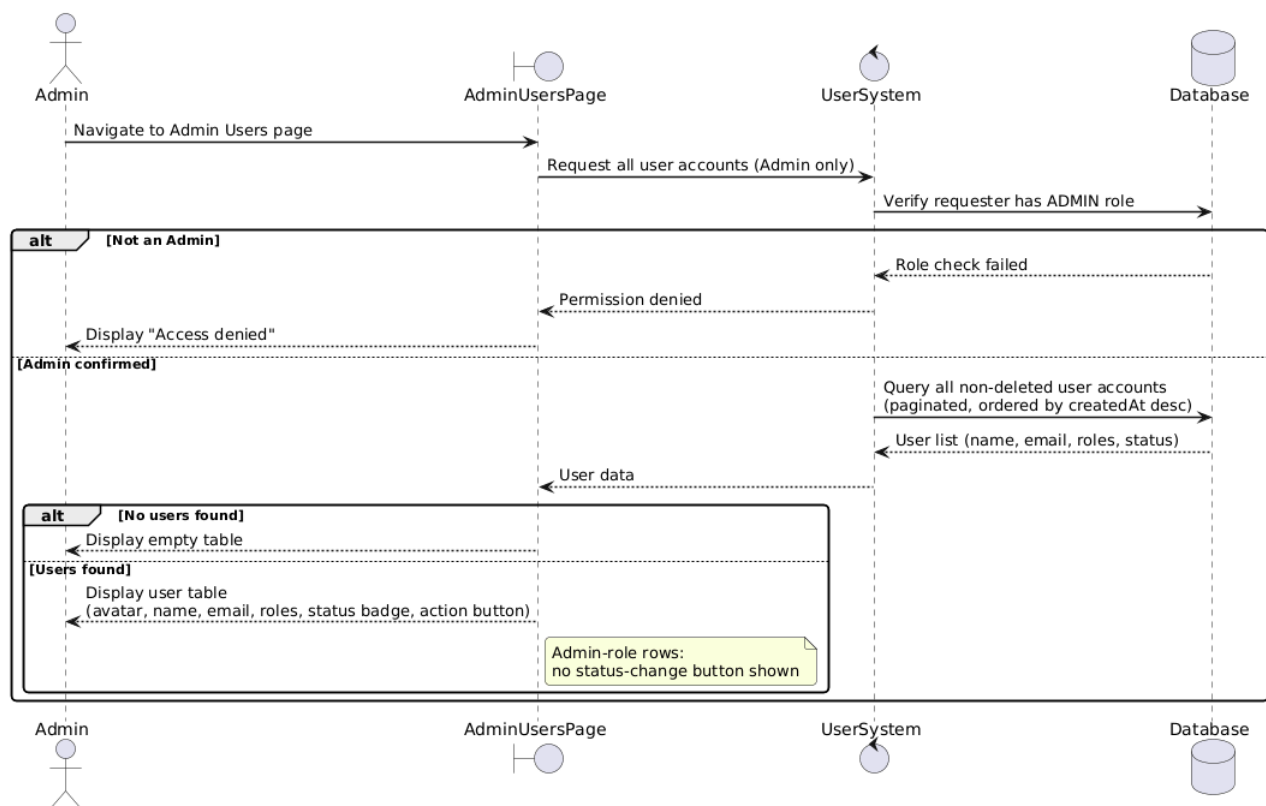


UC-A-05: List Users

Item	Description
Use Case ID	UC-A-05
Use Case Name	List Users

Description	Allows an Administrator to view a paginated list of all registered user accounts on the platform, including each user's name, email address, assigned roles, and current account status.
Actor(s)	Admin
Priority	High
Trigger	Admin navigates to the Admin Users page.
Pre-Conditions	1. The Admin is authenticated.
Post-Conditions	1. A list of all registered user accounts is displayed. 2. No data is modified.
Basic Flow	1. Admin navigates to the Admin Users page. 2. System retrieves all registered user accounts. 3. System displays each account in a table row showing the user's avatar, full name, email address, assigned roles, and current status badge (Active, Inactive, or Suspended). 4. Admin-role rows do not display a status-change action button. 5. Admin may scroll through the list or act on individual users.
Alternative Flow	<i>No Users:</i> 1. No user accounts exist in the system. 2. System displays an empty table.

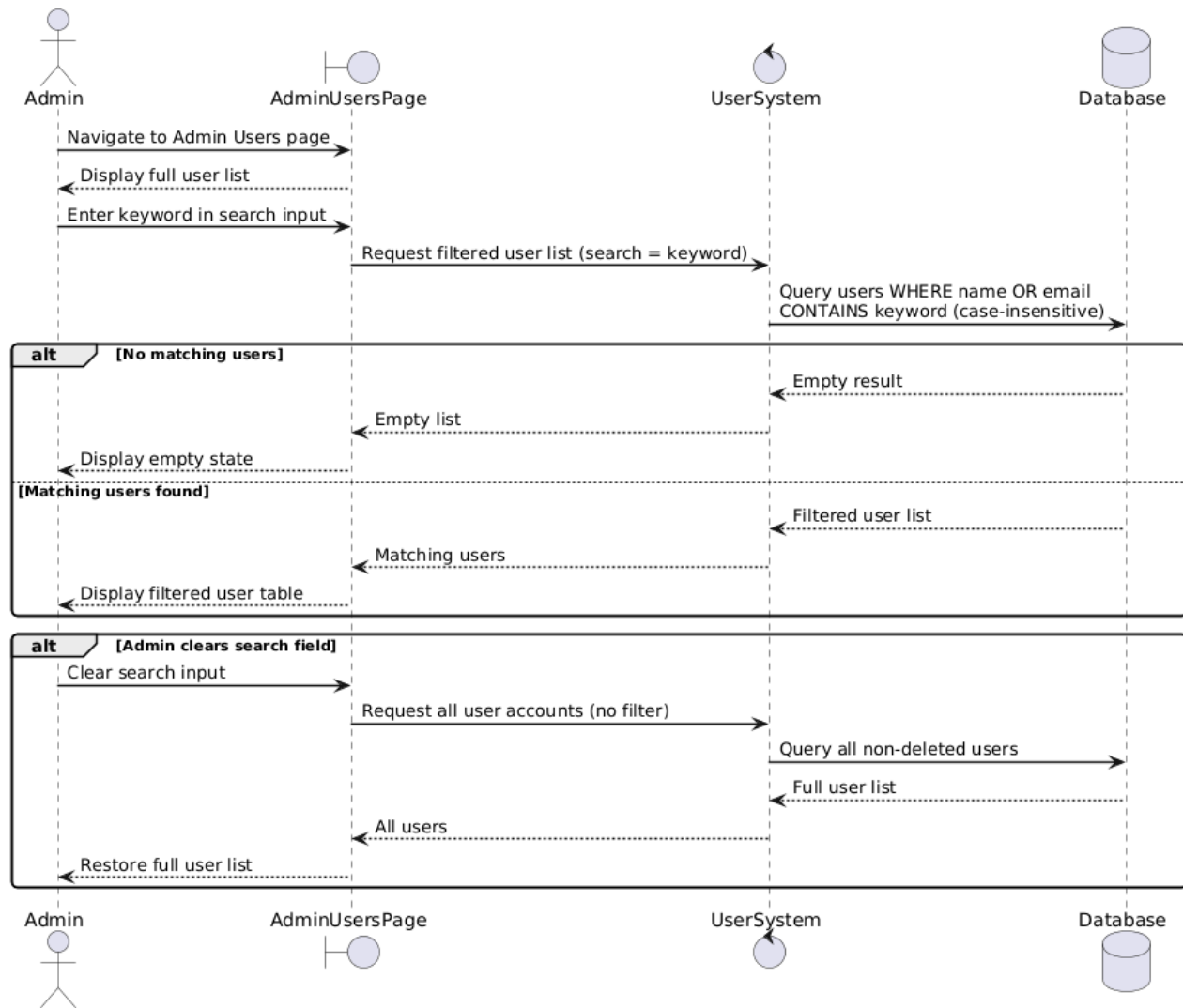
Exception Flow	<i>Not an Admin:</i> 1. A non-Administrator user attempts to access the Admin Users page. 2. System denies access and displays a permission error. <i>System Error:</i> 1. System fails to retrieve user data. 2. System displays an error or loading state. 3. Admin may refresh.
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UC-A-06: Search Users

Item	Description
Use Case ID	UC-A-06
Use Case Name	Search Users
Description	Allows an Administrator to filter the user list by entering a keyword that matches against user names or email addresses, narrowing the results to find a specific account quickly.
Actor(s)	Admin
Priority	Medium
Trigger	Admin enters a keyword in the search field on the Admin Users page.
Pre-Conditions	1. The Admin is authenticated. 2. The Admin Users page is open and displaying the user list.
Post-Conditions	1. The user list is filtered to show only accounts whose name or email matches the entered keyword. 2. No data is modified.
Basic Flow	1. Admin navigates to the Admin Users page. 2. System displays the full user list. 3. Admin enters a keyword in the search input.

	4. System filters the displayed users to show only those whose full name or email address contains the entered keyword (case-insensitive). 5. System updates the table in real time as the Admin types.
Alternative Flow	<i>Clear Search:</i> 1. Admin clears the search field. 2. System restores the full unfiltered user list.
Exception Flow	<i>No Matching Users:</i> 1. The entered keyword does not match any user. 2. System displays an empty table or "No results" message. <i>System Error:</i> 1. System fails to retrieve filtered results. 2. System displays an error message.

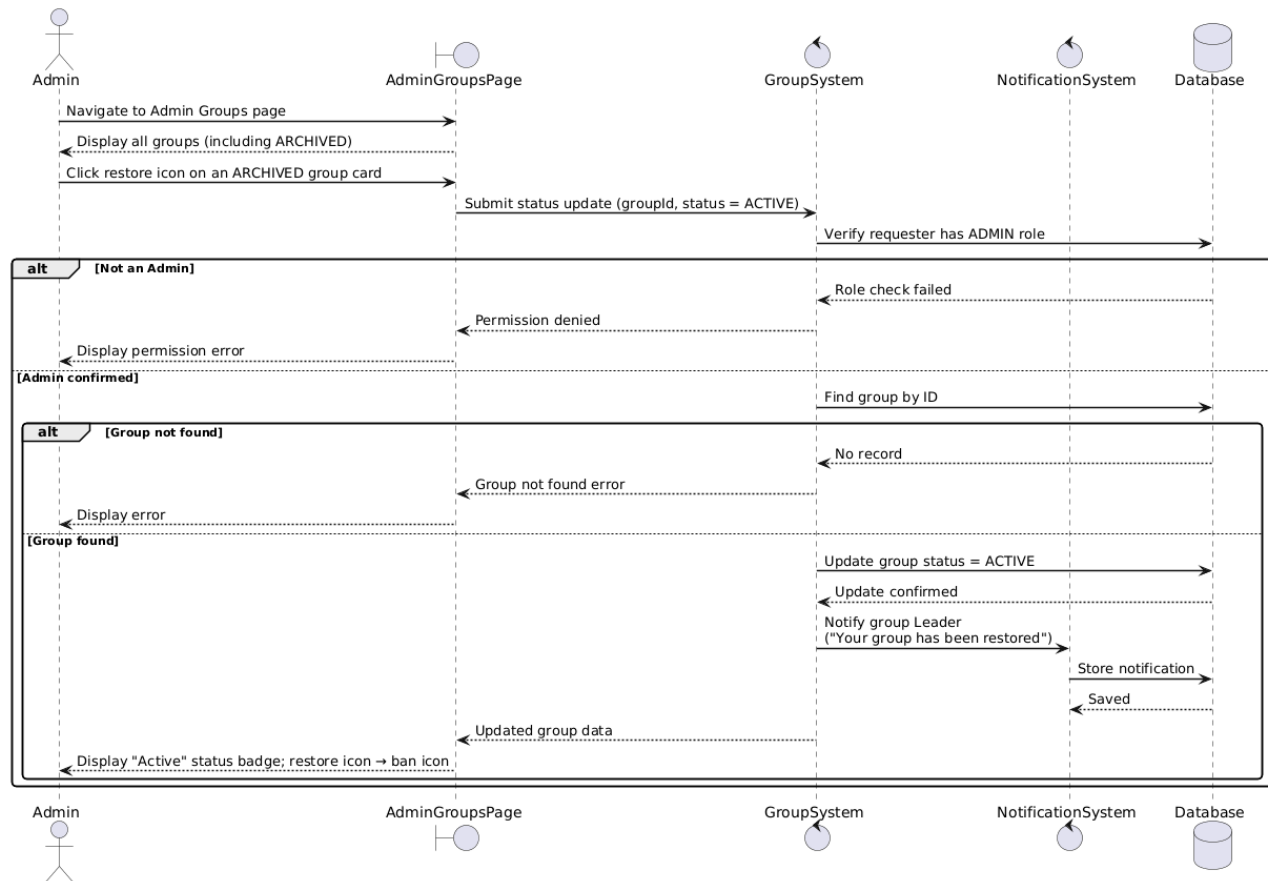


UC-A-07: Restore Group

Item	Description
Use Case ID	UC-A-07
Use Case Name	Restore Group
Description	Allows an Administrator to restore a previously archived study group, returning its status to Active and re-enabling it

	for normal use by members and Leaders. The group's Leader receives a notification when the group is restored.
Actor(s)	Admin
Priority	Medium
Trigger	Admin clicks the restore action for an archived group on the Admin Groups page.
Pre-Conditions	1. The Admin is authenticated. 2. The target group exists and currently has an Archived status.
Post-Conditions	1. The group's status is updated to Active. 2. The group is accessible to members and visible in the group listings. 3. The group's Leader receives an in-app notification that the group has been restored.
Basic Flow	1. Admin navigates to the Admin Groups page. 2. System displays all groups, including archived ones, with their current status. 3. Admin locates the archived group and clicks the restore (shield-check) icon. 4. System verifies the Admin's authority. 5. System updates the group's status to Active. 6. System sends an in-app notification to the group's Leader informing them the group has been restored.

	7. System confirms the change and displays an "Active" status badge on the group card.
Alternative Flow	None.
Exception Flow	<i>Group Not Found:</i> 1. The target group no longer exists. 2. System displays a "Group not found" error. <i>Not an Admin:</i> 1. A non-Admin user attempts to restore a group. 2. System denies the action and displays a permission error. <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message.

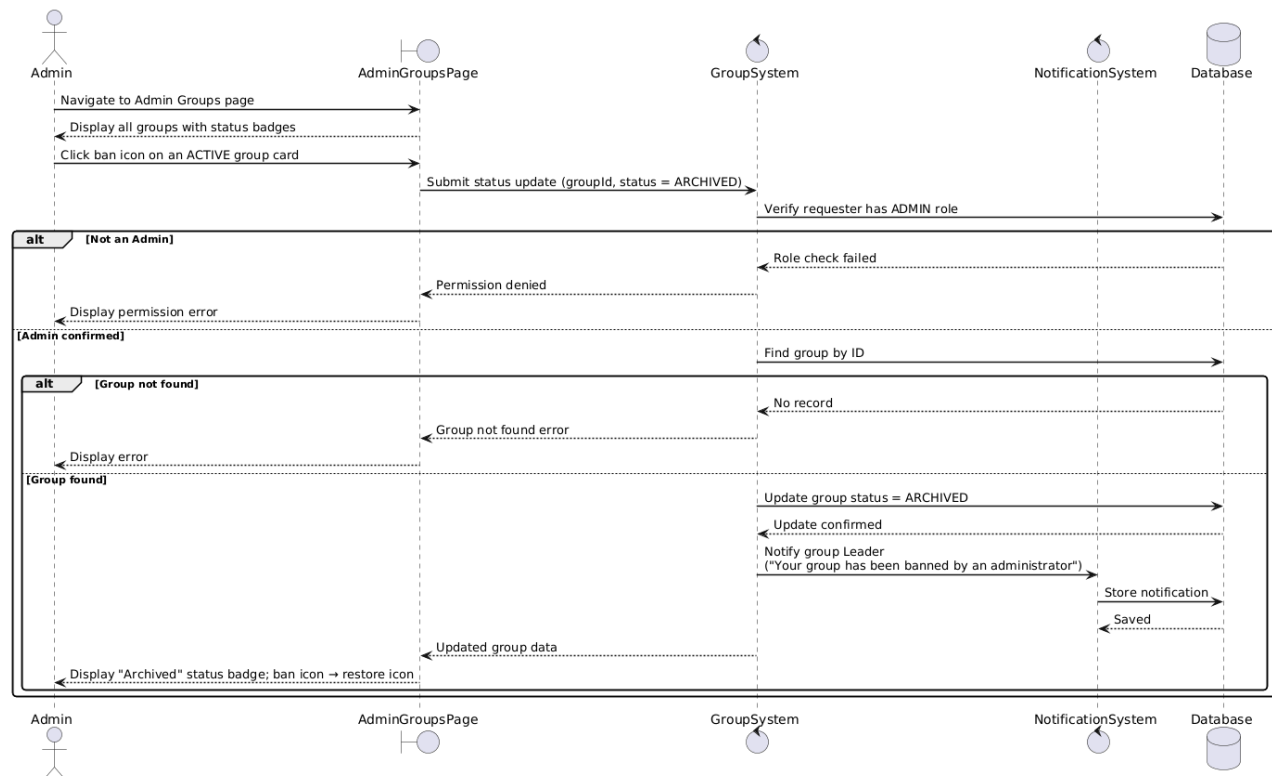


UC-A-08: Archive Group

Item	Description
Use Case ID	UC-A-08
Use Case Name	Archive Group
Description	Allows an Administrator to archive a study group that violates platform policies or is no longer appropriate. Archived groups are hidden from the general group listing and cannot accept new members. The group's Leader receives a notification when the group is archived.

Actor(s)	Admin
Priority	High
Trigger	Admin clicks the archive (ban) action for an active group on the Admin Groups page.
Pre-Conditions	1. The Admin is authenticated. 2. The target group exists and currently has an Active status.
Post-Conditions	1. The group's status is updated to Archived. 2. The group is hidden from the general group listing and is no longer accessible to non-Admin users. 3. The group's Leader receives an in-app notification that the group has been archived.
Basic Flow	1. Admin navigates to the Admin Groups page. 2. System displays all groups with their current status. 3. Admin locates the target active group and clicks the ban (archive) icon. 4. System verifies the Admin's authority. 5. System updates the group's status to Archived. 6. System sends an in-app notification to the group's Leader informing them the group has been archived. 7. System confirms the change and displays an "Archived" status badge on the group card.
Alternative Flow	None.

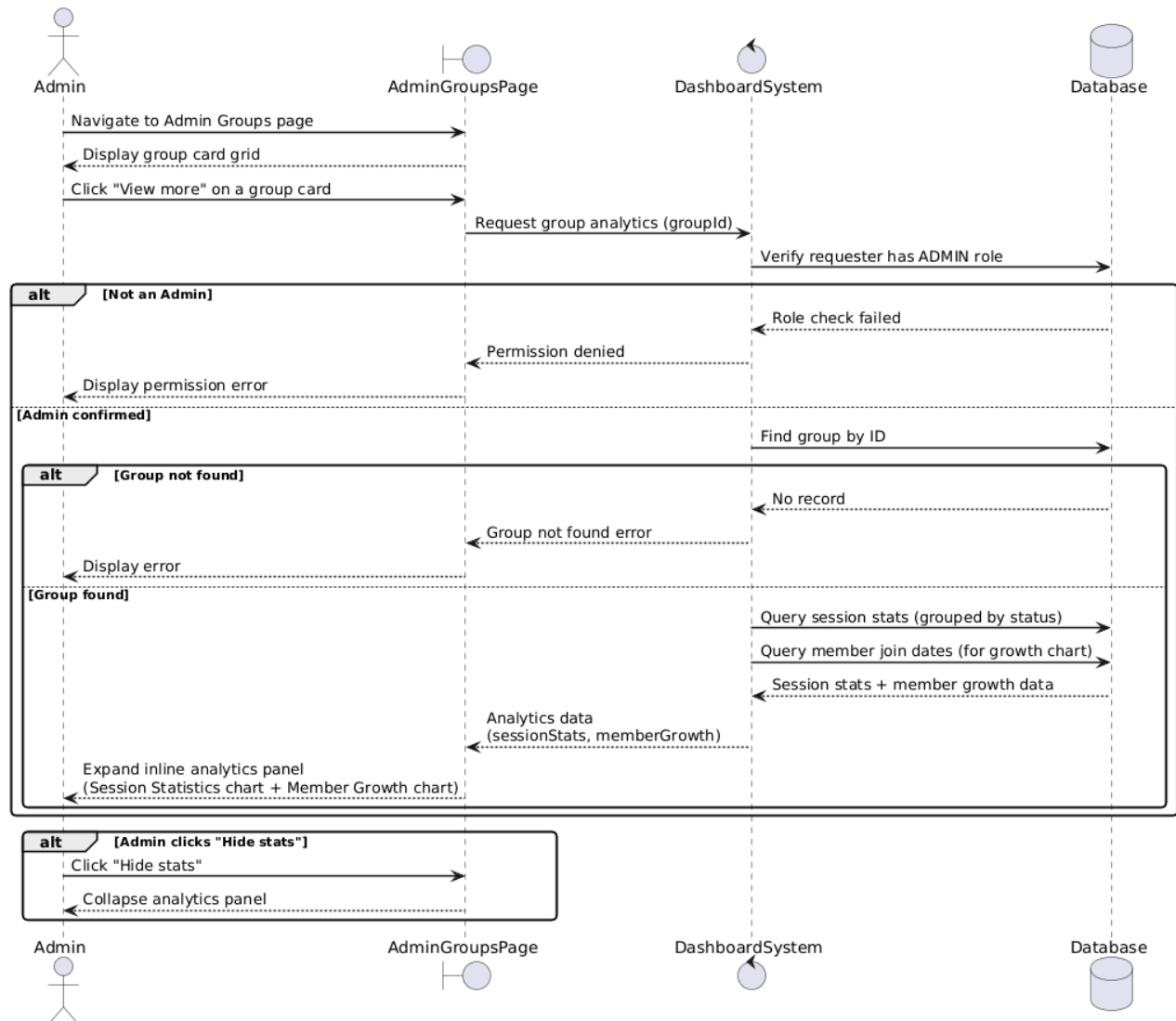
Exception Flow	<i>Group Not Found:</i> 1. The target group no longer exists. 2. System displays a "Group not found" error. <i>Not an Admin:</i> 1. A non-Admin user attempts to archive a group. 2. System denies the action and displays a permission error. <i>System Error:</i> 1. System encounters an unexpected failure. 2. System displays an error message.
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UC-A-09: Group Analytics

Item	Description
Use Case ID	UC-A-09
Use Case Name	Group Analytics
Description	Allows an Administrator to view detailed activity analytics for a specific study group, including a breakdown of session statuses and a chart showing cumulative member growth over time.
Actor(s)	Admin
Priority	Medium
Trigger	Admin clicks the "View more" toggle on a group card in the Admin Groups page to expand its analytics section.
Pre-Conditions	1. The Admin is authenticated. 2. The target group exists on the platform.
Post-Conditions	1. Session statistics and member growth charts for the selected group are displayed inline beneath the group card. 2. No data is modified.
Basic Flow	1. Admin navigates to the Admin Groups page. 2. System displays all groups as a card grid. 3. Admin clicks the "View more" button on the group card they want to inspect.

	4. System retrieves analytics data for the selected group. 5. System displays a Session Statistics bar chart (breakdown by session status) and a Member Growth bar chart (cumulative members by month) directly below the selected group card. 6. Admin may click "Hide stats" to collapse the analytics panel.
Alternative Flow	<i>Collapse Analytics:</i> 1. Admin clicks "Hide stats" on the expanded group card. 2. System hides the analytics panel.
Exception Flow	<i>Group Not Found:</i> 1. The target group no longer exists when analytics are requested. 2. System displays a "Group not found" error. <i>Not an Admin:</i> 1. A non-Admin user attempts to access group analytics. 2. System denies the request and displays a permission error. <i>System Error:</i> 1. System fails to retrieve analytics data. 2. System displays an error or loading state.

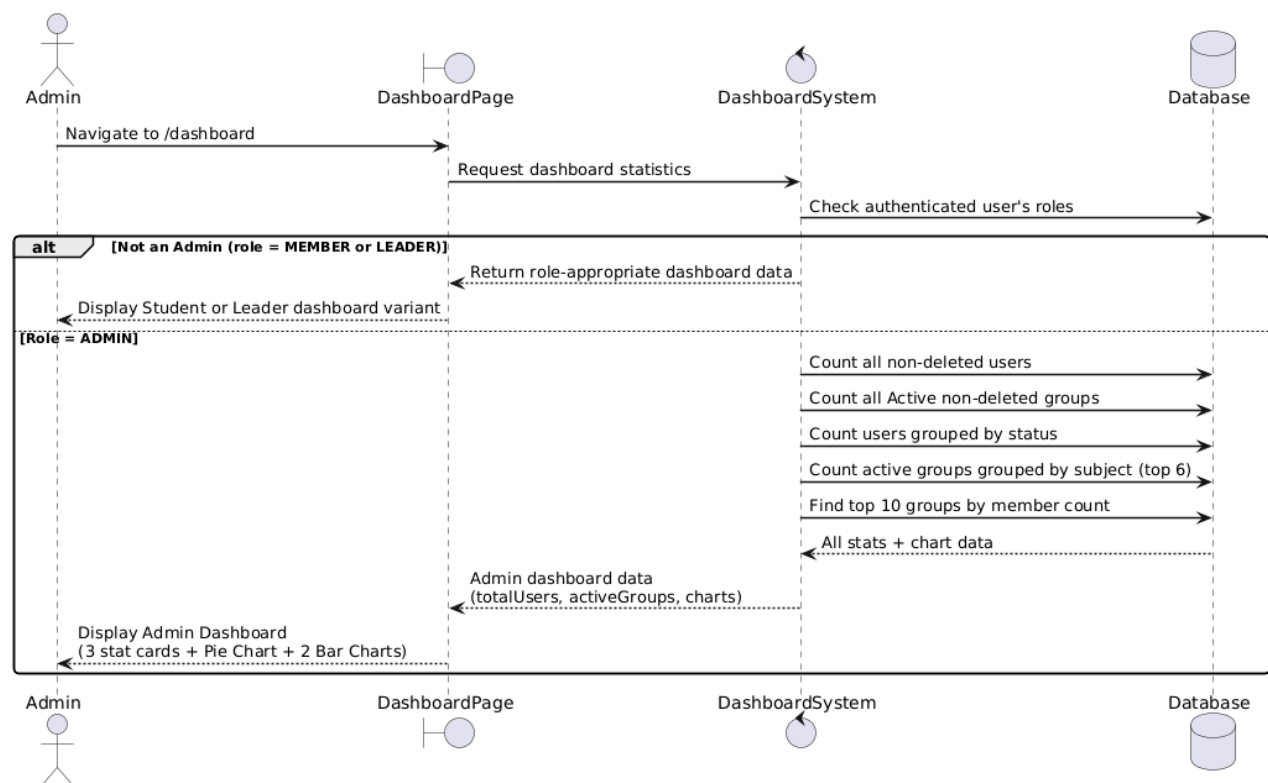


UC-A-10: Platform Dashboard

Item	Description
Use Case ID	UC-A-10
Use Case Name	Platform Dashboard
Description	Allows an Administrator to view a high-level overview of the platform's current state, including total user count,

	active group count, a platform health indicator, and visual charts showing user status distribution and group subject distribution.
Actor(s)	Admin
Priority	High
Trigger	Admin navigates to the Dashboard page.
Pre-Conditions	1. The Admin is authenticated.
Post-Conditions	1. Platform-level statistics and charts are displayed on the Admin Dashboard. 2. No data is modified.
Basic Flow	1. Admin navigates to the Dashboard page. 2. System detects the Admin role and retrieves platform-wide statistics. 3. System displays three summary statistic cards: Total Users, Active Groups, and Platform Health (showing "All systems operational"). 4. System displays a Pie Chart showing user accounts broken down by status (Active, Inactive, Suspended). 5. System displays a Bar Chart showing the number of active groups per subject area (top 6 subjects). 6. System displays a Bar Chart showing the top 10 groups ranked by member count.
Alternative Flow	None.

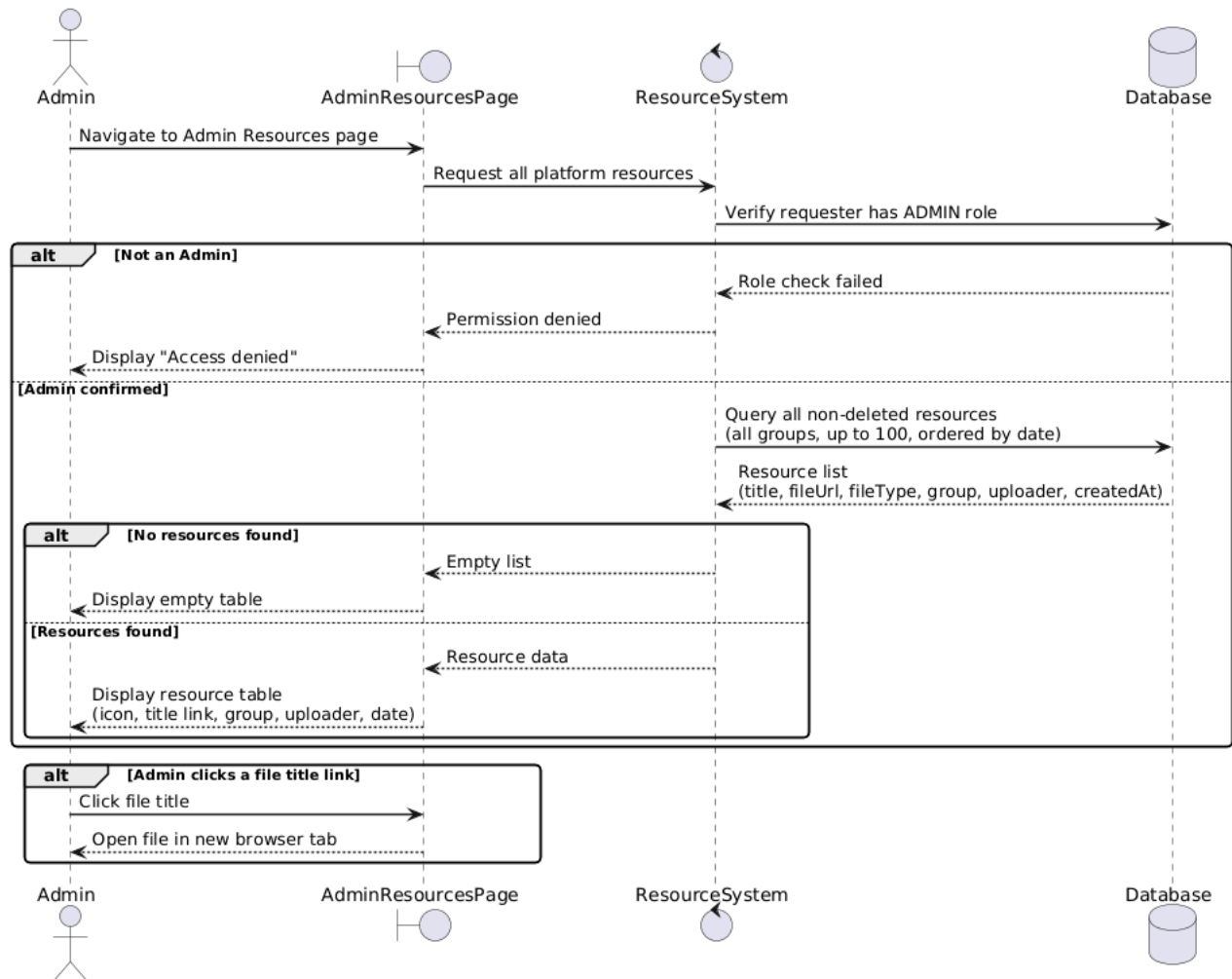
Exception Flow	<i>Not an Admin:</i> 1. A non-Admin user navigates to the Dashboard. 2. System renders the Member or Leader dashboard variant instead. <i>System Error:</i> 1. System fails to retrieve platform statistics. 2. System displays an error or loading state. 3. Admin may refresh.
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UC-A-11: Monitor Resources

Item	Description
Use Case ID	UC-A-11
Use Case Name	Monitor Resources
Description	Allows an Administrator to view a platform-wide listing of all shared resources across all study groups, enabling oversight of uploaded content including the file title, associated group, the user who uploaded it, and the upload date.
Actor(s)	Admin
Priority	Medium
Trigger	Admin navigates to the Admin Resources monitoring page.
Pre-Conditions	1. The Admin is authenticated.
Post-Conditions	1. All platform resources are displayed in a table view. 2. No data is modified.
Basic Flow	1. Admin navigates to the Admin Resources monitoring page. 2. System retrieves all resource files uploaded across all groups on the platform. 3. System displays each file in a table row showing: a file-type icon, the file title as a clickable link opening in a

	new tab, the associated group's name, the uploader's name and avatar, and the upload date. 4. Admin may click any file title to open and inspect the file.
Alternative Flow	<i>No Resources:</i> 1. No files have been uploaded to the platform. 2. System displays an empty table.
Exception Flow	<i>Not an Admin:</i> 1. A non-Admin user attempts to access the Resources monitoring page. 2. System denies access and displays a permission error. <i>System Error:</i> 1. System fails to retrieve resource data. 2. System displays an error or loading state. 3. Admin may refresh.

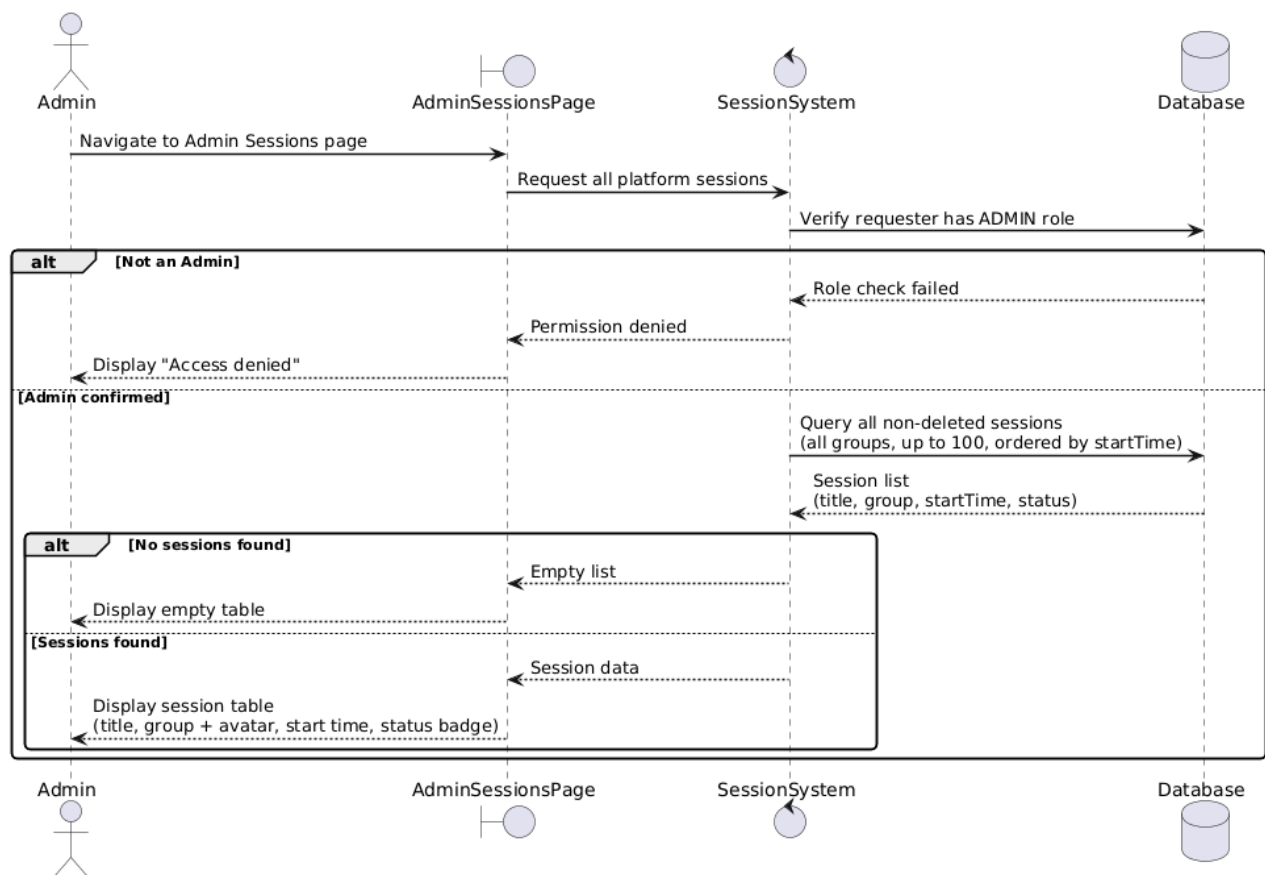


UC-A-12: Monitor Sessions

Item	Description
Use Case ID	UC-A-12
Use Case Name	Monitor Sessions
Description	Allows an Administrator to view a platform-wide listing of all study sessions across all groups, showing each session's

	title, associated group, scheduled start time, and current status, enabling oversight of platform activity.
Actor(s)	Admin
Priority	Medium
Trigger	Admin navigates to the Admin Sessions monitoring page.
Pre-Conditions	1. The Admin is authenticated.
Post-Conditions	1. All platform sessions are displayed in a table view. 2. No data is modified.
Basic Flow	1. Admin navigates to the Admin Sessions monitoring page. 2. System retrieves all sessions across all groups on the platform. 3. System displays each session in a table row showing: the session title, the associated group's name with its avatar, the formatted start date and time, and a color-coded status badge (Scheduled, In Progress, Completed, or Cancelled). 4. Admin may scroll through the list to review session activity across the platform.
Alternative Flow	<i>No Sessions:</i> 1. No sessions have been created on the platform. 2. System displays an empty table.
Exception Flow	<i>Not an Admin:</i> 1. A non-Admin user attempts to access the Sessions

	monitoring page. 2. System denies access and displays a permission error. <i>System Error:</i> 1. System fails to retrieve session data. 2. System displays an error or loading state. 3. Admin may refresh.
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4. External Interface Requirements

4.1 User Interfaces

The SGMS is primarily a web-based application, requiring a responsive and intuitive interface for both Patrons (Customers) and Administrators.

ID	Requirement	Description
UI-1.1	Public Interface & Authentication	The system shall provide a public Landing Page that showcases the platform's key features, active study groups, upcoming sessions, and student testimonials, accessible without an account. The system shall provide a Login screen and a Registration screen where users can sign in or create a new account by entering their name, email, and password. Input errors shall be shown immediately beneath the relevant fields, and all buttons shall visually indicate when an action is in progress.
UI-1.2	Navigation Structure and Layout	The system shall provide a consistent navigation structure on all authenticated pages, consisting of a top bar and a left sidebar. The top bar shall display the current page name, a theme toggle, a notification bell, the signed-in user's name and role, and a logout

		button. The sidebar shall list the main sections: Dashboard, Groups, Sessions, Resources, and Profile, with an additional Admin section visible only to administrators. On smaller screens, the sidebar shall be hidden by default and open as an overlay when the menu button is tapped.
UI-1.3	Role-Based Dashboard	The system shall display a personalised Dashboard that automatically adapts to the user's role. Students shall see their group count, upcoming sessions, attendance rate, and recent resources, along with summary charts. Group Leaders shall see statistics and charts for the groups they manage. Administrators shall see platform-wide totals for users and active groups, along with system health status. Each dashboard view shall present the most relevant information for that role without requiring any manual configuration.
UI-1.4	Group Management Interface	The system shall provide a Study Groups page where users can browse, search, and filter all available groups and request to join them. Each group shall have a dedicated detail page with tabbed sections for Overview, Sessions, Resources, and Discussions. Group Leaders shall have access to an additional

		Requests tab to review and respond to pending join requests, as well as tools to edit the group's information and view group activity charts.
UI-1.5	Session Management and Live Collaboration	The system shall provide a Sessions page where users can view their upcoming and past study sessions in either a list or a calendar layout. Each session shall display its title, group, scheduled time, and current status. Group Leaders shall be able to create and manage sessions, send notifications to members, and mark a session as ended. When a session is in progress, members shall be able to join a live video meeting room directly from the session page. The session detail page shall also display an attendance record showing each participant's status and check-in time, which Leaders can manually update.
UI-1.6	Resource Management Interface	The system shall provide a Resources page where users can browse all study materials shared across their groups, organised into folders. Within each group, members shall be able to create folders, upload files using a drag-and-drop interface, and view uploaded files as a list with their type, title, uploader, date, and star rating. Deleting a file shall require an explicit confirmation step. The upload interface shall

		show the progress of each file individually and report success or failure upon completion.
UI-1.7	Discussion and Voting Interface	The system shall provide a Discussions page where members can read and participate in forum-style conversations across their groups. Members shall be able to create new posts with a title, content, and optional file attachments, and sort posts by most recent or highest voted. Each post's detail page shall display the full content with upvote and downvote controls, and a threaded comment section where members can reply, mention other members using the @ symbol, and vote on individual comments.
UI-1.8	Notifications Interface	The system shall display a notification bell button in the top navigation bar that shows a badge with the number of unread notifications. Clicking the bell shall open a panel listing the user's recent notifications, each with a title, short description, and timestamp; unread items shall be visually highlighted. Users shall be able to mark all notifications as read at once, or mark individual ones by clicking them. Clicking a notification with a linked destination shall navigate the user to the relevant page. The notification count

		shall update automatically in the background every 30 seconds.
UI-1.9	Profile Management Interface	The system shall provide a Profile page where users can view their account summary and update their personal information. The page shall display an overview of the user's activity (groups joined, attendance rate, resources uploaded), followed by an editable section for their name, avatar, and bio. It shall also list the groups they belong to and the files they have uploaded. Changes shall be saved with a single button, which shall confirm success or display an error message accordingly.
UI-1.10	Administration & Search Interface	The system shall provide an Administration panel, accessible only to Administrators, for managing platform users and study groups. The Users page shall display all registered accounts in a table with their name, email, roles, and account status, and shall allow Administrators to suspend or restore individual accounts. The Groups page shall display all study groups with the option to archive or restore them and view per-group activity charts. Across the platform, keyword search shall be available on the Groups,

		Resources, and Admin Users pages, allowing users to filter displayed results by typing in a search field.
UI-1.11	Data Visualization & Loading States	The system shall present activity statistics as interactive Bar Charts and Pie Charts on Dashboard and administration pages, with readable tooltips that match the platform's visual theme. While any page content is being loaded from the server, the system shall display animated placeholder blocks that mirror the shape of the expected content, providing a smooth loading experience without blank screens or abrupt layout shifts.
UI-1.12	Design Standards, Theme & Responsiveness	The system shall maintain a consistent visual style across all pages using a shared set of colours, spacing, and shape values, with smooth visual transitions on interactive elements. The system shall support both a light and a dark colour theme, switchable at any time from the top navigation bar, with the user's preference saved across sessions. All pages shall be fully usable on screen sizes from mobile phones to desktop monitors, with layouts that adapt their column structure to the available screen width. Any changes made on the platform (such as creating or deleting

		content) shall be reflected immediately in the interface without requiring a manual page refresh.
UI-1.13	Non-Functional UI Standards	The system shall meet the following interface quality standards. All buttons that display only icons shall include descriptive labels for screen reader users. Form fields shall be properly labelled so that assistive technologies can identify their purpose. Status information shall always be communicated through text in addition to colour. All interface text, labels, and messages shall be written in English. Dates and times shall appear in a consistent, human-readable format throughout the platform. The application shall operate correctly on current versions of Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari.

4.2 Software Interfaces

The SGMS is built as a Server-Side Rendered (SSR) monolithic application, and relies on several internal and external software interfaces.

ID	Requirement	Interface Detail

SI-2.1	API Structure and Response Format	The system shall expose all features through a unified API. Every response shall include a status indicator, a message, and the relevant data. Error responses shall additionally list the specific issues encountered. Results from listing endpoints shall be returned in pages, with information about the total number of records available. A publicly accessible status endpoint shall allow monitoring tools to confirm the server is running.
SI-2.2	Authentication & Token Management	The system shall allow users to register, log in, and log out. Upon login, the server shall issue a short-lived credential for making requests and a longer-lived credential for renewing access without logging in again. The server shall refuse access to requests with missing or expired credentials and shall deny login to suspended accounts. When a user logs out, their renewal credential shall be invalidated immediately.
SI-2.3	Access Control	The system shall restrict each action to users who have the appropriate role. Platform-wide administrative actions shall only be available to Administrators. Group-level actions such as approving members or managing sessions shall only be available to the Leader of that specific group or the person who

		created the resource. Actions that require group membership shall be blocked for users who are not active members of the relevant group.
SI-2.4	Group Management API	The system shall provide endpoints to browse, create, update, and manage the status of study groups. Anyone may view the group listing and group details. Authenticated users may create groups and submit or withdraw join requests. Group Leaders may respond to pending join requests and edit their group's information. Administrators may archive or restore any group. The server shall prevent duplicate join requests and reject requests to join groups that are full or inactive.
2.5	Session Management and Live Meeting API	The system shall provide endpoints to create, update, and end study sessions. Group members may schedule sessions for a future time or start one immediately. Leaders and session creators may notify all group members about a session and end it when finished. The server shall provide a secure access credential so that members can join the live video room for a session that is currently in progress. The server shall

		reject live meeting requests if the session is not active or if the live meeting service is unavailable.
2.6	Attendance API	The system shall provide endpoints to record and view session attendance. When a member joins a live session, their attendance shall be automatically recorded as Present. Session creators may also manually record or update an attendance status for any group member. The server shall provide the full attendance list for a session, showing each participant's name, status, and the time they joined.
2.7	Resource & File Upload API	The system shall provide endpoints to manage study material folders and files within groups. Members may create folders and upload files into them. Leaders may edit or remove folders. Uploaded files shall be stored in an external cloud storage service; the server facilitates the upload by issuing a temporary permission, and only stores a reference to the file after upload. Members may star files they find useful. Only the person who uploaded a file or a group Leader may delete it.

2.8	Discussion and Voting API	The system shall provide endpoints for forum-style discussions within groups. Members may create, edit, and vote on posts and comments. Voting the same way twice shall remove the vote. Members may reply to comments and mention other members by name. The server shall automatically notify mentioned members and the author of a comment when someone replies to it.
2.9	Notification API	The system shall provide endpoints to retrieve and manage a user's in-app notifications. Users may view their notification list, check how many they have unread, and mark one or all notifications as read. The server shall generate notifications automatically when key events occur on the platform, such as a session announcement, a join request decision, a reply to a comment, or a change to a group's status.
2.10	Dashboard & Reporting API	The system shall provide a single dashboard endpoint that returns a personalised summary adapted to the requesting user's role — showing personal activity data for Members, group management data for Leaders, and platform-wide statistics for Administrators. The system shall also provide a reporting endpoint where any user may flag content

		for review, and Administrators may view all submitted reports and record their resolution.
2.11	Data Validation & Error Handling	The system shall check all submitted data for completeness and correctness before acting on it. If data is missing, in the wrong format, or outside allowed limits, the server shall return a clear error response listing the specific problems so the user can correct them. Attempts to create duplicate records shall be rejected with a conflict error. Requests to non-existent endpoints shall return a not-found response. Unexpected server failures shall return a generic error message in production to avoid exposing internal details.
2.12	Third-Party Service Integrations	The system shall connect to two external services to support its core features. For live video meetings, it shall integrate with LiveKit to manage meeting rooms and issue participant access credentials. For file storage, it shall integrate with Cloudinary, allowing files to be uploaded directly by the client while the server controls the authorisation. Both services shall be configured privately on the server and their credentials shall never be shared with the client.

2.13	Security & Environment Configuration	The system shall apply security protections to all responses and only accept requests from the authorised client application. All sensitive credentials — including the database connection and signing secrets — shall be loaded from environment variables, and the server shall refuse to start if any required value is absent. The server shall reject request payloads larger than 10 MB.
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4.3 Hardware Interfaces

The primary hardware interface requirements relate to the client devices and the payment/entry infrastructure at the cinema.

ID	Requirement	Description
HI-3.1	Client Devices	The system shall be fully accessible and usable on standard consumer hardware, including desktop computers, laptops, tablets, and smartphones. It shall not require users to purchase or install any specialised hardware to operate the platform.

HI-3.2	Audio & Video Capture	The system shall connect to standard webcams and microphones built into or attached to the user's device to facilitate live study sessions. Access to these hardware peripherals shall be managed respectfully through standard browser permission prompts.
HI-3.3	Audio Playback	The system shall route sound through the device's default audio output hardware, such as built-in speakers or connected headphones. This ensures users can hear other participants clearly during live meetings and video conferences.
HI-3.4	Server Infrastructure	The system's central backend services shall be capable of running on standard, commercially available cloud hosting hardware. It shall not depend on bespoke, proprietary, or highly specialized server equipment to function correctly.
HI-3.5	Network Connectivity	The system shall operate reliably over standard internet connections, including Wi-Fi, wired ethernet, and mobile cellular networks (4G/5G). It shall be designed to function adequately across the typical varying

		network speeds found in homes and educational institutions.
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4.4 Communications Interfaces

ID	Requirement	Description
CI-4.1	Network Protocol	All client-server communication must use the secure HTTPS protocol.
CI-4.2	Bandwidth	The system shall function effectively and meet performance requirements (see Section 5.1) over a typical consumer broadband connection.

5. Other Nonfunctional Requirements

This section specifies the qualitative attributes the SGMS must possess.

5.1 Performance Requirements

ID	Requirement	Description
PE-1	Page Load Speed	The system shall load interface pages quickly to provide a smooth and frustration-free user experience. Under normal network conditions, the initial page load should complete within a few seconds, and subsequent page navigations should feel instantaneous.
PE-2	Server Response Time	The system shall process standard user actions—such as saving a profile update, submitting a comment, or uploading a small file—rapidly. The platform should aim to acknowledge and complete these common tasks in under a second to keep users engaged.
PE-3	Live Meeting Performance	The system shall maintain smooth video and audio quality during live study sessions. A single meeting room should be capable of comfortably accommodating a standard study group size (e.g., up to 20 active participants) without experiencing severe lag or dropped connections.

5.2 Security

ID	Requirement	Description
SEC-1	Data Encryption in Transit	The system shall protect all information traveling between the user's device and the server using industry-standard secure connections (HTTPS). This ensures that passwords, personal details, and study materials cannot be intercepted or read by unauthorised parties over the internet.
SEC-2	Secure Password Storage	The system shall never store user passwords in plain text. All passwords must be transformed using strong, industry-standard cryptographic hashing before being saved to the database, ensuring they remain safe even if the data storage is compromised.
SEC-3	Session Security	The system shall protect unattended user accounts by automatically expiring login sessions after a prolonged period of inactivity. It shall also provide a reliable logout feature that immediately invalidates the user's access, ensuring security on shared or public devices.

5.3 Extensibility

ID	Requirement	Description
AVL-1	Modular Design for Extensibility	The system shall be built using a flexible, modular software architecture. This ensures that new features—such as calendar integrations, new resource types, or automated grading tools—can be added by developers in the future without requiring a complete rebuild of the existing platform.
AVL-2	Scalable Infrastructure	The system shall be designed to grow seamlessly alongside its user base. The underlying server setup must allow for easy expansion, meaning additional server capacity can be quickly added to handle increased traffic during peak periods, such as university exam seasons.

6. Business Rules

These are policies, constraints, or computations that govern the business domain of movie ticketing.

ID	Rule	Description
BR-1	Account Status Enforcement	Users with an Inactive or Suspended account status cannot log in to the platform or refresh their existing sessions.
BR-2	Suspension Consequence	Suspending a user account must immediately revoke all of the user's active login sessions across all devices.
BR-3	Admin Immunity	An Administrator's account status cannot be changed, deactivated, or suspended by another Administrator.
BR-4	Join Request Exclusivity	A user cannot submit a join request to a study group if they are already an active member or currently have a pending request for that group.

BR-5	Join Request Approval	A user only becomes an official group member after a Group Leader explicitly approves their pending join request.
BR-6	Group Creator Ownership	The user who creates a study group is automatically assigned the role of Leader within that group.
BR-7	Group Capacity Limit	A group cannot accept new members or new join requests if its active member count has reached its defined maximum capacity.
BR-8	Group Archival Consequence	An Archived group is hidden from general listings and cannot accept new members or new activity (sessions, posts, uploads).
BR-9	Group Management Authority	Only a Leader of a group can modify the group's details, change its status, or approve/reject pending join requests.
BR-10	Admin Group Override	Administrators have the authority to archive or restore any group on the platform, regardless of whether they are a member of that group.

BR-11	Session Scheduling	A scheduled session must have a defined start time in the future and a explicitly defined end time.
BR-12	Live Session Initiation	A session can be started immediately (status set to In Progress) without requiring a predefined end time.
BR-13	Session Modification	Only the user who created a session or a Group Leader can update its details, end it prematurely, or cancel it.
BR-14	Live Meeting Access	Only active members of the associated group can join a live video meeting room for a session currently in progress.
BR-15	Session Completion States	Ending an "In Progress" session marks its status as "Completed". Ending a "Scheduled" session marks its status as "Cancelled".
BR-16	Attendance Auto-Tracking	A member's attendance is automatically recorded as "Present" when they successfully join the live meeting room.

BR-17	Resource Upload Permission	Only active members of a group can upload files to that group's resource folders.
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